Coworker Influence on Job Choice: Information, Connection, and Industry Switching

Xinyue Lin Armando Miano Sophia Mo*

October 10, 2026

Abstract

How do coworker networks shape workers' perceived outside options and sectoral mobility? To answer this question, we design and implement a survey with 2,721 full-time U.S. wage-and-salary workers ages 25–60, reweighted to CPS benchmarks. Stronger coworker ties to a destination sector are associated with more accurate beliefs about wages in that sector, higher perceived probabilities of receiving an offer, and greater intentions to apply. In the conjoint experiment, connection with a former coworker at the hiring firm raises the probability that workers perceive to receive an offer from that firm by the most compared to all other job attributes. Using these empirical findings, we calibrate a partial-equilibrium sector-choice model under past and current coworker influence. The two channels work in opposite directions: coworker-provided wage information improves the allocation of workers across sectors, whereas coworker-linked perceived access tilts applications toward destinations paying closer to what workers already earn. Measured on wages, the two effects are comparable in size, so that neither the reduced-form estimates nor their sum reveals the net consequence of coworker networks. Correcting cross-sector wage misperceptions entirely would be worth 2.23 percent of permanent consumption, and coworker-provided information accounts for approximately 3 percent of that gain. Because what matters is the direction of ties rather than their number, redirecting the networks of the lowest-paying sectors toward higher-paying destinations, without adding any connections, is worth roughly six times as much as broadening them.

Keywords: Job Mobility, Job Search, Coworker Networks, Industries, Survey, Subjective Expectations.

JEL Codes: J01, J62, D91, D83, E71

*Lin: Harvard (e-mail: xinyue_lin@g.harvard.edu); Miano: University of Naples Federico II, CSEF, and CEPR (e-mail: armando.miano@unina.it); Mo: Harvard (e-mail: yiningmo@fas.harvard.edu). We gratefully acknowledge support from the Chae Family Research Fund and the Molly and Dominic Ferrante Economics Research Fund. This research is also funded by the European Union - Next Generation EU, in the framework of the GRINS - Growing Resilient, Inclusive and Sustainable Project (GRINS PE00000018 – CUP E63C22002140007). The views and opinions expressed are solely those of the authors and do not necessarily reflect those of the European Union and the European Union cannot be held responsible for them. Harvard IRB approval IRB24-0384.

1 Introduction

Workers spend a substantial portion of their lives alongside their coworkers, sharing experiences, information, and skills that shape their career trajectories. While a rich literature has documented the role of social networks in shaping labor market outcomes, less attention has been given to how interactions with coworkers specifically influence job mobility, and through which channels those effects operate. This gap is particularly notable given that the average American spends more time at work than in almost any other social setting, that job-to-job transitions are key drivers of wage growth and career advancement, and that a great number of public policies aim to promote labor mobility and workplace integration.[1]

Understanding the extent and mechanisms through which coworkers shape job mobility is particularly relevant in today's labor market. These insights are essential for designing effective policies to support mobility, improve labor allocation, and reduce inequality in access to opportunities. In this paper, we leverage novel survey data to study how both current and former coworkers influence workers' job search and mobility decisions. We explore several core questions: Do coworkers affect whether workers switch jobs or industries? Through which channels, informational or referral-based, do these influences operate? Which types of coworkers, past or current, are most influential? Finally, once the informational and access channels are separated, what is each one worth for worker welfare, and do they reinforce or offset one another?

The challenge in addressing these questions lies in the fact that coworkers, wage beliefs, job or industry switching decisions, and job-offer probabilities are all difficult to observe in standard datasets. To overcome this, we designed and administered an online survey to a quota-balanced sample of 2,721 full-time wage and salaried workers in the U.S., reweighted to match CPS benchmarks.[2] The survey responses provide new insights into the extent and nature of coworker influence on job choice, shedding light on when and how coworkers shape mobility decisions. Specifically, we find that having more coworkers previously employed in an industry increases belief accuracy on the wages one could earn in that industry, as well as the perceived probability of receiving an offer if applying to a job in that industry. Having more past coworkers currently employed in an industry also raises the perceived job-offer probability when applying to jobs in that industry. In addition to influencing the perceived job-offer rate, having more past coworkers currently employed in an industry also leads to a higher intention to apply to those industries. Finally, the influence of coworkers on perceived job-offer rates not only applies to the industry choice dimension but also extends to the job choice level. Having connections at a job that one can reach out to is associated with a significant increase in the perceived job-offer rate, holding other job attributes constant.

[1] For example, U.S. mobility-supporting policies include retraining programs and tax incentives that facilitate job transitions. Equal employment laws, affirmative action, immigration programs, and workplace safety regulations also promote integration of diverse populations into the workforce.

[2] We do not sample workers in the healthcare industry because the skills of these workers are generally very specific to that industry and less transferable.

In the survey, we first investigate how coworkers influence respondents' *future* industry and job choices. We examine the role of current and past coworkers *at the current job* in shaping workers' perceived outside options, including their perceived wages and job-offer rates, and their intentions to switch industry. To begin, we ask each respondent to report the number of current and past coworkers with whom they interact, and whose previous or current industries of employment they know. We then present each respondent with five industries selected using the LEHD job-to-job transition matrix: the three industries with the highest out-transition flows from the respondent's current industry—which include the respondent's own industry, since remaining in one's current industry is the most common outcome—and the two industries with the lowest out-transition flows. We request that respondents list the number of their current coworkers who were previously employed or their previous coworkers who are currently employed in one of these industries. Furthermore, we ask respondents about their beliefs regarding the median wage and their expected starting wage in each of these five industries. We also ask them to indicate whether they would consider applying to a job in each of the five industries and to estimate the likelihood of receiving an offer if they applied to a typical job in those industries.

We find that having more current coworkers who were previously employed in a given industry is associated with more accurate beliefs about the median wage in that industry, even after controlling for how commonly workers move between industries – a proxy for industries' actual similarity or connection. Additionally, having more past coworkers who are currently employed in a particular industry raises respondents' perceived likelihood of receiving a job offer in that industry. This effect holds even after accounting for the typical difficulty of transitioning between the respondents' current industry and the reference industry, as well as the respondents' expected starting wage in the target industry. Quantitatively, having all past coworkers currently employed in a given industry increases the perceived job-offer probability by 11 percentage points, roughly a 29% increase relative to the average perceived probability of 38%. It also raises the likelihood of expressing an intention to apply to that industry by 29 percentage points without controls, and by 23 percentage points in our preferred specification, compared to an average application intention of 50%.

Given that the relationship between coworkers and future job-switching intentions reflects stated preferences, which may be subject to reporting or hypothetical bias, we next examine the relationship between individuals' past coworkers from previous jobs and their current job choices, using a realized outcome: their current wage. Controlling for individual characteristics and the difficulty of transitioning between the worker's previous and current industries (for their own occupation), we find that having at least one past coworker who mentioned the respondent's current industry of employment is associated with a 13 log-point higher current wage. Receiving a direct referral from a past coworker is associated with a 20 log-point higher wage in the unconditional specification (approximately 22%, or roughly $18,500 in additional annual earnings evaluated at our sample mean of $83,840); this association falls to 0.05 log points and is no longer statistically distinguishable from zero

once the full set of controls and fixed effects is included.

While our previous questions allow us to conclude that individuals associate industries where more of their past coworkers are currently employed with higher job-offer probabilities, we cannot directly quantify the perceived increase in job-offer probabilities due to specific coworker connections. Since we ask about job-switching decisions at the industry level, the effect of coworker connections, which often applies to specific jobs rather than entire industries, remains unclear. To address this gap, we designed a conjoint experiment where each respondent is presented with four pairs of hypothetical job profiles that vary by wage, required skill level, flexibility in work arrangements, and the degree of connection they have through former coworkers.

In this conjoint experiment, respondents are asked to indicate their perceived probability of receiving a job offer for different jobs, which vary in terms of wages, skill requirements, schedule and workplace flexibility, and, crucially, personal connections through previous coworkers. We incorporate these by presenting respondents with realistic scenarios in a fictional job application setting. Our findings reveal a strong positive association between the degree of coworker connection (low, medium, or high) and perceived job-offer probabilities. Specifically, in the forced-choice specification, having a high degree of personal connection raises the probability that a job is judged more likely to yield an offer by 11.4 percentage points, making it the most influential factor among all attributes tested; in the slider-probability specification, the corresponding effect is 3.9 to 4.3 percentage points. This association holds across demographic groups, though it is somewhat stronger among active job seekers compared to those who are not currently looking for jobs.

To study the implications of coworkers' influence on search behavior and welfare, we develop a stylized model of job choice that incorporates the influence of coworkers. Departing from standard labor market models that assume workers have full information about wages and job-offer probabilities, we instead allow both perceived wages and job-offer beliefs to depend on the composition of a worker's coworker network. Specifically, we model perceived wages as more accurate in sectors where an individual has more current coworkers with prior experience, and we allow the perceived probability of receiving a job offer to increase with the number of both the past coworkers currently employed at a sector and the current coworkers previously coming from a sector. These assumptions are directly supported by our reduced-form analysis, and the key parameters governing the strength of coworker influence on wage beliefs and job-offer probabilities are estimated using our survey data. This allows us to quantify the relative importance of the two channels, information about wages and referral-based access, and to assess their implications for job mobility and worker welfare through the lens of the model.

Finally, we use the model to quantify the partial-equilibrium, steady-state welfare effects of coworker networks. Survey estimates discipline how coworker ties shape wage beliefs and perceived access, while destination-specific switching costs are calibrated to LEHD Job-to-Job Flows and held fixed across counterfactuals. The two coworker channels have opposite model-implied effects. Remov-

ing coworker-provided wage information reduces welfare by 0.07 percent of permanent consumption and lowers welfare for workers in all 19 origin sectors. Removing coworker-linked perceived access, by contrast, raises welfare by 0.18 percent and benefits workers in all 19. Both counterfactuals flatten a matrix and therefore raise the option value of workers' choice sets mechanically, so we read the channels through their wage components, which isolate reallocation: removing information costs 0.14 percent in wages, while removing access gains 0.10 percent. We report the two channels separately rather than netting them, since the residual between effects estimated from different moments is not a quantity the model determines. Reduced-form estimates alone cannot reveal these welfare consequences. Although coworker ties are associated with both more accurate wage beliefs and greater perceived access,the allocative effect of access also depends on the destinations toward which it directs applications. Separating these forces is therefore a central contribution of the model. The information coworkers provide accounts for approximately 3 percent of the gain from correcting cross-sector wage misperceptions entirely, which is worth 2.23 percent of permanent consumption.

The welfare effect of coworker-linked access depends not only on the total intensity of workers' ties but also on their destination composition. Because ties are not spread evenly across destinations, coworker-linked access raises perceived offer probabilities unevenly across the moves a worker might make. To isolate the role of destination composition, we permute link destinations while preserving, within each origin sector, both total tie intensity and the empirical distribution of link weights. The observed LEHD-flow-based network proxy produces a stronger wage gradient in applications than all 999 destination-randomized networks (randomization $p = 0.001$), so it is the composition of ties rather than their density that matters. A model counterfactual prices what changing that composition would be worth. Redirecting the ties of the six lowest-wage origin sectors toward the six highest-wage destinations, holding total tie intensity fixed so that no worker gains connections, generates a welfare gain roughly six times as large as broadening those same networks in proportion to employment, and the gain is almost entirely a wage effect rather than a change in the option value of workers' choice sets. It raises the annual transition rate from the low-wage group to the high-wage group from 2.95 to 3.07 percent, a 4 percent increase, and reallocates 0.4 percent of the workforce upward in the long run. The advantage of directing over broadening widens as programmes shrink: at a reach of one in ten workers, broadening delivers essentially nothing while directing still delivers a positive gain.

These findings speak to several literatures. Models of job search and mobility (e.g., Jovanovic 1979; Burdett and Mortensen 1998; Mortensen and Pissarides 1999; Cahuc, Postel-Vinay and Robin 2006) treat job switching as a decision based on wage differentials and match quality. Recent work has expanded this framework to incorporate non-wage job characteristics, providing evidence on workers' valuation of non-wage amenities (e.g., Sullivan and To 2014; He, Neumark and Weng 2021; Sockin 2022). These studies reveal that workers consider multiple dimensions when evaluating employment opportunities. Our contribution to this literature lies in emphasizing the role of non-job-related factors, such as coworkers, which can also significantly influence job-switching decisions.

Closest to our setting is the literature on how social networks influence labor market decisions. Building on foundational works by Rees (1966), Granovetter (1973), and Montgomery (1991), more recent studies (e.g., Ioannides and Loury 2004; Calvo-Armengol and Jackson 2004; Topa 2001; Dustmann et al. 2016; Caldwell and Harmon 2019) have demonstrated the importance of social ties in job search and labor market outcomes. Glitz (2017) and Lin and Mo (2024) specifically examine the role of coworker networks in job mobility and wages. We extend this line of research by providing a comprehensive analysis of how different types of coworkers—past and present—affect job-switching decisions, and by exploring the specific channels through which this influence operates, such as information sharing and networking opportunities.

A closely related body of empirical work shows that personal contacts not only facilitate job finding but also improve the quality of the resulting matches. Granovetter (1995) documents that most workers find jobs through personal contacts rather than formal channels, and that weak ties are particularly effective because they bridge otherwise-disconnected groups. Jobs found through contacts pay more and end less often (Mortensen and Vishwanath 1994; Simon and Warner 1992), and administrative data sharpen the pattern: workers in the same census block are more likely to share an employer (Bayer, Ross and Topa 2008), job-finding rates rise when former coworkers sit at firms with vacancies (Cingano and Rosolia 2012), and employers prefer applicants connected to incumbents, whose hires show better match quality (Hensvik and Skans 2016). We build on this evidence by isolating perceived job-offer probability as the channel through which coworker connections operate, and by quantifying its importance relative to other job characteristics in a broad sample of U.S. workers.

The management and human resources literature has separately documented how employee referrals shape hiring. Referred applicants are hired at higher rates, survive more screening stages and stay longer (Fernandez, Castilla and Moore 2000), and are more productive and less likely to quit even conditional on observables (Burks et al. 2015), consistent with referrers transmitting private information about candidate quality. Field experiments confirm that connections raise offer probabilities holding qualifications fixed (Pallais and Sands 2016), and that referral quality depends on the incentives referrers face (Beaman and Magruder 2012). Our paper complements this literature by showing how workers themselves perceive the role of coworker connections in their chances of receiving an offer, and by benchmarking connections against wages and other job attributes in a conjoint design.

A complementary perspective from strategy and organizational behavior examines how network position generates differential access to information and opportunity. Social capital is a resource embedded in relationships (Coleman 1988); managers who bridge otherwise-disconnected groups gain more varied information and advance faster (Burt 2004); within firms, information-bearing ties drive upward mobility while cohesive support ties may constrain it (Podolny and Baron 1997); and survey evidence links network-based access to information and sponsorship to wages, promotions and career satisfaction (Seibert, Kraimer and Liden 2001). We operationalize these mechanisms at the labor-market level, estimating how coworker networks generate informational advantages and perceived

access, and using a calibrated model to price them.

Our evidence also bears on peer effects in the workplace. Prior research has primarily examined productivity spillovers (e.g., Falk and Ichino 2006; Mas and Moretti 2009; Jackson and Bruegmann 2009; Bandiera, Barankay and Rasul 2009; Azoulay, Graff Zivin and Wang 2010) and compensation-related decisions (e.g., Card et al. 2012), often within laboratory settings or within individual firms. In contrast, relatively few studies draw on representative samples of workers. A notable exception is Cornelissen, Dustmann and Schönberg (2017), which documents peer effects in productivity using administrative data from a representative sample. However, the literature has paid less attention to how coworkers shape job search behavior and mobility decisions across sectors or occupations. Our paper addresses this gap by providing direct survey evidence on peer effects in job mobility, using data from a large, quota-balanced sample of U.S. wage and salaried workers reweighted to CPS benchmarks.

Because we elicit expected wages and job opportunities both within and outside a worker's sector, the paper also joins a growing literature on how expectations shape job search. Manski (2004) provides a foundational framework for using survey-based elicitation of subjective probability distributions to measure agents' beliefs, validating the approach we adopt in our survey. Building on this foundation, recent studies (e.g., Conlon et al. 2018; Mueller, Spinnewijn and Topa 2021; Mueller and Spinnewijn 2023; Miano 2023; Jäger et al. 2024) have established that workers' job search decisions involve expectation errors regarding both potential wages and the probability of receiving job offers. We extend this literature by providing direct survey evidence on workers' inaccuracies in perceiving outside options—both in terms of wages and job-offer probabilities—and emphasize that coworker influence can help mitigate these misperceptions.

Taken together, these literatures treat coworker networks as a channel for both information and access, but rarely observe either directly. Our contribution is to measure both in the same sample, eliciting beliefs about wages and perceived offer probabilities for the same worker across sectors, and to price them in a model that separates their allocative consequences. In doing so we connect the economics of job search to the organizational literatures on social capital and employee referrals, which document the same mechanisms from the employer's side.

The rest of the paper proceeds as follows. Section 2 details the procedures for data collection and survey construction. Section 3 documents the relationship between past and current coworker shares and perceived wages and probabilities of receiving an offer. Section 4 explains our conjoint analysis on the importance of coworkers in influencing one's perceived job-offer rate and presents the results. Section 5 presents a stylized sector-choice model in which past and current coworkers affect workers' job and sectoral choices through information and connection channels, and uses the calibrated model to compute welfare counterfactuals. Section 6 concludes.

2 Survey Design

We administered the survey in the United States between August and September 2024, with a final sample of 2,721 respondents.[3] The survey targeted full-time and salaried workers aged 25 to 60. Self-employed individuals were excluded due to differences in coworker interactions, and healthcare workers were also excluded because of the low transition rate to other sectors due to the specificity of their skills. We designed the survey using Qualtrics, and distribution was handled by the commercial company Bilendi and its partner panels. Quotas were set for gender, age, household income, education, and census region to target the composition of the U.S. employee population aged 25 to 60. Respondents were first screened, and those whose quotas were full were excluded. Respondents were compensated for completing the survey, with an average incentive of $4. The average completion time was 26 minutes, with a median of 19 minutes.[4]

Table 1: Sample Characteristics

	Survey	CPS – March Supplement
Male	0.59	0.60
Age		
25-34 years old	0.27	0.30
35-44 years old	0.31	0.30
45-60 years old	0.42	0.40
Household income		
<$60,000	0.24	0.17
$60,000-$125,000	0.48	0.37
>$125,000	0.28	0.46
4-year college degree or more	0.76	0.46

Note: This table reports summary statistics for the survey, in the first column, and corresponding statistics for the target population in the US, in the second column. Population statistics come from the 2023 March Supplement of the Current Population Survey (2023 CPS ASEC, Flood et al. (2023)). Target population: full-time and salaried workers not in the healthcare sector, between 25 and 60 years old.

The final sample matches the target population of full-time wage and salaried workers aged 25 to 60 in the U.S. closely on some dimensions and less closely on others. Table 1 compares the sample's characteristics with those of the U.S. population, based on the 2023 Current Population Survey Supplement. The survey sample tracks the CPS closely on gender (59 versus 60 percent male) and on age composition, but it oversamples college graduates (76 versus 46 percent) and middle-income house-

[3] We filtered out individuals that completed the survey too fast, or that report wages or work hours unreasonably low or high, or coworker numbers unreasonably high.

[4] The distribution of the time taken for completion can be found in Figure B1.

holds ($60,000–$125,000; 48 versus 37 percent), and undersamples high-income households (over $125,000; 28 versus 46 percent). To correct for this over- and under-sampling, we compute survey weights using raking, an iterative proportional fitting procedure recommended in Stantcheva (2023).[5] We use these weights for all the analysis in the paper. Because the education gap in particular is sizable, our estimates should be read as representative of the U.S. workforce only to the extent that reweighting on these observables corrects for selection into the panel. In addition to demographics, we compare the industry composition of survey respondents to the CPS data. Figure B3 shows that industry distribution is comparable to the CPS, except for the absence of healthcare workers and a higher share of "other services." When respondents selected "other services," they were prompted for more details, revealing employers in industries like accommodation, education, and public administration. Finally, the geographical distribution of our sample is similar to the one in CPS. Appendix Figure B2 presents the geographic distribution of respondents by state.

Table 2 presents key labor market statistics for the survey respondents. 15% work more than one job, 69% work in person, 21% work from home for part of the week, and 10% work fully remotely. This indicates that most respondents have some in-person coworker interaction. Regarding job search status, 26% are active job searchers—defined as those actively seeking full-time or part-time employment or due to potential layoffs—and 44% are passive job searchers, open to opportunities but not actively seeking employment.

Table 2: Summary Statistics of Employment Characteristics

	Mean	Median	P25	P75	Obs.
Work hours per week	41.95	40.00	40.00	45.00	2721
Gross annual earnings	83,840.24	74,000.00	46,000.00	100,000	2,721
Gross hourly earnings	40.62	32.69	21.37	48.08	2,721
Tenure at current job (in yrs.)	2.30	1.65	0.75	3.29	2,707
Working at multiple jobs	0.15	0.00	0.00	0.00	2,721
Working fully in-person	0.69	1.00	0.00	1.00	2,721
Working remotely some time	0.21	0.00	0.00	0.00	2,721
Active job searcher	0.26	0.00	0.00	1.00	2,721
Passive job searcher	0.44	0.00	0.00	1.00	2,721

Note: Table reports some labor market related summary statistics for the main survey sample. The variables, working at multiple jobs, working fully in person, working remotely some time, active job searcher, and passive job searcher are dummies equal to 1 if the respondent is, respectively, has more than one job, is working fully in person or remotely sometime at his main job, is actively or passively looking for a new job according to the BLS definition of active search methods.

Table 3: Coworker Connections by Type and Purpose

	Share ($\geq$1)	Mean count	N
Panel A: Current coworkers at current job — information channel			
Mentioned wages at previous job	0.54	4.11	2,721
Mentioned non-wage amenities at previous job	0.47	3.78	2,720
Mentioned job satisfaction at previous job	0.57	4.61	2,721
Previous industry of employment known	0.78	8.07	2,721
Panel B: Past coworkers at current job — referral channel			
Reachable for career-related advice	0.63	3.43	2,721
Current industry of employment known	0.68	4.23	2,721
Panel C: Past coworkers at previous job — referral channel			
Maintain regular contact for networking or referrals	0.65	3.49	2,328
Mentioned current industry of employment	0.27	–	2,329
Mentioned current firm of employment	0.28	–	2,329
Provided direct referral to current position	0.18	–	2,329

Note: Table reports the share of respondents with at least one coworker of each type (Share $\geq$1), the mean count of such coworkers, and the number of observations. Panel A covers current coworkers at the current job, whose information about prior sectors feeds the wage-belief accuracy channel. Panel B covers past coworkers who have left the current employer, whose knowledge of the respondent feeds the referral channel. Panel C covers past coworkers from the previous job, who influence the respondent's current job choice through industry mentions and direct referrals. Statistics are weighted using rake-normalized survey weights. N varies in Panel C because these questions are asked only of respondents with a previous employer. Mean count is reported as – for binary indicators in Panel C where the survey elicited a yes/no response rather than a count; for these rows the share equals the mean.

2.1 Survey Overview

Table 3 documents the prevalence of each type of coworker connection, organized by the channel through which it operates. Coworker networks are pervasive: 78 percent of respondents know the previous industry of at least one current coworker, providing a basis for the information channel, and 63 percent have at least one past coworker at their current job they feel comfortable approaching for career advice, providing a basis for the referral channel. Information sharing about wages and working conditions is also widespread: 54 percent of respondents have a current coworker who mentioned their previous wages, 47 percent mentioned non-wage amenities, and 57 percent mentioned job satisfaction. Among workers with a previous employer, 27 percent report that a past coworker mentioned their current industry of employment, 28 percent mentioned their current firm, and 18 percent provided a direct referral to their current position. These statistics establish that both the information and referral channels are active for a substantial share of the workforce, and provide benchmark estimates of coworker network prevalence that generalize beyond the specific outcomes studied in this paper.

We now provide an overview of our survey. The full questionnaire can be found in Appendix A.

[5]Specifically, we assign equal initial weight to each observation. In each iteration we alternately scale the weights within each category so that the weighted shares match the CPS targets for all categories listed in the Table 1.

Background socioeconomic questions and employment characteristics At the beginning of the survey, we collect demographic information from respondents, including gender, age, race and ethnicity, household income, education, zip code of residence, and current employment status. This information is used for screening purposes and to establish quotas. After the screening questions, we ask about their current job, including when they started, how many hours they work per week, their annual earnings, occupation, employer's industry, whether the job is in-person, remote, or hybrid, and the benefits they receive, if any. Respondents holding more than one job are asked to provide information about their "main" job, defined as the job where they work the most hours per week. We also ask respondents whether they were employed elsewhere or not employed when they found their current job, as workers without a previous job will not be able to answer questions about former coworkers. Additionally, we ask respondents to indicate the number of employees in their current establishment and in all locations of their current firm, as workers in small versus large firms may have different interactions with their co-workers.

Coworkers, Wage Beliefs, and Job-Switching Decisions We dedicate three blocks of the survey to ask about "current coworkers at the current workplace," "past coworkers at the current workplace" and "past coworkers at a previous workplace" Before these blocks, we explain how we define these three types of coworkers to ensure clarity.[6] To test respondents' understanding of these categories, we presented a simple scenario:[7]

"Suppose that Ben and Mary both worked at the same Walmart store in 2022. Ben quit his job at Walmart and started working at Target in January 2023, while Mary continued working at Walmart. For Mary, what type of coworker is Ben?"

Panel (A) of Figure B4 shows the distribution of responses. While 56.2% of participants correctly identified Ben as a "past coworker at the current job," 39.7% mistakenly categorized him as a "past coworker at the previous job." For those who selected the incorrect answer, we provided the correct answer and then asked a follow-up question about what type of coworker Mary is for Ben under the same scenario. Panel (B) of Figure B4 illustrates the distribution of responses for this follow-up question, showing evidence of respondent learning. Of all respondents who answered the first question incorrectly, 68.2% were able to answer the second question correctly. We now explain in more details what each one of the three blocks regarding coworker types entail.

We begin by asking respondents about their current coworkers at their present job. First, we elicit the number of coworkers they regularly interact with, along with how many of these coworkers provide relevant information about their situations, including wages, amenities, and job satisfaction at their previous jobs. Next, we ask respondents to indicate how many of their current coworkers were previously employed in each of five listed industries. These five industries are held fixed for a

[6] We asked respondents to categorize individuals who used to work with them at the same office but have since transferred to a different branch or location within the same company as "current coworkers at the current workplace," rather than "past coworkers at the current workplace."

[7] To aid understanding, we provided a graphical illustration depicting the timeline of the scenario, shown in Figure B5.

given respondent throughout the survey and are selected using the LEHD job-to-job transition matrix: the three industries with the highest out-transition flows from the respondent's current industry—which include the respondent's own industry, since remaining in one's current industry is the most common outcome—together with the two industries with the lowest out-transition flows.[8] Finally, to test whether having more coworkers enhances one's understanding of the compensation levels in different industries, we ask respondents to estimate the median annual salary of workers with similar characteristics to their own for each of the five industries, and to rate their confidence in these estimates.

Next, we assess whether participants are in touch with their past coworkers at their current job by asking how many coworkers they feel comfortable reaching out to for career-related advice, and how many of these coworkers are aware of their current industry of employment. We then ask respondents to indicate the number of past coworkers they have at their current job who are currently employed in each of the five listed industries.

Finally, with the block "past coworkers at previous job", our goal is to understand how individuals' past coworkers at previous jobs have influenced their current industry and job choices. To achieve this, we asked respondents about the number of past coworkers from their previous job who are currently working in the same industry and job. We also inquired about the number of coworkers who mentioned or recommended their current industry or job, as well as whether any of them directly referred them to their current position.

Future employment To study the influence of current coworkers on job and industry-switching decisions, we use this block to elicit respondents' future employment plans. We begin by asking whether respondents plan to switch jobs or industries within the next year. For those considering switching industries, we present the same list of five industries described above. We then ask respondents to estimate the probability of receiving a job offer if they applied to each of the five industries. Finally, we ask them to predict their starting wage in each of these industries, along with how confident they are in their wage estimates.

Conjoint experiment on job-offer probabilities Although we ask survey respondents to estimate their job-offer probability for a typical job in a list of five industries different from their own, the role of personal connections through previous coworkers at the firm level remains relatively unclear. To address this, we include a conjoint experiment, asking respondents to assess the probability of receiving a job offer for four pairs of jobs that differ in characteristics such as wage, skill levels, work arrangement flexibility, and the connection they have with incumbent workers. Details regarding the conjoint experiment are outlined in Section 4.

[8] The one exception is the perceived job-offer probability question, which is asked only for the four listed industries other than the respondent's own.

3 Coworkers, Wage Beliefs, and Job-Offer Probabilities

In this section, we describe the sectoral distribution of respondents' current and past coworkers, their beliefs about wages in their own sector and other sectors, as well as their beliefs about job-offer rates.

3.1 Coworker Distribution

We begin by examining the number of current and past coworkers at respondents' current jobs with whom they are familiar or interact regularly. Table C2 shows that workers have extensive interactions and familiarity with their coworkers. On average, respondents in our sample interact with 20 coworkers each month, approximately 20% of whom share details about their past jobs, such as wages, amenities, and job satisfaction levels. Additionally, respondents know the previous industry of employment for an average of 8 current coworkers. Regarding past coworkers at their current jobs, respondents feel comfortable reaching out to an average of 3 individuals, and they are aware of the current industry of employment for about 4 coworkers.

Do respondents accurately perceive the previous and current industries of employment of their coworkers, both before these coworkers entered and after they left the reference firm? To investigate this question, we examine the relationship between respondents' perceived inflows and outflows at their current firms and the actual cross-sector flows observed in the LEHD data. In Panels (A) and (B) of Figure C1, we plot responses to questions on coworker movements across industries. Panel (A) shows respondents' estimates of how many current coworkers were last employed in a specific industry against the LEHD inflow from that industry into the respondent's industry. Panel (B) plots respondents' estimates of how many past coworkers are currently employed in each industry against the LEHD outflow from the respondent's industry into each other industry.[9] These plots demonstrate a positive correlation between perceived and actual industry-level transitions, suggesting that respondents' estimates of coworker movements are generally reliable.

Finally, we explore whether it is necessary to distinguish between different types of coworkers by examining the relationship between past and current coworkers. If both types of coworkers simply reflect the difficulty of transitioning across industries, then distinguishing between them might be redundant. Moreover, the coworker network effects we study rely on deviations from the law of large numbers, meaning there is some variation in coworker shares across firms. Panel (C) of Figure C1 examines the relationship between inflows to an establishment from an outside industry and outflows from the establishment to that industry. We study this by residualizing the number of past coworkers currently employed in each industry based on the number of current coworkers last employed in that industry and controlling for respondent fixed effects. The results in Panel (C) indicate substantial variation in the residualized number of past coworkers across industries, supporting our distinction

[9] Because both directions of flow appear in what follows, we fix notation here. $OutTransition_{is}$ denotes the LEHD job-to-job transition rate from respondent i's current industry *to* listed industry s, and $InTransition_{is}$ denotes the LEHD job-to-job transition rate *from* industry s into respondent i's current industry. Unless stated otherwise, the transition-flow control included in our regressions is $OutTransition_{is}$, and the five industries shown to each respondent are selected on $OutTransition$.

between different types of coworker connections.

3.2 Wage Beliefs

How accurate are workers' beliefs about wages across industries? To answer this question we conduct two analyses using the answers collected for both respondents' beliefs on the median wage in a sector and their beliefs on their personal starting wage if they switch into a sector. First, we compare workers' beliefs about the median wage in each of the listed industries with the actual median wage for those industries, obtained from the CPS. Next, we compare workers' beliefs about their potential starting wage in each of the destination industries with the wage we predict they could have earned.[10] We also examine the difference in wage beliefs by worker characteristics, including gender, income, and job search status.

Figure D1 shows the distribution of the difference between the logged perceived median wage and the logged actual median wage by industry. Although the distribution centers around zero, there is substantial heterogeneity in the prediction error. The 25th and 75th percentiles of the logged prediction error for the median industry wage are -0.42 and 0.22, respectively, which correspond to perceived median wages that are 66% and 125% of the actual median wages.

Figure C2 presents binscatter plots from regressions of survey participants' logged predicted median wage against the logged actual median wage.

$$\log(PredictedMedianWage)_{is} = \alpha_0^{median} + \alpha_1^{median} \log(ActualMedianWage)_s + \gamma_i + \epsilon_{is} \tag{1}$$

where i indexes the individual and s indexes the industry that the respondent is being asked about. We include individual fixed effects to isolate each participant's tendency to over- or under-estimate wages across all industries. On average, individuals have a reasonable sense of median salaries across sectors, with notably more accurate predictions for their own sector. This conclusion holds when we additionally control for the "similarity" between origin and destination industries—measured by job-to-job transition shares—or when we exclude individual fixed effects to allow for variation in average prediction levels across individuals.

Although perceived median wages are highly correlated with actual median wages, the slope of the regression is 0.55. This implies that, on average, a 10% higher actual median wage in an industry is associated with only a 5.5 percentage point increase in perceived wages. The slope suggests substantial underestimation of wage levels in higher-paying sectors (and overestimation in lower-paying ones). When controlling for job-to-job transition shares and excluding respondents' own industries, the slope remains stable. More details are shown in Figure D2. This relationship is robust to controlling for respondents' occupation at the 2-digit SOC code level, as seen in Figure C3.

[10] The predicted wage is derived using a Mincer-style earnings regression (Mincer 1958). Specifically, for each individual in the CPS, we regress their wage on age, squared age, an indicator for college attendance, state, industry (at the two-digit NAICS code level), and occupation. We then use the estimated coefficients from this regression to predict the wage for each respondent in our survey based on their personal characteristics.

Apart from the median wage in an industry, respondents' beliefs about their own starting wages are positively correlated with wage predictions derived from data. Figure C4 shows binscatter plots from a regression of survey participants' log-predicted starting wages against the log-predicted wage in a reference industry, estimated using a Mincer-style regression on CPS data. On average, respondents' expected starting wages in various industries are positively correlated with the predictions based on CPS data. This holds even after accounting for transition difficulty between the respondent's current industry and the industry in question by including the transition share between each industry pair as a control variable, and the slope increases to being close to one. Nevertheless, there is still lower variation among perceived personal starting wages compared to the actual wages observed in the data.

3.2.1 Heterogeneity in Wage Beliefs

How does the perception of wages in other sectors vary with individual characteristics? Understanding heterogeneity in misperceptions is important for identifying which groups may face greater barriers to efficient job transitions. We now examine whether these misperceptions in the median wage of other industries differ by gender, income level, and job search status. The results are displayed in Table C3.

We find that the slope coefficient, α_1^{median}, which captures the responsiveness of perceived sectoral wages to actual wages, is smaller for males compared to females, and smaller for active job seekers compared to those with a lower willingness to switch jobs within the next year. In contrast, the intercepts, α_0^{median}, are larger for both males and job seekers. This pattern indicates that, conditional on their own level of wage beliefs, males and job seekers are less responsive to true cross-sector wage variation. Moreover, both groups tend to overestimate wages in lower-paying sectors, consistent with the higher intercepts.[11] Unlike heterogeneity by gender and job search status, we find that both slope and intercept are relatively stable across income groups.

The Relationship with Coworker Shares While perceived and actual sectoral wages are positively correlated, the estimated coefficient remains well below one, indicating that individuals do not fully internalize available wage information. This suggests that people may rely on incomplete or selective sources when forming beliefs. One plausible source of sector-specific information is their coworkers, who may transmit wage knowledge through daily interactions or shared experiences. We then investigate whether having more coworkers is associated with lower prediction error in the following

[11] One possible interpretation is that job seekers face greater uncertainty and possess less accurate or more optimistic wage beliefs, especially in unfamiliar sectors. For the gender difference, it may reflect differences in labor market experience or the types of jobs and sectors men and women are typically exposed to. If women are more concentrated in a narrower range of sectors or have more stable sectoral trajectories, they may form more accurate beliefs about cross-sector wage differences. Alternatively, the observed gap may reflect differences in the way men and women gather or use wage information during job search.

regression:

$$AbsolutePredictionError_{is} = \alpha_0^{pred} + \alpha_1^{pred} PastCoworkerShare_{is} + \alpha_2^{pred} CoworkerShare_{is} + \boldsymbol{\beta}^{pred} \boldsymbol{X}_{is}' + \gamma_i + v_{is}^{pred} \quad (2)$$

We define $AbsolutePredictionError_{is}$ as the absolute value of either the difference between the values or the logged values of individuals' perceived median wage in an industry and the actual median wages observed in the CPS.[12] *PastCoworkerShare* represents the fraction of individual i's past coworkers who are currently employed in industry s, and $CoworkerShare_{is}$ represents the proportion of individual i's current coworkers who were last employed in industry s. To control for individual tendencies toward optimism or pessimism about job prospects, we include individual fixed effects. Furthermore, factors such as the difficulty of transitions between industry pairs, which may influence respondents' perceived wages in other industries, could also vary with their shares of current coworkers last employed in an industry. To address this potential correlation thoroughly, we include a set of controls, $\boldsymbol{X}_{is}$, either separately or in combination across different specifications. First, we control for the proportion of individuals transitioning from the respondent's current industry to the industry in question. Second, we incorporate the perceived probabilities of job offers from individuals for themselves if they were to apply to the reference sector. Third, we control for occupational group fixed effects at the 2-digit SOC level. Fourth, to check whether past coworkers or current coworkers exert more influences on workers' wage prediction errors, we include both the share of current coworkers who were previously employed in the reference sector and the share of past coworkers at the current job who are currently employed in that sector as regressors. Finally, we look into the differential impacts of coworkers on positive vs. negative wage prediction errors, by including an indicator variations *PositiveError* that indicates overestimation of the median wage in an industry, as well as an interaction between the indicator variable and the *CoworkerShare*.

Table 4 shows that a higher share of current coworkers who previously worked in an industry is associated with smaller errors when respondents predict that industry's median wage. In Column (1), moving from zero to all such coworkers reduces the prediction error by \$7,466. This estimate remains similar in magnitude and statistically significant after adding controls for the perceived probability of receiving a job offer in that industry and occupation fixed effects. In Column (4), we include the share of past coworkers who moved into the industry (*PastCoworkerShare*) and find a statistically insignificant coefficient. Column (5) tests for asymmetric effects by interacting *CoworkerShare* with an indicator for overestimation (*PositiveError*). Although overestimators err by a larger amount on average, the interaction term is statistically insignificant, suggesting no evidence of differential impact of coworker exposure on optimistic versus pessimistic respondents.

Table C4 defines *AbsolutePredictionError* as the difference between the logged values of individuals' perceived median wage in an industry and the actual median wage observed in CPS. Across

[12] We also select our sample to exclude answers that fall above the level that is being top-coded by CPS or the ones that fall below the federal minimum wage.

Table 4: Current Coworkers' Prior Industries and Prediction Errors in Median Wages

	Wage Error	Wage Error	Wage Error	Wage Error	Wage Error
CoworkerShare	-7,465.85*	-7,174.27*	-7,327.36*	-8,051.10*	-7,528.77*
	(4,214.11)	(4,222.24)	(4,246.77)	(4,148.59)	(3,931.96)
OutTransition	-513.22**	-493.11**	-469.18**	-478.10**	-741.63***
	(236.07)	(240.00)	(239.09)	(238.96)	(248.93)
PerceivedProb		-18.79	-19.76	-22.60	0.22
		(28.69)	(28.80)	(28.44)	(29.33)
PastCoworkerShare				4,215.96	
				(4,676.20)	
PositiveError					10,312.55***
					(2,874.40)
PositiveError × *CoworkerShare*					1,874.86
					(5,628.42)
Occupation FE	No	No	Yes	Yes	Yes
Mean of Dep. Var	25,434.43	25,450.92	25,415.73	25,415.73	25,415.73
Adjusted R^2	0.24	0.24	0.24	0.24	0.25

Note: Table displays the coefficient estimates for equation (2). The outcome variable in all five columns is the absolute difference, in dollars, between the respondent's predicted median wage in a listed industry and the actual median wage for that industry observed in the CPS. All columns include individual fixed effects and control for *OutTransition*, the LEHD job-to-job transition rate from the respondent's current industry to the listed industry. Occupation fixed effects are absorbed by the individual fixed effects, since each respondent reports a single occupation; the row documents the specification estimated, and the reported coefficients are unchanged by its inclusion. Column (2) adds the respondent's perceived probability of receiving an offer in that industry; column (3) adds occupation-group fixed effects at the 2-digit SOC level; column (4) adds the share of past coworkers currently employed in the listed industry; column (5) adds an indicator for wage overestimation (*PositiveError*) and its interaction with *CoworkerShare*. The corresponding specification with the outcome measured in log points is reported in Appendix Table C4.

specifications that add respondents' perceived offer probability for the industry, occupation-group fixed effects, the share of past coworkers currently employed in the industry, and an indicator for wage overestimation (with its interaction with coworker share), the coefficient on *CoworkerShare* is stable at -0.05 to -0.06. Quantitatively, a 10 percentage point increase in *CoworkerShare* is associated with roughly a 0.5–0.6 percentage point improvement in accuracy.

With respect to starting wage, Table C5 shows that having more coworkers last employed in an industry reduces the gap between the predicted starting wage from observations in the CPS and the perceived own starting wage indicated by each respondent, with the caveat that the estimate is no longer statistically significant.[13]

[13]Since our predicted starting wage is obtained using a Mincer-style regression with observables, there may be some noise in imputation, leading to the statistical insignificance.

3.3 Job-Offer Probabilities

Other than perceived wages for a given sector, perceived job-offer probabilities are also important for determining workers' job-switching decisions, as they factor into the option value of potentially switching jobs in the future. Before examining the role of coworkers in influencing perceived outside options, we assess how well calibrated those beliefs are on average. Unlike industry-level wages, there is no directly observable moment in the data corresponding to the actual job-offer rate. Therefore, we compare perceived job-offer probabilities with realized flows between the respondent's origin industry and the industry for which they are predicting the job-offer rate.

Admittedly, transition flows between industry pairs may capture factors beyond job-offer probabilities, such as transition difficulties due to skill or labor demand differentials between industries. However, given the absence of data on actual job-offer probabilities, using in- and out-transition flows between industry pairs as a proxy is a practical approach. These transition flows capture the frequency of successful moves between industries, which indirectly reflects job-offer rates. Since individuals can only switch industries if they receive job offers, these flows provide a reasonable benchmark for evaluating perceived job-offer probabilities.

Figure C6 shows the distribution of perceived job-offer rates in other industries, pooling responses across all industries and participants. The distribution is relatively uniform, with notable peaks at around 0%, 50%, and 100%, indicating that respondents often group job opportunities into broad categories. Figure C7 plots the perceived probability of receiving a job offer in an industry against the actual transition flows into or out of that industry, controlling for individual fixed effects. Overall, respondents predict higher job-offer probabilities for industries that experience higher inflows into their current industry or outflows from it, as well as for industries that workers in their current industry tend to transition into.

Additionally, for each 1 percentage point increase in the outflow transition rate, respondents predict a 1.09 percentage point increase in the job-offer probability. Conversely, they predict a 0.86 percentage point increase in job-offer probability for industries with higher inflows. This pattern aligns with the idea that higher outflow transition rates signal that the target industry is easier to move into, leading respondents to predict higher job-offer rates for industries that are perceived as easier to transition into.

The Relationship with Coworker Shares We look into whether having more coworkers is associated with higher perceived job-offer probabilities, by examining the following specification:

$$PerceivedProb_{is} = \alpha_0^{offer} + \alpha_1^{offer} PastCoworkerShare_{is} + \alpha_2^{offer} CoworkerShare_{is} + \boldsymbol{\beta}^{offer} \mathbf{X}_{is}' + \gamma_i + v_{is}^{offer} \tag{3}$$

When applying for jobs, workers tend to reach out to both their connections from the past and from the present to enhance the probabilities that they receive an offer. We therefore include both

the past and the current coworker shares in the specification. More specifically, $PastCoworkerShare_{is}$ represents the share of individual i's past coworkers at the current job who are currently employed in industry s, and $CoworkerShare_{is}$ represents the share of individual i's current coworkers who were previously employed in sector s. Similar to our specification on perceived wages, we control for individual fixed effects to account for an individual's tendency to be overly optimistic or pessimistic about their job prospects in general. Factors such as the transition difficulty between industry pairs, which may influence perceived job-offer probabilities, could also correlate with the share of past coworkers currently employed in a given industry. To address this issue as comprehensively as possible, we include a set of controls, $\boldsymbol{X}_{is}$. These controls are included either separately or in combination across different specifications. First, we control for the fraction of individuals transitioning from the respondent's current industry to the industry they were asked about. Second, we include individuals' perceived starting wage for themselves if they were to transition into the reference sector. Third, we control for individuals' occupational group fixed effects.

Table 5 presents the results for specification (3). According to the first column, having more past coworkers currently employed in an industry and more current coworkers who were last employed in an industry both increase the perceived job-offer probability for a job in that industry. This relationship remains positive and significant even when controlling for the average transition difficulty to that industry, the perceived logged starting wage in that industry, and the occupational group fixed effects. Quantitatively, according to Column (4), increasing the share of past coworkers who move to an industry from 0 to 1 will increase the perceived probability of receiving an offer from that industry by 11 percentage points, and increasing the share of current coworkers who were last employed in the industry from 0 to 1 will increase the perceived probability by 17 percentage points.

3.4 Job-Switching Intentions

Section 3.3 explores the relationship between individuals' perceived job-offer probabilities and their coworkers. However, another key question in studying industry transitions, which remains unexplored, is whether individuals intend to apply to these industries in the first place. We now examine the relationship between individuals' intention to apply to a job in another industry and their coworker composition. For each respondent, we elicit the probability that they are voluntarily looking for a new job at a new different employer. For respondents who indicated a nonzero probability of voluntarily switching to another employer, we asked whether they would consider switching to a job outside of their own current industry. For those who answered "yes" to potentially switching to a job in a different industry, they were then asked to answer "yes" or "no" for a list of five industries other than their own, indicating whether they would consider applying for a job in each industry.

We follow our survey design to examine the influence of coworkers on three outcome variables related to job- and industry-switching. The outcome variables are 1) $LookforJob$, the percentage each respondent puts down for voluntarily looking for a different job, and 2) an indicator variable, $LookforJobIndustry$, for looking for a job in a certain industry, when the respondent answers "yes"

Table 5: Past Coworkers' Current Industries, Current Coworkers' Prior Industries, and Perceived Job-Offer Probabilities

	Offer Probability	Offer Probability	Offer Probability	Offer Probability
PastCoworkerShare	0.14***	0.11***	0.11***	0.11***
	(0.02)	(0.02)	(0.03)	(0.03)
CoworkerShare	0.20***	0.17***	0.17***	0.17***
	(0.02)	(0.02)	(0.02)	(0.02)
OutTransition		0.01***	0.01***	0.01***
		(0.00)	(0.00)	(0.00)
log(*StartWage*)			-0.01	-0.01
			(0.01)	(0.01)
Occupation FE	No	No	No	Yes
Mean of Dep. Var	0.38	0.38	0.38	0.38
Adjusted R^2	0.54	0.57	0.55	0.55

Note: Table displays the coefficient estimates for equation (3). *PastCoworkerShare* is defined as the share of past coworkers who have left the respondent's current job and move to a certain industry, out of all past coworkers at the current job who the respondent knows the industry where they are currently employed. *CoworkerShare* is defined as the share of current coworkers for the respondent that were previously employed in a certain industry, out of the coworkers who the respondent knows the industry where they were previously employed. All columns include individual fixed effects. Column (2) controls for *OutTransition*, the CPS transition rate (in percent) from the respondent's own industry to the listed industry. Column (3) controls for the respondent's logged starting wage that they believe they could have earned by moving into the industry. Column (4) additionally controls for the occupational group fixed effects.

for considering switching to a different industry. The first regression is at the individual level:

$$LookforJob_i = \alpha_0^{look} + \alpha_1^{look} PastCoworker_i + \alpha_2^{look} Coworker_i + \boldsymbol{\beta}^{look} \boldsymbol{X}_i' + v_{is}^{look} \tag{4}$$

$PastCoworker_i$ is the total number of coworkers the current industry of whom individual i knows from her past job, and $Coworker_i$ is the total number of coworkers the past industry of whom individual i knows. And the second regression is at the level of individual for each potential destination industry:

$$LookforJobIndustry_{is} = \alpha_0^{ind} + \alpha_1^{ind} PastCoworkerShare_{is} + \alpha_2^{ind} CoworkerShare_{is} + \boldsymbol{\beta}^{ind} \boldsymbol{X}_{is}' + \gamma_i + v_{is}^{ind} \tag{5}$$

Analogous to the regression examining coworker influences on wage perception, $PastCoworkerShare_{is}$ represents the share of individual i's past coworkers who are currently employed in industry s, and $CoworkerShare_{is}$ is the share of current coworkers previously employed in industry s. We control for individual fixed effects to account for individual tendencies toward optimism or pessimism regarding job prospects. Additionally, we control for the out-transition probability from the individual's current industry to a potential destination industry and the perceived logged starting wage in that industry

to capture both the transition difficulty and the attractiveness of a job in industry s.

Table C6 presents results for specification (4). An additional past coworker whose current job the respondent knows increases the probability of switching jobs within a year by 0.50 percentage points on average, an effect that remains statistically significant and stable across specifications controlling for out-transition probabilities, current wages, and occupational group fixed effects. In contrast, the effect of current coworkers is much smaller: one additional current coworker whose past industry the respondent knows raises the probability of switching jobs by about 0.10 percentage points, and only becomes statistically significant once the controls are included.

Table 6 presents the results for specification (5). If the share of past coworkers currently employed in an industry increases from 0 to 1, the intention to apply for a job in that industry increases by 29 percentage points. This relationship remains positive and significant even when controlling for average transition difficulty to that industry, by including the out-transition rate and the logged perceived starting wage, and when controlling for occupation fixed effects. Importantly, the coefficient estimate for *PastCoworkerShare* remains positive and significant, only 6 percentage points lower than in the specification that excludes both the perceived wage and out-transition probabilities. Current coworkers also influence workers' intentions but to a lesser extent: if the share of current coworkers previously employed in an industry increases from 0 to 1, the intention to apply for a job in that industry increases by 17 percentage points. This coefficient shrinks by around 6 percentage points.

Taken together, Tables 5 and 6 reveal a differential pattern for the two types of coworkers. Past coworkers *currently* employed in an industry are more strongly associated with workers' application decisions (0.29, versus 0.17 for current coworkers), consistent with destination-side contacts who can provide referral access and firm-specific leads lowering the perceived barrier to applying. Current coworkers *previously* employed in an industry, by contrast, are more strongly associated with the perceived probability of receiving an offer conditional on applying (0.17, versus 0.11 for past coworkers), consistent with these coworkers transmitting information about hiring practices and requirements in the sector they came from. This pattern suggests that coworker networks operate both through referral-based access, which shapes where workers choose to apply, and through information, which shapes how likely workers believe they are to succeed.

Other factors influencing job-switching decisions Other than their intention to search and their perceived outside options, we elicit participants' answers to questions about the general reasons why they chose their current job, as well as the factors they consider most important when deciding which industry other than their own to switch to. Participants were asked to choose three options from a list of job attributes.[14]

In Figure C8, we show the distribution of respondents choosing each of the provided categories.

[14]The options listed include: wage and benefits; probability of receiving an offer; costs (ease) of application; match of your own skills and qualifications; career growth prospects; networking and connection opportunities; job security, prestige, and work-life balance; convenient location; flexibility in work hours or ability to work from home; company culture and friendly work environment; other (please specify).

Table 6: Past and Current Coworkers and the Intention to Apply, by Industry

	Switch Industry	Switch Industry	Switch Industry	Switch Industry
PastCoworkerShare	0.29***	0.28***	0.23***	0.23***
	(0.06)	(0.07)	(0.07)	(0.07)
CoworkerShare	0.17***	0.15***	0.11*	0.11*
	(0.06)	(0.06)	(0.06)	(0.06)
OutTransition		0.01***	0.01***	0.01***
		(0.00)	(0.00)	(0.00)
log(*StartingWage*)			0.18***	0.18***
			(0.04)	(0.04)
Occupation FE	No	No	No	Yes
Mean of Dep. Var	0.50	0.50	0.50	0.50
Adjusted R^2	0.24	0.24	0.25	0.25

Note: Table displays the coefficient estimates for equation (5). The outcome variable across all columns is an indicator variable equal to 1 if the respondent is willing to consider to switch to a certain destination industry. *PastCoworkerShare* is defined as the share of past coworkers who have left the respondent's current job and move to a certain industry, out of all past coworkers at the current job who the respondent knows the industry where they are currently employed. *CoworkerShare* is defined as the share of current coworkers for the respondent that were previously employed in a certain industry, out of the coworkers who the respondent knows the industry where they were previously employed. Column (2) controls for the out transition probability in percentage from the individual's own industry to the industry they were asked about. Column (3) controls for the respondent's logged starting wage that they believe they could have earned by moving into the industry. Column (4) additionally controls for the occupational group fixed effects.

Seventy-four percent of the participants chose "wage" as one of the three reasons why they chose their current job, while 83% of the respondents selected "wage" as one of their major considerations for choosing the next industry. Other reasons that were most frequently selected were "job security, prestige, and work-life balance" and "career growth prospects."

Overall, participants consider the same factors when they chose their current job and when making future decisions about which industry to switch into next. The only exceptions are that respondents place more importance on the flexibility of work arrangements for the industries they are planning to move into compared to when they chose their current job. They also place slightly less importance on the extent of skill match when considering future options.

To sum up, the analysis of workers' job-switching decisions emphasizes the importance of wages in influencing these choices and consequently highlights the role of coworkers. By enhancing workers' accuracy of wage beliefs associated with their outside options, coworkers can propel individuals to select into industries that better align with their considerations.

3.5 Coworkers from the Previous Job and Current Employment Outcomes

A note on terminology before proceeding. Up to this point, "past coworkers" has referred to *former coworkers from the respondent's current employer*—people who worked alongside the respondent at her

current job and have since left, whose destination industries define *PastCoworkerShare* in Tables 5 and 6. This subsection instead concerns *coworkers from the respondent's previous job*, who may have shaped how she came to hold her current job. The two groups are distinct, and we use the fuller labels wherever confusion is possible.

The previous analysis on coworkers at current jobs and individuals' future employment decisions relies on stated preferences and may therefore be subject to reporting bias. Survey responses do not always align with realized behavior, particularly when they involve hypothetical scenarios or lack real economic consequences.[15] To complement this evidence, we turn to revealed preferences and examine whether past coworkers from previous firms are associated with individuals' realized job choices.

Table C9 reports summary statistics on respondents' interactions with former coworkers and the potential role of these contacts in their current employment. On average, respondents remain in contact with 2.60 coworkers from their previous job; 8.67 of these former coworkers had previously worked in the respondent's current industry, and 1.63 at the respondent's current firm. Moreover, 27% of respondents report that a past coworker mentioned their current industry of employment and 28% their current firm, while 18% report receiving a direct referral to their current position. While these statistics do not isolate specific mechanisms, they point to substantial information flows within coworker networks and suggest that coworkers shape realized industry and firm choices.

What is the relationship between workers' wages at their current jobs and their past coworkers? To explore answers to this question, we study the specification that regresses the respondents' current logged wages on indicator variables for whether the respondent has at least one past coworkers at her past job that worked in the current industry, mentioned to her the current industry, or referred her to the current job:

$$\ln(wage)_i = \alpha_0^{wage} + \alpha_1^{wage} \mathbb{1}(coworker)_i + \boldsymbol{\beta}^{wage} \boldsymbol{X}_i' + \upsilon_{is}^{wage} \tag{6}$$

We control for a comprehensive set of variables that may influence individual i's wage, including personal characteristics such as age, gender, tenure at the current job, education level, and state of work. Additionally, we include occupation-by-previous-industry-by-current-industry fixed effects to capture the transition difficulty for individuals with specific skill sets moving from their previous industry of employment to their current industry. Given that our occupation codes are categorized at the six-digit SOC level, these fixed effects should provide a reasonably detailed representation of skill variation.

Workers who report that a coworker mentioned opportunities in their current industry earn significantly higher wages in the unconditional specification. Specifically, the simple regression shows that these workers earn about 22 percent higher wages on average. After controlling for worker characteristics such as age and tenure and including detailed fixed effects for gender, education, state,

[15] For example, see Ben-Akiva, Morikawa and Shiroishi 1992, Cummings, Harrison and Rutström 1995, List and Gallet 2001, Harrison and Rutström 2008.

Table 7: Past Coworkers and Wages at the Current Job

	(1)	(2)	(3)	(4)
Past coworkers mention current industry	0.22***	0.13***		
	(0.07)	(0.04)		
Past coworker referral			0.20**	0.05
			(0.08)	(0.07)
Controls	No	Yes	No	Yes
Mean of Dep. Var	11.00	11.00	11.00	11.00
R^2	0.01	0.26	0.00	0.25

Note: Table displays the coefficient estimates for equation (6). Columns (1) - (2) define *coworker* as having at least one past coworker who has mentioned to the individual her current industry of employment. Columns (3) - (4) define *coworker* as having at least one past coworker who referred the reference individual to their current position. Columns (1) and (3) include no additional controls. Columns (2) and (4) control for age, age squared, and tenure at the current job, together with the share of workers transitioning from the respondent's previous industry to their current industry, and include fixed effects for gender, education, state of work, and the interaction of 6-digit SOC occupation with previous and current industry.

and the interaction of current occupation, previous occupation, and industry, the estimated premium remains positive and statistically significant, though smaller in magnitude at about 13 percent. In contrast, workers who report receiving a referral exhibit a positive wage difference of roughly 20 percent in the unconditional specification, but this association becomes small and statistically insignificant once the full set of controls and fixed effects is included. This pattern suggests that while referrals are correlated with higher wages in raw comparisons, this relationship largely reflects sorting into different occupations and industries rather than higher wages within narrowly defined job categories, whereas information transmission from coworkers about industry opportunities remains associated with higher wages even after accounting for these factors.

4 The Importance of Coworkers in Influencing Perceived Job-Offer Probabilities

Section 3.3 delineates the correlation between the percentage of coworkers in a sector and the perceived probabilities of job offer once applied to a job in that sector. However, it remains unclear how important coworkers are compared to other characteristics that impact the probabilities of receiving an offer. To quantify the effect of coworker connections on job-offer probabilities, we designed a conjoint experiment for survey respondents aimed at assessing their beliefs regarding job offers within their current sector. The introductory text prompts respondents to imagine seeking a new job in their current sector and applying to two different companies. They are then asked to evaluate short descriptions of these positions and select the one they believe is more likely to result in a job offer.

Specifically, respondents are presented with the following instructions: "We are going to show

you four pairs of jobs. These jobs are identical in all aspects except for the characteristics presented in the tables you will see. We will ask you to indicate how likely you believe you are to receive a job offer if you apply to each of the jobs presented. Remember, a job offer is not necessarily a job you will accept. Please respond without considering whether the job is ideal for you. You will be able to select a percentage chance of receiving a job offer for each job by moving the sliders below the tables, where 0 on the far left means you believe it is absolutely impossible you will receive an offer, and 100 on the far right means you are certain to receive an offer."After reading the instructions, respondents are shown a table comparing the two jobs, each with sliders ranging from 0 to 100, allowing them to indicate their perceived probability of receiving an offer for each job separately. Figure B6 contains a screenshot of one example of the conjoint experiment according to how it was presented to survey respondents.

To avoid overwhelming respondents with too many variables, we follow Folke and Rickne (2022) by limiting the comparison to four key job attributes: monthly wage, skill level required, schedule flexibility (including the option to work from home), and personal connections with other workers. Other than these attributes, the jobs are identical.

We introduce randomization by varying both the order of the rows and the values of the job attributes in each description.[16] The personal connection attribute is presented through vignettes outlining different levels of coworker connections, with four possible conditions: (i) Null condition: "You do not know any other current employees at the job you are applying for." (ii) Slightly positive condition: "You know at least one employee at the job you are applying for but have not reached out for any discussion of the job." (iii) More positive condition: "You know and have reached out to at least one employee at the job. They have a good opinion of you but are unable to assist with your application in any concrete way." (iv) Active effort condition: "You know and have reached out to at least one employee at the job, and they are actively helping you with your application (e.g., providing a referral or advocating on your behalf)." This last condition is oversampled to have a 33% probability of occurrence, while the other three conditions each have a 22% probability.

Quantifying personal connections can be challenging, so we use vignettes to illustrate different scenarios in which personal connections may influence job-offer probabilities. Conditional on the active-effort condition being drawn, one of three vignettes is shown: (i) Referral agreement: "You had a coffee chat with a previous colleague currently employed at the target firm, and they have agreed to provide a referral." (ii) Networking support: "You had a coffee chat with a previous colleague at the target firm. While they cannot provide a referral themselves, they have agreed to ask their other connections at the firm to do so." (iii) Informal mention: "You had a coffee chat with a previous

[16]The values for the first three job traits are randomly selected from predefined lists with equal probabilities. For example, wage options include: "5% less than your current wage," "approximately the same as your current wage," "5% more than your current wage," or "10% more than your current wage." Skill requirements are presented as: "5% lower than your current job," "approximately the same as your current job," or "5% higher than your current job." Schedule flexibility options include: "entirely flexible and able to work from home," "flexible hours but unable to work from home," or "no personal influence over work hours and unable to work from home."

colleague at the target firm. While they cannot formally refer you, they will informally mention you to recruiters." The design of the connection attribute is detailed in Appendix Table C1.

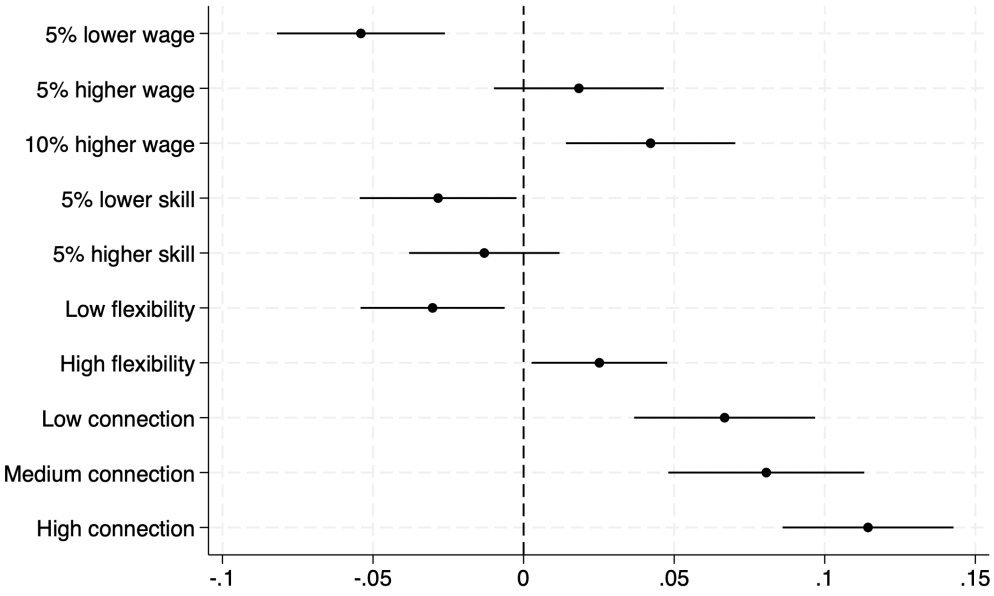
4.1 Calculating the perceived impact of connection on job-offer probability

Following existing survey studies with conjoint analysis (Hainmueller, Hopkins and Yamamoto 2014, Folke and Rickne 2022), we measure the importance of connection through coworkers on individuals' perceived probabilities of receiving a job offer. We create a dummy variable $ChooseJob_{ijt}$ to indicate whether individual i believes she is more likely to receive an offer from job j in table t. We regress this indicator variable on one dummy for each value of each job trait:

$$ChooseJob_{ijt} = \alpha + \beta_{connection} Connection_{ijt} + \boldsymbol{\beta}_x \boldsymbol{X}'_{ijt} + \gamma^k_{table} table^k_{ijt} + e_{ijt} \quad (7)$$

where we omit one dummy for each value of each job trait. $Connection_{ijt}$ is equal to 1 if one of the three "active effort" of establishing personal connection vignettes is shown. To simplify notation, we put the other categories of personal connection, as well as other job traits, including wages, job flexibility, and skills required in the **X** vector. In case that the order in which a job is shown matters for respondents' selection, we also control for a dummy variable $table^k_{ijt}$, which is equal to 1 if the table is the k-th one being shown to the respondents. The coefficients γ^k_{table} are all close to 0 and statistically insignificant, showing that individuals tend to make consistent choices regardless of the order in which they are being shown the different pairs of jobs.

Figure 1: Estimates of the Importance of Job Features in influencing Job-Offer Probability



Note: 95% confidence intervals are adjusted for clustering by respondent. Each respondent provided responses to four experiments on job-offer probabilities. The dependent variable is equal to 1 if the respondent selects a higher probability of receiving an offer from a job, compared to other job in the table. N = 2,721.

We pool the three "high connection" scenarios into a single dummy variable, $Connection_{ijt}$. The coefficient $\beta_{connection}$ on this variable shows the percentage point difference in the proportion of people who associates a job with higher job-offer rate compared to the reference category, "no connection." For the wage characteristic, we set the reference category to "approximately the same as your current wage." Interestingly, individuals on average think that the probability they will receive an offer from a job *increases* with the compensation level of that job. The coefficient estimates for "5% less", "5% more" and "10% more" are $-0.04, 0.01, 0.05$, respectively.[17] The fact that respondents tend to associate higher-wage jobs with high job-offer rate despite that such positions are typically more competitive and our reminder on how having a job offer doesn't indicate acceptance of the job may be due to several reasons. First, optimism bias may lead respondents to overestimate their chances of securing high-paying positions, driven by confidence in their own qualifications. Additionally, wage levels may serve as a signal of job desirability, making respondents believe that companies offering higher wages are more eager to fill roles, despite increased competition. Finally, workers may be overqualified for lower-paid positions. In our study, we do not attempt to delve into the exact reasons why respondents associate higher-wage jobs with higher job-offer probabilities. We note that this positive association concerns hypothetical jobs *within* the respondent's own sector, where the wage differences shown are small (from 5% below to 10% above the current wage) and may plausibly signal firm generosity or urgency to fill a vacancy. It is therefore not in tension with the negative wage coefficient we estimate in Section 5 for *cross-sector* destinations, where a higher destination wage largely reflects the greater selectivity of a different and better-paying sector.

Figure 1 presents the full set of estimates on the importance of job attributes in influencing perceived job-offer probabilities among all respondents. On average, respondents believe that having connections plays a significant and positive role in increasing their likelihood of receiving a job offer. In fact, respondents perceive a strong connection to be the most important factor in improving their chances of getting an offer. Having someone who is willing to make an effort—whether through a direct referral or by communicating with the hiring team—raises the probability that a job is judged more likely to yield an offer by 11.4 percentage points. Even without active outreach, simply knowing at least one employee at the firm where the individual is applying raises that probability by 8.1 percentage points, more than any other job attribute unrelated to connection. We emphasize that these magnitudes are effects on the *forced-choice* outcome—which of the two jobs the respondent judges more likely to produce an offer—rather than on the level of the perceived offer probability itself; the corresponding estimates for the elicited probability are reported below and are smaller.

We examine heterogeneity in perceived job-offer probabilities across workers with different characteristics. First, we run specification (7) separately for male and female workers. Second, we explore heterogeneity by age. As shown in Figure C10, job-offer probabilities do not vary significantly by gender or wage, with a few exceptions. Female workers tend to associate jobs offering higher

[17] The t-stats associated with the three categories are -4.87, 1.58, and 5.15, respectively.

wages with higher job-offer probabilities, jobs requiring higher skill levels than their current position with lower job-offer probabilities, and they perceive having a low degree of connection as slightly more detrimental to securing an offer. Finally, we account for wage level differences by classifying workers into low (annual wage of $0–$49,999), middle ($50,000–$99,999), and high wage ($100,000 or more) groups. Although wage groups generally do not strongly influence perceptions of how job attributes affect job-offer rates, high-wage workers tend not to associate high wages with higher job-offer rates—consistent with the idea that high-wage jobs are more competitive and feature lower offer rates. However, even among high-wage workers, having a connection is still considered the most important factor in improving job-offer probabilities. This mitigates concerns that respondents confuse job-offer probability with their likelihood of accepting a job.[18]

Since we ask about the specific job-offer probabilities associated with each one of the two jobs in a table, we can examine as a robustness check an additional specification that is otherwise identical to equation (7) but includes the perceived job-offer probability, $JobOfferProbability_{ijt}$ for individual i's perceived probability that she will receive an offer from job j in table t, as the dependent variable:

$$JobOfferProbability_{ijt} = \alpha + \beta_{connection} Connection_{ijt} + \boldsymbol{\beta}_x \boldsymbol{X}'_{ijt} + \gamma^k_{table} table^k_{ijt} + e_{ijt} \tag{8}$$

Figure C9 plots the estimates from this specification. The coefficient estimates for the personal connection categories are still positive and significant, but smaller in magnitude: respondents on average perceive that having some connection raises their probability of receiving an offer by 3.9 to 4.3 percentage points. The two specifications answer different questions—the forced-choice estimate measures how often a connection tips the comparison between two otherwise-similar jobs, while the slider estimate measures how much it shifts the perceived offer probability in levels—and we report both throughout. We further discuss the reliability of our estimates in the conjoint experiment in Appendix C.2.

5 A Stylized Model on Coworkers and Job Choices

The empirical results in Sections 3 and 4 establish that current and past coworkers shape what workers believe about wages elsewhere in the economy and how accessible they believe those sectors to be. Both associations are positive: workers with more coworker ties to a sector hold more accurate beliefs about its wages *and* expect a higher chance of receiving an offer there. Read on their own, these estimates suggest that coworker networks are unambiguously useful.

That reading is incomplete, because it prices only the sign of each channel and not where each channel points. Coworker ties are not spread evenly across destinations, and information travels along those ties but so does access. Where a worker's ties happen to point, better access directs her applications toward destinations she would not have chosen on wage grounds alone. The two

[18] Despite overlapping confidence intervals across wage groups, the "high connection" category remains the largest point estimate among all attributes for high-wage workers as well, underscoring the importance of connections for perceived job-offer probabilities even among those least likely to misinterpret the survey question.

channels can therefore work against each other, and no cross-sectional regression can separate them: both enter with a positive coefficient.

This section builds a partial-equilibrium model of sector choice that can. The model prices one margin: reallocation across 19 broad sectors. It is silent on mobility across firms, occupations and match qualities within a sector, which is where most wage dispersion resides, and the magnitudes below should be read as the sectoral component alone. Workers know their own sector's wage but must infer wages elsewhere, relying partly on current coworkers who arrived from those sectors. They also face a perceived probability of receiving an offer that depends on their coworker ties. We calibrate the model to the survey and to CPS transition flows, and use it to answer three questions. First, how costly is the belief friction as a whole? Second, once the informational and access roles of coworker networks are separated, what is each worth, and what is their net effect? Third, if the wage composition of coworker ties is what determines where access sends workers, what would a policy that changed that composition be worth?

5.1 Setup

The model focuses on employed workers. Time is infinite and discrete. An agent can supply one unit of labor to $n = 1, ..., N$ sectors. Agents receive the competitive market wage w_t^n if employed in sector n at time t. We consider the partial equilibrium for a sequence of wages, $\{w_t^n\}$, that follow any exogenous stochastic process, in each industry.

Denote the number of workers in industry n at the beginning of period t by L_t^n. At the end of each period, workers choose which sector to direct an application toward based on adjustment costs, their perceived wages differentials, idiosyncratic taste shocks, and the job-offer rate. Denote the fraction of workers that reallocate from industry n to k $\mu_t^{n,k}$. If a worker moves from n to k, she incurs an adjustment cost $\kappa^{n,k}$, which is the same for all workers in all periods and is publicly known. The adjustment cost can be thought of as the cost of applying to and preparing for a move from the origin sector to the destination sector—search effort, retraining, and the time spent assembling an application. It is a *search* cost rather than a moving cost: it is incurred whenever the worker directs her application toward sector k, whether or not an offer materializes.

In addition to the adjustment costs, workers are subject to idiosyncratic taste shocks $\nu\epsilon_t^n$ each period. The idiosyncratic taste shocks are independently and identically distributed across individuals, and drawn from a Type-I Extreme-Value (Gumbel) Distribution with zero mean and scaled by a parameter ν.

We assume that workers in the same industry all have the same beliefs. The belief of workers in industry n at the end of period t about the wage in industry k at time $t' \geq t$ is characterized by the following equation:

$$\mathbb{E}_t^n \ln w_{t'}^k = \omega^{nk} \ln w_{t'}^k + (1 - \omega^{nk}) \ln w_{t'}^n, \qquad \omega^{nk} = \phi_0 + \phi_{cw}\, s_{cw}^{nk} \tag{9}$$

where s_{cw}^{nk} denotes the share of workers currently in sector n whose most recent prior employer was in sector k. Equation (9) assumes that agents form beliefs of wages in other sectors as a weighted average of the correct industry wage and their own industry wage.[19] The weight ω^{nk} placed on the accurate belief has two parts. The intercept ϕ_0 is the accuracy a worker achieves with no coworker ties to sector k at all—from public information, media coverage, and general labor-market knowledge. The slope ϕ_{cw} is the additional accuracy each unit of current-coworker share delivers; at the mean coworker share of 0.10 this amounts to about 1.3 percentage points of extra weight, and moving from the 25th to the 75th percentile of the share distribution adds about 4 points. This separation matters for what follows: ϕ_0 determines how large the belief friction is, while ϕ_{cw} determines how much of it coworkers close. A model that imposes $\phi_0 = 0$ forces coworkers to carry all cross-sector wage knowledge and mechanically overstates their informational role, so we estimate both. While one could allow past coworkers—those who left sector n and are now in sector k—to also transmit wage information back (with a coefficient ϕ_{past}), our estimates indicate that this reverse channel is not statistically significant; accordingly, we set $\phi_{past} = 0$ and retain only the current-coworker channel in equation (9).

When workers in sector n apply for jobs in sector k, they expect to receive an offer with probability $\Gamma^{n,k}$, which depends on their social connections: with $\Gamma^{n,k} \in [0, 1]$ and $\Gamma^{n,n} = 1$, so that remaining in one's own sector is always feasible. If they do not receive an offer, they remain in the same sector for the next period. Consistent with the interpretation above, $\kappa^{n,k}$ is sunk once the application is directed and is incurred regardless of whether the move succeeds; this is why $\kappa^{n,k}$ enters the value function below outside the $\Gamma^{n,k}$ weighting.

Crucially, $\Gamma^{n,k}$ is the worker's *subjective* probability of receiving an offer. It governs which sector she directs her application toward and therefore shapes the model-implied transition flows; we do not separately model the true arrival of offers, so $\Gamma^{n,k}$ should be read as a belief that structurally governs mobility decisions rather than as a measured hiring rate. We make the additional assumption that $\Gamma^{n,k}$ follows the specific form:

$$\log\left(\frac{\Gamma^{n,k}}{1-\Gamma^{n,k}}\right) = \psi_0 + \psi_{past}\, s_{past}^{nk} + \psi_{cw}\, s_{cw}^{nk} + \psi_{wage}\,(\ln w^k - \ln w^n) \tag{10}$$

where s_{past}^{nk} is the share of past coworkers of workers in sector n currently employed in sector k, s_{cw}^{nk} is the share of current coworkers who were last employed in sector k (both distinct from the application probabilities μ^{nk} defined below), and $\ln w^k - \ln w^n$ is the destination-to-origin log wage differential. The perceived job-offer probability therefore rises with referral ties through past coworkers (ψ_{past}) and information ties through current coworkers (ψ_{cw}). Depending on the sign of ψ_{wage}, perceived offer probabilities may rise or fall with the destination-to-origin wage differential: a positive value would indicate that workers view better-paying destinations as more accessible, while a negative

[19] We assume that beliefs of other industries' wages depend on own wages because agents could use their own compensation level to infer the temporary shocks hitting the entire economy. In addition, recent works such as Jäger et al. (2022) have shown that workers' beliefs of outside options are correlated with their own wage.

value would indicate that they view them as more selective. We estimate this parameter rather than sign it a priori, and report the estimate in Section 5.2.

We normalize $\psi_0 = 0$, so that a worker with no coworker ties to a sector paying the same wage as her own perceives an even chance of receiving an offer there. This is a normalization rather than a substantive restriction: the *level* of $\mathbf{\Gamma}$ is not separately identified from the level of the adjustment cost κ, since raising every offer probability and raising every search cost both reduce mobility. Section 5.2 pins the joint level down with an aggregate exit-rate moment, which fixes the product of the offer-probability level and the cost level even though neither is separately identified from the transition data alone. We note that the offer-probability regression does estimate an intercept from the elicited levels, and that imposing $\psi_0 = 0$ in the model therefore discards that information; the sensitivity analysis in Section 5.7 varies the mapping from perceived to actual access at a fixed mean, and so does not speak to the level restriction itself.

It is worth being precise about what $\mathbf{\Gamma}$ does in the model, because it does two jobs. It governs where a worker directs her application—this is the sense in which it is a belief, and it is the object the survey measures. It also governs whether that application succeeds, which determines realized flows. Writing $\boldsymbol{\mu}$ for application probabilities, the realized transition matrix is

$$\boldsymbol{P} = \boldsymbol{\mu} \odot \mathbf{\Gamma}, \qquad P^{nn} = 1 - \sum_{k \neq n} \mu^{nk} \Gamma^{nk} \tag{11}$$

so that unsuccessful applicants remain in their origin sector. Because we observe only perceived offer probabilities, using the same $\mathbf{\Gamma}$ for both jobs is a maintained assumption—that perceptions are calibrated. Section 5.7 relaxes it by letting actual access be a monotone but not one-to-one transformation of perceived access, and reports how much the results move.

Therefore, the value of an agent in sector n at time t is assumed to be given by:

$$V_t^n = \ln w_t^n + \max_k \left\{ \left(\mathbb{E}_t^n [\Gamma^{n,k} \beta V_{t+1}^k + (1 - \Gamma^{n,k}) \beta V_{t+1}^n - \kappa^{n,k}] \right) + \nu \epsilon_t^k \right\}$$

Taking the expectation of V_t^n with respect to the vector ϵ, we can obtain the expected value of being employed in sector n:

$$\mathbb{E}_t^n [V_t^n] = \mathbb{E}_t^n \ln w_t^n + \mathbb{E}_\epsilon \left[\max_k \left\{ \mathbb{E}_t^n [\Gamma^{n,k} \beta V_{t+1}^k + (1 - \Gamma^{n,k}) \beta V_{t+1}^n - \kappa^{n,k}] + \nu \epsilon_t^k \right\} \right]$$

Note that the superscript n for the expectation operator indicates that the expected value of being in a sector depends on the workers' current sector. The coworker composition in industry n affects both the beliefs on wages and the job-offer rate.

Proposition 1. *The estimation equation can be written as:*

$$\begin{aligned}\ln \mu_t^{n,m} - \ln \mu_t^{n,n} = \frac{\beta}{\nu}\Gamma^{n,m}\mathbb{E}_t^n[\ln w_{t+1}^m - \ln w_{t+1}^n] \\ - \beta\Gamma^{n,m}\mathbb{E}_t^n \ln \mu_{t+1}^{m,m} + \beta(\Gamma^{n,m}-1)\mathbb{E}_t^n \ln \mu_{t+1}^{n,n} + \beta\mathbb{E}_t^n \ln \mu_{t+1}^{n,m} \\ - \frac{1-\beta}{\nu}\Gamma^{n,m}\kappa^{n,m}\end{aligned} \tag{12}$$

Proof. See Appendix E.2. □

Welfare Implication Realized welfare is evaluated under the true wage process, not under workers' beliefs: a worker who applies to a sector because she overestimates its wage is paid what the sector actually pays. Beliefs therefore enter welfare only through the choices they induce. Three terms make up the flow payoff in sector n. The worker collects the true log wage $\ln w^n$. She pays the search cost of whichever application she directs, in expectation $\sum_k \mu^{nk}\kappa^k$; this is incurred when the application is sent, not when the move succeeds, which is why it is weighted by $\boldsymbol{\mu}$ rather than by $\boldsymbol{P}$. Finally, because the taste shocks are Gumbel, she enjoys an option value equal to ν times the entropy of her application probabilities,

$$H(\boldsymbol{\mu}^n) = -\sum_k \mu^{nk}\ln \mu^{nk},$$

which is the standard logit inclusive-value term and captures the welfare a worker derives from having genuine alternatives.[20] Continuation values are discounted along *realized* transitions $\boldsymbol{P}$, since an unsuccessful applicant continues in her origin sector. At the steady state,

$$\begin{aligned}\vec{W} &= \ln \vec{w} - (\boldsymbol{\mu}\odot\boldsymbol{\kappa})\mathbf{1} + \nu H(\boldsymbol{\mu}) + \beta \boldsymbol{P}\vec{W} \\ &= (\mathbb{I} - \beta\boldsymbol{P})^{-1}\big(\ln \vec{w} - (\boldsymbol{\mu}\odot\boldsymbol{\kappa})\mathbf{1} + \nu H(\boldsymbol{\mu})\big).\end{aligned} \tag{13}$$

We report counterfactuals as consumption-equivalent variation. Writing $\vec{\pi}$ for the stationary employment distribution, the welfare change from a counterfactual that delivers $\vec{W}'$ in place of the baseline $\vec{W}$ is

$$\text{CEV} = 100 \times \left[\exp\left\{(1-\beta)\,\vec{\pi}^\top\left(\vec{W}' - \vec{W}\right)\right\} - 1\right], \tag{14}$$

the permanent percentage change in consumption a worker would accept in exchange for the change in environment. The expression follows from the flow payoff being logarithmic in the wage: scaling consumption permanently by $1+x$ adds $\ln(1+x)/(1-\beta)$ to $\vec{W}$, so that inverting gives (14). Two features matter for interpretation. The factor $(1-\beta)$ converts a present discounted value into a per-period flow, so the numbers are annual-equivalent rather than lifetime magnitudes. And the weights $\vec{\pi}$ are held at their *baseline* values in every comparison. Comparing $\vec{\pi}'^\top\vec{W}'$ against $\vec{\pi}^\top\vec{W}$ would confound the change in welfare with the reshuffling of workers across sectors, and can report a gain

[20] Omitting this term would understate welfare in exactly the states where the model says networks matter—those in which a worker's applications are spread across many destinations rather than concentrated on one.

purely because population moved toward sectors that were already better off. Fixing the weights asks the economically meaningful question: how does the baseline population fare under the counterfactual?

5.2 Calibration

The model contains five behavioural parameters estimated from the survey, two parameters set externally, and 19 calibrated destination costs. The belief parameters ϕ_0 and ϕ_{cw} and the offer-probability parameters ψ_{past}, ψ_{cw}, ψ_{wage} come directly from the survey. The adjustment costs $\{\kappa^k\}$ are calibrated to CPS transition flows. We set the discount factor $\beta = 0.95$ and the scale of the taste shock $\nu = 2$, both standard in this literature. Table E2 collects the results with bootstrap confidence intervals.

Coworkers and Wage Beliefs Equation (9) says that a worker's belief about the wage in sector k is a weighted average of the true wage there and the wage in her own sector, with the weight $\omega^{nk} = \phi_0 + \phi_{cw} s^{nk}_{cw}$ rising in her coworker ties to k. We estimate both parameters from a within-person least squares regression of the log prediction error on the log wage differential and its interaction with the coworker share:

$$\log(PredictionError)_{ijs} = \phi_0 \log(WageDiff)_{ijs} + \phi_{cw}\, CoworkerShare_{ijs} \times \log(WageDiff)_{ijs} + \gamma_i + \epsilon_{ijs} \tag{15}$$

where $CoworkerShare_{ijs}$ is the share of current coworkers for respondent i—currently employed in sector j—who were last employed in sector s, and $\log(WageDiff)_{ijs}$ is the log difference between the CPS median wage in sector s and in sector j. Individual fixed effects γ_i absorb systematic over- or under-estimation of wages that varies by respondent. We proxy for the coworker share using respondents' estimates of the number of current coworkers from each industry, divided by the total number of coworkers whose previous industry they know.

One feature of the elicitation deserves comment. The survey asks respondents for the median wage of workers *with their own characteristics*, whereas we score their answers against the unconditional industry median. These are different objects, and the mismatch is deliberate. The model assigns a single wage w^k to each sector, identical across workers; the parameter ϕ must therefore measure how accurately a worker knows what a sector pays, which is what the unconditional median records. Scoring against a person-specific conditional median would estimate a quantity the model has no place for. Because the regression includes individual fixed effects, any respondent-level gap between "workers like me" and the industry average is absorbed; what remains is the within-person variation across the industries each respondent was shown.

We estimate $\hat{\phi}_0 = 0.515$ (95% CI to be re-bootstrapped) and $\hat{\phi}_{cw} = 0.129$ (interval to be re-bootstrapped).[21] Both are precisely estimated and the interpretation is straightforward. A worker

[21] Full regression results are in Appendix Table C7. Confidence intervals throughout this section are percentile bootstrap intervals from 200 resamples over survey respondents; each replicate re-estimates the entire chain from survey moments through $\{\kappa^k\}$ to welfare.

with no coworker ties to a sector already places about two-thirds of the weight on that sector's true wage, so cross-sector wage beliefs are substantially but not fully informed at baseline. Each additional unit of coworker share adds about a fifth of the remaining weight. Since the average current-coworker share for a given destination sector is small, the marginal contribution of coworkers to belief accuracy at observed tie densities is correspondingly modest—a fact that will matter for the decomposition in Section 5.4.

Restricting $\phi_0 = 0$, as in a specification where coworkers carry all cross-sector wage information, yields a single coefficient of 0.50. That estimate is not wrong, but it conflates the two roles: it is what a worker with an average coworker share knows in total, not what her coworkers contributed. Attributing all of it to the network would roughly double the informational role we assign to coworkers. We therefore use the two-parameter version throughout.

Equation (9) also assumes a symmetric contribution of the origin- and destination-industry wages to the prediction error. To test this, we estimate a specification that allows differential weighting:

$$\begin{aligned}\log(PredictionError)_{ijs} =& \phi_1 CoworkerShare_{ijs} \times \log(DestWage)_{ijs} + \\ & \phi_2 CoworkerShare_{ijs} \times \log(OrigWage)_{ijs} + \gamma_i + \epsilon_{ijs}\end{aligned} \tag{16}$$

The results are presented in Appendix Table C8. The estimates for ϕ_1 and ϕ_2 are comparable, confirming that individuals form beliefs about other-industry wages using a weighted average of their own-industry wage and the other-industry wage.

Coworkers and Beliefs on Job-Offer Probabilities We estimate ψ_{past}, ψ_{cw}, and ψ_{wage} in equation (10) using a binomial generalized linear model (fractional logit) with survey sampling weights:

$$\begin{aligned}\log\left(\frac{PerceivedProb_{is}}{1 - PerceivedProb_{is}}\right) =& \psi_0 + \psi_{past}\, PastCoworkerShare_{is} + \psi_{cw}\, CoworkerShare_{is} \\ & + \psi_{wage}\left(\ln StartWage_{is} - \ln OwnWage_i\right) + \boldsymbol{\alpha}^\top \bar{\boldsymbol{Z}}_i + \epsilon_{is}\end{aligned} \tag{17}$$

where $PerceivedProb_{is}$ is the perceived probability of receiving an offer for individual i in industry s, scaled to $[0, 1]$. $PastCoworkerShare_{is}$ is the share of past coworkers currently employed in industry s, and $CoworkerShare_{is}$ is the share of current coworkers previously employed in industry s. $\ln StartWage_{is} - \ln OwnWage_i$ is the respondent's log predicted starting wage in sector s relative to their current wage. The vector $\bar{\boldsymbol{Z}}_i$ collects the respondent's mean past-coworker and current-coworker shares across all listed industries; these Mundlak-style controls absorb correlated random effects arising from unobserved individual propensities to perceive many (or few) coworkers across all sectors.[22]

Table 8 reports the estimates: $\hat{\psi}_{past} = 0.32$, $\hat{\psi}_{cw} = 0.53$, and $\hat{\psi}_{wage} = -0.27$. Two features are noteworthy. First, the current-coworker channel is larger than the past-coworker channel, suggesting that

[22] This correlated random effects approach follows Mundlak (1978) and is the cross-sectional analogue of including individual fixed effects in a linear model.

information transmitted by coworkers about their previous sectors matters more for perceived access than referral connections maintained by past coworkers at the destination. Second, the negative sign on $\hat{\psi}_{wage}$ implies that workers perceive higher-wage destinations as harder to enter: conditional on coworker ties, a more attractive wage differential is associated with a lower perceived offer probability, consistent with workers recognizing that high-wage sectors hire more selectively. This is the first hint of the mechanism developed in Section 5.5. Coworker ties raise perceived access, but a worker's ties point where her network points, and the wage gradient works against her.

Table 8: Estimated Effect of Coworker Networks on Perceived Job-Offer Probabilities

Parameter	Estimate	Bootstrap SE
Past coworker share, ψ_{past}	0.3158	0.1526
Current coworker share, ψ_{cw}	0.5256	0.1756
Log wage differential, ψ_{wage}	-0.2667	0.1508
Mean of Dep. Var.	0.40	

Note: Table reports estimated coefficients from equation (17). The dependent variable is the perceived job-offer probability scaled to $[0, 1]$. The regression uses a binomial GLM (fractional logit) with survey sampling weights and Mundlak controls for unobserved individual heterogeneity. Bootstrap standard errors from 200 resamples over survey respondents are reported. These are the standard errors of the regression coefficients alone. Table E2 reports larger standard errors for the same parameters because each replicate there propagates the draw through the full calibration chain, including the re-solved cost vector $\{\kappa^k\}$.

Adjustment Costs The remaining parameters are the adjustment costs. Rather than assuming a common scalar, we estimate a destination-specific vector $\{\kappa^k\}_{k=1}^{19}$, where κ^k applies to all moves into sector k regardless of origin.[23] We decompose the vector as $\kappa^k = \kappa_0 + \tilde{\kappa}_k$ with $\sum_k \tilde{\kappa}_k = 0$, separating the overall level of mobility costs from their cross-sector profile, because the two are identified by different moments.

The profile $\{\tilde{\kappa}_k\}$ is calibrated to the LEHD *conditional* destination distribution: for each origin sector, the shares of cross-sector movers going to each destination. The level κ_0 is calibrated separately to the aggregate annual cross-sector exit rate of 0.10. This split is necessary rather than cosmetic. The transition data record where movers go conditional on changing sectors—each origin row sums to one—so they contain no information about how many workers move at all. Treating them as unconditional transition probabilities, as is sometimes done, implies exit rates several times larger than any observed in the CPS, and in most origin rows implies a flow that no admissible offer-probability

[23] Skill requirements and entry barriers vary primarily by destination. Identifying a separate cost for each of the $19 \times 18 = 342$ origin-destination pairs would require additional assumptions; destination-specific costs are parsimonious but still flexible.

matrix can generate.[24] With the split, the model matches the aggregate exit rate exactly and fits the conditional shares with a root mean squared error of 0.029 and a correlation of 0.84 across the 342 off-diagonal pairs (Table E3).

Figure E2 plots the estimated costs. The mean is 8.71 and the range is [5.20, 13.45], with no sector at a bound. The ordering is economically sensible. Entry into specialized, capital-intensive sectors is most expensive—Utilities (13.45), Mining (12.38), and Agriculture (11.25)—consistent with high occupational skill requirements and long training periods. High-turnover service sectors are cheapest: Administrative Support (5.20), Retail Trade (5.25), and Accommodation and Food Services (5.72).

One methodological point deserves emphasis, since it drives the results that follow. We calibrate $\{\kappa^k\}$ *once*, on the coworker baseline, and hold it fixed in every counterfactual. Adjustment costs are structural: they describe retraining and search effort, not the network environment, and there is no reason for them to change when we remove a coworker channel. Re-estimating them under each counterfactual—so that each scenario is forced to reproduce the observed CPS flows—would mechanically absorb the very channel being removed into the cost vector, leaving nothing for the counterfactual to reveal. This is the standard treatment in the dynamic discrete-choice mobility literature (Artuç, Chaudhuri and McLaren, 2010; Caliendo, Dvorkin and Parro, 2019), and it is what makes the decomposition in Section 5.4 interpretable.

5.3 The Cost of the Belief Friction

Before separating the coworker channels, it is useful to know how much is at stake in beliefs at all. We compute a benchmark in which workers know every sector's wage exactly—setting $\omega^{nk} = 1$ for all pairs—holding adjustment costs and offer probabilities at their calibrated values.

Correcting cross-sector wage beliefs is worth 2.23% of permanent consumption (95% CI [0.97, 1.47]). The gain is positive in all 19 sectors, ranging from +1.95% to +2.88%, and positive in every bootstrap draw. In the long run the stationary average wage rises from \$52,775 to \$60,659.

That last figure requires care, and we report the full transition path rather than the steady state alone. Figure E3 traces the average wage after beliefs are corrected at date zero, starting from the baseline employment distribution. The adjustment is slow: workers reallocate one sector-switch at a time, so the average wage rises by \$851 in the first year—10.8% of the eventual gain—and reaches half the total only after about seven years. Quoting the \$7,884 long-run gap as though it were an immediate wage increase would overstate the near-term value of the intervention by more than an order of magnitude. It is a long-run stationary average, and we describe it that way throughout.

This benchmark also fixes the scale for everything that follows. It is the value of eliminating the belief friction *entirely*, by any means. Coworker networks are one imperfect mechanism for closing part of it, so 2.23% is a ceiling: no result in the next section can exceed it, and the question is what

[24] Formally, matching an unconditional target T^{nk} requires $\sum_{k \neq n} T^{nk}/\Gamma^{nk} \leq 1$, since $\mu^{nk} = T^{nk}/\Gamma^{nk}$ must be a probability. With the target read unconditionally, 16 of the 19 origin rows violate this bound for any $\mathbf{\Gamma} \leq 1$. Splitting the target into a conditional shape and a separate level moment removes the problem.

fraction of it the network delivers.

5.4 Two Channels, Oppositely Signed

We now separate what coworker networks do. The model contains two channels that the reduced-form estimates cannot tell apart, because both enter with a positive coefficient. The *information* channel operates through ϕ_{cw}: coworkers make wage beliefs more accurate. The *access* channel operates through ψ_{past} and ψ_{cw}: coworkers raise the perceived probability of receiving an offer. We shut each down in turn, holding $\{\kappa^k\}$ fixed, and report the consumption-equivalent variation relative to the coworker baseline.

Interpreting these counterfactuals requires care, because each of them flattens a matrix. Setting $\phi_{cw} = 0$ makes the belief weight ω^{nk} identical across destinations; setting $\psi_{past} = \psi_{cw} = 0$ does the same to $\mathbf{\Gamma}$. A flatter matrix spreads application probabilities more evenly, and the logit option value $\nu H(\boldsymbol{\mu})$ rewards that dispersion mechanically, whatever it does to wages. Table E4 therefore reports the welfare difference alongside its three components: the wage flow, the search costs paid on applications, and the option value. Only the first reflects reallocation.

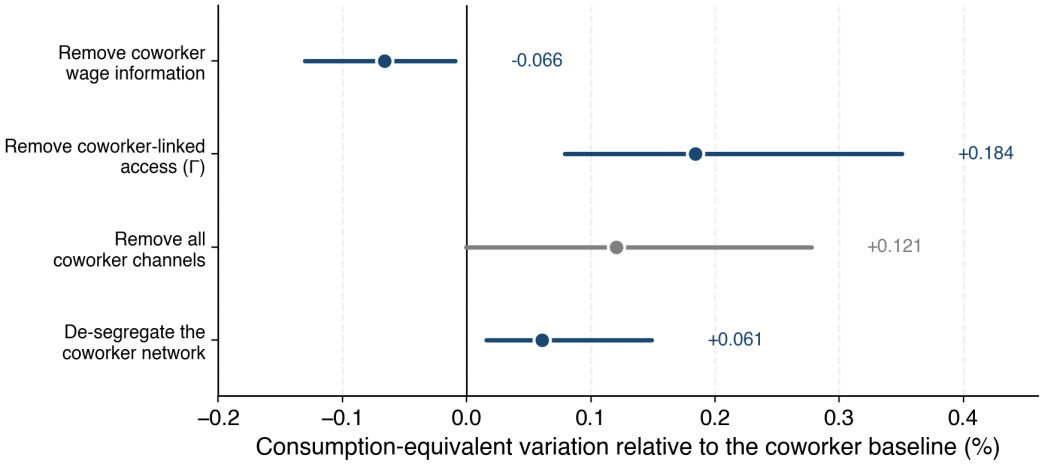
The option-value term is large in both counterfactuals and similar across them, $+0.360$ for information and $+0.264$ for access, and it is an artefact of how each experiment is constructed rather than a statement about coworkers. The wage components are what distinguish the channels, and they are oppositely signed and comparable in size. Removing coworker wage information costs -0.138 in wages: without it, workers direct applications toward sectors that pay less well than they would have with the information. Removing coworker-linked access *gains* $+0.102$: the access coworkers provide points applications toward destinations that pay less well than the alternatives.

This is what the model contributes. Coworker networks improve the allocation of workers across sectors through what they reveal, and worsen it through where they direct applications, and the two effects are close enough in magnitude that neither the reduced-form estimates nor their sum discloses the net consequence. We therefore report the channels separately and do not net them: the residual between two effects estimated from different moments is not a quantity the model determines, and nothing that follows depends on it.

Distributional Effects It is natural to ask whether coworker networks change who ends up where, and in particular whether they hold low-wage workers in place. In the calibrated model, they do not, at least not through retention. The correlation between log sector wage and the model-implied stayer rate is 0.992 both with and without coworker networks; the average stayer rate is 0.26 percentage points *lower* with networks than without, and the difference in the six lowest-wage sectors is -0.37 percentage points. Networks do not compress the wage-stayer gradient. Nor do they push population toward the bottom of the wage distribution: the correlation between a sector's log wage and its change in stationary employment share when networks are added is $+0.13$, so if anything networks shift workers slightly toward better-paying sectors.

The access channel documented above therefore operates not on *whether* workers move but on *where* they move. Coworker ties do not lock workers into their current sectors; they steer the moves that do occur toward destinations selected by network composition rather than by wages. That is a claim about the destination margin, and it is the one the next section takes up directly.

Figure 2: Welfare Effects of Coworker Channels



Note: Consumption-equivalent variation relative to the coworker baseline for each coworker counterfactual, with 95% percentile bootstrap intervals from 400 resamples over survey respondents. The perfect-beliefs benchmark of +2.229 (interval [+1.967, +2.547]) is omitted from the figure, since plotting it on the same axis compresses the coworker channels to illegibility; it is reported in the text and in Table E4. Positive values indicate that the counterfactual world is better than the world with coworker networks. Intervals excluding zero are shown in dark blue; intervals including zero in gray. The information and access channels are individually signed and oppositely directed, so their net effect is small and not precisely signed.

5.5 Composition, Not Extent

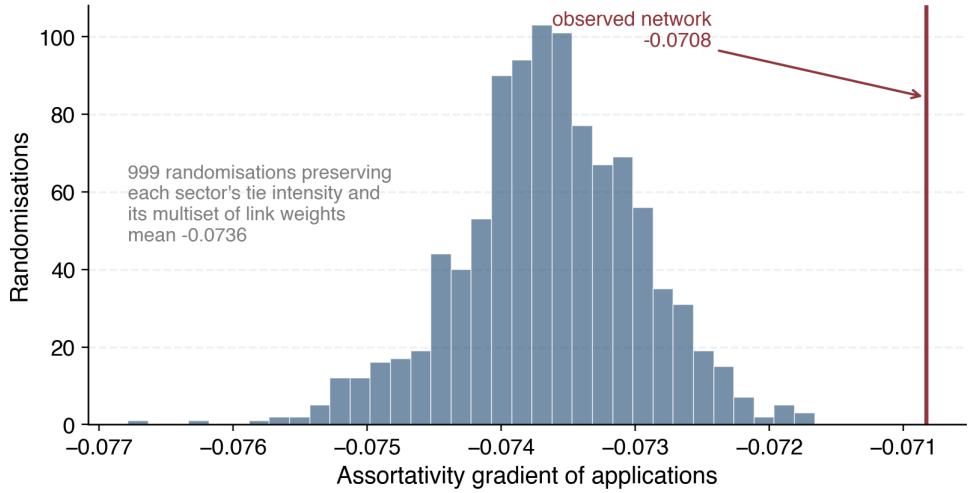
Why should access and information differ in sign? Better information improves a worker's belief about any sector she considers, so it helps regardless of which sector that is. Access does not have that property. It raises the perceived chance of an offer only where a worker has ties, and ties are not distributed at random across destinations. Access therefore tilts applications toward the particular sectors a worker is already connected to, at the expense of applications she would otherwise have directed elsewhere. Whether that tilt helps or hurts depends on where those ties point, which the permutation test below settles.

The tilt is small, and we do not overstate it. Removing coworker access shifts the average wage of the destinations workers apply to by less than 0.002 log points. What matters is not that coworker networks send workers to dramatically worse places, but that the direction of the tilt is adverse while the direction of the information channel is favourable, and that the two roughly offset.

That the tilt reflects the network's composition rather than its density can be tested directly. For each of 999 randomisations we hold fixed each origin sector's total tie intensity *and* the multiset of its individual link weights, permuting only which destinations those links point to. Every randomised network is exactly as concentrated as the observed one; only its wage composition is scrambled. The

observed network produces a stronger wage gradient in applications than all 999 randomisations, a one-sided p-value of 0.001 (Figure 3). Composition, not extent, is what makes coworker access allocatively distinctive—and it is therefore composition that a policy could change.

Figure 3: Permutation Placebo: Composition, Not Extent



Note: Distribution of the wage gradient of applications across 999 randomisations of the coworker network. The gradient is the slope, across origin sectors, of the average log wage of applied-to destinations on the origin's own log wage. Each randomisation preserves every origin sector's total tie intensity and the multiset of its individual link weights, permuting only the destinations those links point to, so each randomised network is exactly as concentrated as the observed one. The vertical line marks the observed network, which exceeds all 999 randomisations (one-sided $p = 0.001$).

5.6 Directing Coworker Networks

If composition is what makes coworker access allocatively distinctive, then a policy that changes composition should be valuable, and one that merely adds connections should not. This section prices both. Every intervention holds each treated sector's *total* tie intensity fixed and changes only the destination mix, so no worker gains connections; and each is scaled by the share of workers a programme reaches, since rewiring an entire sector's network is a frontier no mentorship or exposure programme attains.

We treat the six lowest-wage sectors, which employ 44% of workers, under two designs. A *directed* programme points ties toward the six highest-paying sectors. A *broad* programme spreads them in proportion to destination employment, which is what generic networking amounts to. Table E5 reports both at reach from 10 to 100%.

Direction dominates breadth at every scale, and by more the smaller the programme. At full reach, directing is worth +0.252% of permanent consumption against +0.042% for broadening, a factor of six. At half reach the ratio is nine, at a quarter it is twenty, and at a tenth—the scale at which real programmes operate—broadening delivers essentially nothing while directing still delivers a positive gain. Generic networking works only if it reaches nearly everyone, because spreading ties in proportion to employment points workers at *large* sectors, which are mid-paying; what pays is pointing them

at better ones.

The gains are convex in reach. Directing rises from $+0.007$ at a tenth to $+0.031$ at a quarter, $+0.088$ at a half and $+0.252$ at full coverage, so the gain per point of employment treated more than triples between the smallest and largest programmes. Small pilots therefore understate what a scaled programme would deliver, which is an uncomfortable message for evaluation designs built around them.

These are genuine reallocation gains rather than artefacts of the counterfactual. The wage component accounts for the entire directed effect at full reach, $+0.262$ against a total of $+0.252$, with the option-value term small and frequently negative. Workers move: the annual rate at which workers leave the six lowest-wage sectors for the six highest rises from 2.95 to 3.07%, and the long-run employment-weighted geometric mean wage rises by about \$195 per worker per year.

Two caveats bound the reading. The directed programme assumes a policy can aim ties at particular high-paying destinations, which real mentorship and exposure schemes cannot do cleanly; it is an upper bound on what targeting could achieve rather than a programme evaluation. And these are changes in network composition, not costed interventions: we compute benefits, not net benefits. What the exercise establishes is a ranking robust to both caveats—for a given quantity of network-building effort, direction dominates breadth, and increasingly so at the scales at which programmes actually run.

5.7 Robustness and Limitations

Perceived versus Actual Access The model uses a single $\mathbf{\Gamma}$ both to direct applications and to determine whether they succeed, and the survey identifies only the former. Since the access channel carries essentially all of the de-segregation and policy results, this assumption deserves a direct test. We relax it by letting actual access be a monotone transformation of perceived access,

$$\operatorname{logit} q^{nk} = \alpha + \rho \operatorname{logit} \tilde{\Gamma}^{nk},$$

where applications are still directed by perceived $\tilde{\mathbf{\Gamma}}$ but realized transitions use q^{nk}, and α is chosen at each ρ to preserve the aggregate exit rate. Values of $\rho < 1$ mean workers over-discriminate—actual access is flatter than perceived—and $\rho > 1$ the reverse; $\rho = 1$ recovers the baseline.

Table E6 reports $\rho \in [0.5, 1.5]$. Every result retains its sign. The gain from removing coworker access ranges from $+0.062$ to $+0.207$ and the de-segregation gain from $+0.098$ to $+0.248$, with the belief benchmark nearly invariant at 1.16 to 1.24. Magnitudes vary by a factor of about 2.5 across this range, which we do not minimize; but the qualitative conclusions—that the two channels are oppositely signed, that de-segregation helps, and that the belief friction dominates both—do not depend on treating perceived access as actual access.

Respondent Comprehension A concern specific to this survey is that 39.7% of respondents initially misclassified the current- versus past-coworker taxonomy before the clarification screen. Re-estimating the belief parameters on the 56.2% who classified the coworker taxonomy correctly on the first attempt yields $\hat{\phi}_0 = 0.666$ and $\hat{\phi}_{cw} = 0.193$, against 0.662 and 0.231 in the full sample; the offer-probability estimates move similarly little. The results are not driven by respondents who misunderstood the categories.

The Exit-Rate Target The transition data are conditional on a move and so identify the composition of cross-sector flows but not their rate. The calibration therefore requires one moment stated in levels, for which we target an aggregate annual cross-sector exit rate of 0.13. This figure is not directly observed: the LEHD source is quarterly, and converting a quarterly job-to-job separation rate into an annual probability of changing sector involves offsetting approximations, so plausible values lie between roughly 0.12 and 0.16. Appendix Table E1 varies the target from 0.08 to 0.20. Every channel we report retains its sign across that range, and the de-segregation and policy results retain theirs as well; the access channel is the most sensitive, varying by roughly a factor of two.

6 Conclusion

Coworker networks shape industry mobility through two channels. Current coworkers transmit information about their prior sectors through two mechanisms: they improve wage belief accuracy and raise perceived job-offer probabilities there. Past coworkers employed at destination firms generate additional referral access, further raising perceived offer rates. Using novel survey data from a quota-balanced sample of full-time U.S. workers aged 25 to 60 outside health care, reweighted to CPS benchmarks, we document both channels and estimate their magnitudes. Having all past coworkers employed in a given industry is associated with perceived offer probabilities 11 percentage points higher relative to a mean of 38 percent, and with a stated intention to apply 23 percentage points higher in our preferred specification. In a forced-choice conjoint experiment, active coworker connections raise the probability that a job is judged more likely to yield an offer by 11.4 percentage points, the most influential factor among all job attributes tested, including wages, skill requirements, and schedule flexibility; the corresponding effect on the elicited offer probability is 3.9 to 4.3 percentage points. Workers whose former coworkers mentioned their current industry of employment earn wages approximately 13 log points higher after controls, consistent with information transmission about industry opportunities. The analogous premium for receiving a direct referral is 20 log points unconditionally but falls to 0.05 log points and loses significance once occupation, previous industry, and current industry are controlled for, suggesting that the raw referral premium largely reflects sorting across job categories rather than a within-category wage gain.

Embedding these reduced-form estimates in a calibrated partial-equilibrium sector-choice model separates two things the regressions cannot. In the reduced form, both coworker channels enter positively: ties raise belief accuracy and they raise perceived access. In the model, they point in opposite

directions. Removing the informational channel costs 0.07 percent of permanent consumption, and every sector is worse off. Removing coworker-linked access *raises* welfare by 0.18 percent, and 18 of 19 sectors are better off. Both channels are individually signed, and neither is visible in the reduced form, where they enter with the same sign. We report them separately rather than netting them, since their difference is a residual whose sign is not robustly determined. Set against a benchmark in which cross-sector wage misperceptions are eliminated entirely—worth 2.23 percent—coworkers close roughly 3 percent of the information friction they are often credited with solving.

Where coworker access sends workers is compositional, not a matter of network strength. Raising the odds of an offer along a worker's existing ties raises them only for the sectors those ties reach. A permutation test shows this is what drives the channel: holding each sector's total tie intensity and its distribution of link weights fixed while randomizing where ties point, the observed network produces a stronger wage gradient in applications than all 999 randomizations ($p = 0.001$). We also find that coworker networks do not, in this model, hold workers in place—the wage-stayer gradient is essentially identical with and without them. The effect operates on the destination margin rather than the retention margin: networks do not determine whether workers move, but they do shape where the moves that occur end up.

The policy implication follows from the mechanism and is sharper than a general case for network-building. Direction matters far more than breadth. Redirecting the coworker networks of the six lowest-wage sectors toward the six highest-wage sectors—holding total network intensity fixed, so that no one receives additional connections—is worth about six times as much as broadening those same networks generically, and the advantage widens as programmes shrink: at a reach of one in ten workers, broadening delivers essentially nothing. The gain is a wage effect rather than a change in the option value of workers' choice sets. It raises annual movement from the bottom to the top of the wage distribution from 2.95 to 3.07 percent and reallocates 0.4 percent of the workforce upward in the long run; the two groups of sectors differ in average pay by roughly $47,000 a year in the calibration. Generic breadth accomplishes much less because employment-weighted connections point workers toward *large* sectors, which are mid-wage. Exposure and mentorship programmes that aim at specific higher-paying destinations should therefore dominate general networking initiatives, though we stress that our directed counterfactual assumes a precision of targeting that real programmes are unlikely to achieve, and so bounds rather than predicts what such policies could deliver.

Two limitations bound these conclusions. The model prices reallocation across 19 broad sectors and is silent on mobility across firms, occupations, and matches within a sector, which is where most wage dispersion resides; this is the main reason the aggregate magnitudes are small, and a coworker network that delivers a better job in the same sector is invisible in our accounting. And the offer probabilities are elicited as perceptions rather than measured as realized hiring rates. We show that the results retain their sign when actual access is allowed to differ from perceived access, but the survey alone cannot establish that the two coincide.

Our paper opens several avenues for future research. First, the analysis holds wages fixed in partial equilibrium; a general equilibrium extension would clarify how network-mediated mobility affects wage-setting and industry composition. Second, the growing prevalence of remote work may weaken the information and referral ties that operate through in-person coworker contact, altering the magnitude and distribution of network effects documented here. Third, incorporating the coworker channel into evaluations of specific labor market policies, including wage transparency laws, non-compete agreements, occupational licensing, and immigration programs, would help assess how these regulations interact with network-based mobility. Finally, studying how firm management practices shape the composition and reach of coworker networks could yield valuable insights into the organizational determinants of worker mobility.

References

Artuç, Erhan, Shubham Chaudhuri, and John McLaren. 2010. "Trade shocks and labor adjustment: A structural empirical approach." *American economic review*, 100(3): 1008–45.

Azoulay, Pierre, Joshua S Graff Zivin, and Jialan Wang. 2010. "Superstar extinction." *The Quarterly Journal of Economics*, 125(2): 549–589.

Bandiera, Oriana, Iwan Barankay, and Imran Rasul. 2009. "Social connections and incentives in the workplace: Evidence from personnel data." *Econometrica*, 77(4): 1047–1094.

Bayer, Patrick, Stephen L. Ross, and Giorgio Topa. 2008. "Place of Work and Place of Residence: Informal Hiring Networks and Labor Market Outcomes." *Journal of Political Economy*, 116(6): 1150–1196.

Beaman, Lori, and Jeremy Magruder. 2012. "Who Gets the Job Referral? Evidence from a Social Networks Experiment." *American Economic Review*, 102(7): 3574–3593.

Ben-Akiva, Moshe, Takayuki Morikawa, and Fumiaki Shiroishi. 1992. "Analysis of the reliability of preference ranking data." *Journal of business research*, 24(2): 149–164.

Burdett, Kenneth, and Dale T Mortensen. 1998. "Wage differentials, employer size, and unemployment." *International Economic Review*, 257–273.

Burks, Stephen V., Bo Cowgill, Mitchell Hoffman, and Michael Housman. 2015. "The Value of Hiring through Employee Referrals." *The Quarterly Journal of Economics*, 130(2): 805–839.

Burt, Ronald S. 2004. "Structural Holes and Good Ideas." *American Journal of Sociology*, 110(2): 349–399.

Cahuc, Pierre, Fabien Postel-Vinay, and Jean-Marc Robin. 2006. "Wage bargaining with on-the-job search: Theory and evidence." *Econometrica*, 74(2): 323–364.

Caldwell, Sydnee, and Nikolaj Harmon. 2019. "Outside options, bargaining, and wages: Evidence from coworker networks." *Unpublished manuscript, Univ. Copenhagen*, 203–207.

Caliendo, Lorenzo, Maximiliano Dvorkin, and Fernando Parro. 2019. "Trade and labor market dynamics: General equilibrium analysis of the china trade shock." *Econometrica*, 87(3): 741–835.

Calvo-Armengol, Antoni, and Matthew O Jackson. 2004. "The effects of social networks on employment and inequality." *American economic review*, 94(3): 426–454.

Card, David, Alexandre Mas, Enrico Moretti, and Emmanuel Saez. 2012. "Inequality at work: The effect of peer salaries on job satisfaction." *American Economic Review*, 102(6): 2981–3003.

Cingano, Federico, and Alfonso Rosolia. 2012. "People I Know: Job Search and Social Networks." *Journal of Labor Economics*, 30(2): 291–332.

Coleman, James S. 1988. "Social Capital in the Creation of Human Capital." *American Journal of Sociology*, 94: S95–S120.

Conlon, John J, Laura Pilossoph, Matthew Wiswall, and Basit Zafar. 2018. "Labor market search with imperfect information and learning." National Bureau of Economic Research.

Cornelissen, Thomas, Christian Dustmann, and Uta Schönberg. 2017. "Peer effects in the workplace." *American Economic Review*, 107(2): 425–456.

Cummings, Ronald G, Glenn W Harrison, and E Elisabet Rutström. 1995. "Homegrown values and hypothetical surveys: is the dichotomous choice approach incentive-compatible?" *The American Economic Review*, 85(1): 260–266.

Dustmann, Christian, Albrecht Glitz, Uta Schönberg, and Herbert Brücker. 2016. "Referral-based job search networks." *The Review of Economic Studies*, 83(2): 514–546.

Falk, Armin, and Andrea Ichino. 2006. "Clean evidence on peer effects." *Journal of labor economics*, 24(1): 39–57.

Fernandez, Roberto M., Emilio J. Castilla, and Paul Moore. 2000. "Social Capital at Work: Networks and Employment at a Phone Center." *American Journal of Sociology*, 105(5): 1288–1356.

Flood, Sarah, Miriam King, Renae Rodgers, Steven Ruggles, J. Robert Warren, et al. 2023. "Integrated Public Use Microdata Series, Current Population Survey: Version 11.0 [dataset]."

Folke, Olle, and Johanna Rickne. 2022. "Sexual harassment and gender inequality in the labor market." *The Quarterly Journal of Economics*, 137(4): 2163–2212.

Glitz, Albrecht. 2017. "Coworker networks in the labour market." *Labour Economics*, 44: 218–230.

Granovetter, Mark S. 1973. "The strength of weak ties." *American journal of sociology*, 78(6): 1360–1380.

Granovetter, Mark S. 1995. *Getting a Job: A Study of Contacts and Careers.* . 2nd ed., University of Chicago Press.

Hainmueller, Jens, Daniel J Hopkins, and Teppei Yamamoto. 2014. "Causal inference in conjoint analysis: Understanding multidimensional choices via stated preference experiments." *Political analysis*, 22(1): 1–30.

Harrison, Glenn W, and E Elisabet Rutström. 2008. "Experimental evidence on the existence of hypothetical bias in value elicitation methods." *Handbook of experimental economics results*, 1: 752–767.

He, Haoran, David Neumark, and Qian Weng. 2021. "Do workers value flexible jobs? A field experiment." *Journal of Labor Economics*, 39(3): 709–738.

Hensvik, Lena, and Oskar Nordström Skans. 2016. "Social Networks, Employee Selection, and Labor Market Outcomes." *Journal of Labor Economics*, 34(4): 825–867.

Ioannides, Yannis M, and Linda Datcher Loury. 2004. "Job information networks, neighborhood effects, and inequality." *Journal of economic literature*, 42(4): 1056–1093.

Jackson, C Kirabo, and Elias Bruegmann. 2009. "Teaching students and teaching each other: The importance of peer learning for teachers." *American Economic Journal: Applied Economics*, 1(4): 85–108.

Jäger, Simon, Christopher Roth, Nina Roussille, and Benjamin Schoefer. 2022. "Worker Beliefs About Outside Options." National Bureau of Economic Research.

Jäger, Simon, Christopher Roth, Nina Roussille, and Benjamin Schoefer. 2024. "Worker beliefs about outside options." *The Quarterly Journal of Economics*, qjae001.

Jovanovic, Boyan. 1979. "Job matching and the theory of turnover." *Journal of political economy*, 87(5, Part 1): 972–990.

Lin, Xinyue, and Sophia Mo. 2024. "Does Mobility Beget Mobility? Coworker Networks and the Sectoral Reallocation of Labor." Working Paper.

List, John A, and Craig A Gallet. 2001. "What experimental protocol influence disparities between actual and hypothetical stated values?" *Environmental and resource economics*, 20: 241–254.

Manski, Charles F. 2004. "Measuring Expectations." *Econometrica*, 72(5): 1329–1376.

Mas, Alexandre, and Amanda Pallais. 2020. "Alternative work arrangements." *Annual Review of Economics*, 12(1): 631–658.

Mas, Alexandre, and Enrico Moretti. 2009. "Peers at work." *American Economic Review*, 99(1): 112–145.

Miano, Armando. 2023. "Search Costs, Outside Options, and On-the-Job Search." *Job market paper, Harvard University*.

Mincer, Jacob. 1958. "Investment in human capital and personal income distribution." *Journal of political economy*, 66(4): 281–302.

Montgomery, James D. 1991. "Social networks and labor-market outcomes: Toward an economic analysis." *The American economic review*, 81(5): 1408–1418.

Mortensen, Dale T, and Christopher A Pissarides. 1999. "New developments in models of search in the labor market." *Handbook of labor economics*, 3: 2567–2627.

Mortensen, Dale T., and Tara Vishwanath. 1994. "Personal Contacts and Earnings: It Is Who You Know!" *Labour Economics*, 1(2): 187–201.

Mueller, Andreas I, and Johannes Spinnewijn. 2023. "Expectations data, labor market, and job search." *Handbook of Economic Expectations*, 677–713.

Mueller, Andreas I, Johannes Spinnewijn, and Giorgio Topa. 2021. "Job seekers' perceptions and employment prospects: Heterogeneity, duration dependence, and bias." *American Economic Review*, 111(1): 324–363.

Mundlak, Yair. 1978. "On the pooling of time series and cross section data." *Econometrica: journal of the Econometric Society*, 69–85.

Pallais, Amanda, and Emily Glassberg Sands. 2016. "Why the Referential Treatment? Evidence from Field Experiments on Referrals." *Journal of Political Economy*, 124(6): 1793–1828.

Podolny, Joel M., and James N. Baron. 1997. "Resources and Relationships: Social Networks and Mobility in the Workplace." *American Sociological Review*, 62(5): 673–693.

Rees, Albert. 1966. "Information networks in labor markets." *The American Economic Review*, 56(1/2): 559–566.

Seibert, Scott E., Maria L. Kraimer, and Robert C. Liden. 2001. "A Social Capital Theory of Career Success." *Academy of Management Journal*, 44(2): 219–237.

Simon, Curtis J., and John T. Warner. 1992. "Matchmaker, Matchmaker: The Effect of Old Boy Networks on Job Match Quality, Earnings, and Tenure." *Journal of Labor Economics*, 10(3): 306–330.

Sockin, Jason. 2022. "Show Me the Amenity: Are Higher-Paying Firms Better All Around?"

Stantcheva, Stefanie. 2023. "How to run surveys: A guide to creating your own identifying variation and revealing the invisible." *Annual Review of Economics*, 15(1): 205–234.

Sullivan, Paul, and Ted To. 2014. "Search and nonwage job characteristics." *Journal of Human Resources*, 49(2): 472–507.

Topa, Giorgio. 2001. "Social interactions, local spillovers and unemployment." *The Review of Economic Studies*, 68(2): 261–295.

Appendix

A Survey Design

A.1 Full Survey: link here

A.1.1 Screening and Quotas

1. Do you currently live in the United States?
2. How do you identify your gender?
3. How old are you?
4. In which state is your primary residence (the place where you usually live)?
5. What is the highest level of education you have completed?
6. What was the TOTAL income of your household, before any taxes or deductions, in 2023?
7. What is your current employment status?
8. How many jobs do you have (excluding volunteer or other unpaid work)?
9. What kind of industry is your employer in? If you have more than one job, please consider the employer for whom you work the most hours per week. Please select the most appropriate industry from the dropdown menu. You can find a brief description of each industry below.

A.2 Attention Check

1. This is a question to check that you are paying attention and reading the questions carefully. Please select both "1" and "4" to move to the next page of the survey.

A.3 Personal Characteristics

1. Are you working for a temporary employment agency, a firm that connects businesses to workers for temporary or contract work?
2. Do you consider yourself of Hispanic, Latino or Spanish origin?
3. Which of the following best describes your race?
4. What is the ZIP code of your primary residence (the place where you usually live)?

A.4 Employment Characteristics

1. What was the month and year that you started working at your main job?
2. What is your occupation at your main job? Some examples of occupation titles include electrical engineer, stock clerk, waiter/waitress, typist...

   Please type your occupation in the box below and select one of the suggested options. Try to be specific. For instance, write "preschool teacher" or "high school teacher" rather than just "teacher".

   If none of the options correspond to your occupation, try adding more detail or rephrasing.

3. How many hours per week do you USUALLY work at your main job? Please include overtime hours.

4. How much do you earn before taxes and other deductions at your main job in a year? Please include any bonuses, overtime pay, tips or commissions.

5. Roughly speaking, what are your annual earnings, before taxes and other deductions, at your main job? Please include any bonuses, overtime pay, tips or commissions.

6. Is your job primarily in-person, remote, or a hybrid of both?

7. How many hours per week do you usually work IN PERSON?

8. How many people work for your main employer at your usual place of work?

9. Counting all locations where your employer operates, what is your best guess for the total number of persons who work for your main employer?

10. Are you currently looking for a new job?

11. Were you employed somewhere else when you were hired for your main job?

12. Which of the following statements best describes your situation when you were hired for your main job? [I was employed when I got hired and quit my previous job; I was employed when I got hired but was about to lose my previous job; I was employed when I got hired but a temporary or seasonal job ended; I was employed when I got hired and kept my previous job; I was employed in a temporary job that was converted into a permanent job; Other (please specify)]

13. Why did you apply to your main job? Choose the three most important reasons from the ones listed below.

A.5 Past Employment

1. What kind of industry was your employer at your previous job in? Please select the relevant industry from the dropdown menu below. Industries are defined as in the previous question.

A.6 Types of coworkers - Explainer

1. Now we would like to ask you some questions about your current and past co-workers. We are going to mention three different types of coworkers. Please take some time to read the definitions below for each type, so you can answer the following question referring to the correct type of co-workers. Thanks a lot!

   1. Current coworkers at the current job: These are the coworkers that you have at your current employer and are still employed there. Please also include coworkers who are working for your employer in different offices/locations/branches.

   2. Past coworkers at the current job: These are the coworkers that worked with you at your current employer at some point in the past, but who have then left to work at other employers. Please do not consider those coworkers who transferred to different offices, but still work for your employer. Instead, consider those coworkers as belonging to category 1, "current coworkers at the current job."

3. Past coworkers at the previous job: These are the coworkers that worked with you at your previous employer (the employer where you worked before the current one). Please consider all those coworkers who worked for your previous employer while you were there, regardless of whether they have since then switched to other jobs.

2. To test your understanding of our definitions of the three types of coworkers, please answer the question below:

   Suppose that Ben and Mary both worked at the same Walmart store in 2022. Ben quit his job at Walmart and started working at a Target store in January 2023, while Mary continued working at Walmart. For Mary, what type of coworker is Ben? Please see the image below for a graphical depiction.

3. (If answered previous question wrong) Now think again about Ben and Mary. Remember that Ben and Mary both worked at the same Walmart store in 2022. Ben quit his job at Walmart and started working at a Target store in January 2023, while Mary continued working at Walmart. For Ben, what type of coworker is Mary? See the image below for a graphical depiction.

A.7 Current Coworkers at Current Job

1. Among all those who work at your current employer, approximately how many coworkers do you interact with at least once every month? Include those that you interact with for either work purpose or non-work purpose. These workers do not have to be working in the same team as you do.

2. Among all the coworkers that you interact with at your current job, approximately how many have mentioned the compensation they received at their previous job? Please count all those who mention the compensation at their previous job in a broad sense; they do not need to talk in specific details of their wages.

3. Among all the coworkers that you interact with at your current job, approximately how many have mentioned the non-wage benefits and amenities at their previous job?

4. Among all the coworkers that you interact with at your current job, approximately how many have mentioned how satisfied they were with their previous job? You can count all those who mention their feelings about their previous job in a broad sense.

5. For how many of all your current coworkers do you know the industry where they were working before starting working for your employer?

6. Among the coworkers whose previous jobs you know, how many of them worked in each of the listed industries for their last job? [Show to respondents the five sectors they are the most likely to transition into based on their current sector of employment]

7. In each of the listed industries, what do you believe is the median annual salary for individuals with the same characteristics as you, including age, gender, education, location, etc., before taxes and other deductions, and including extras like bonuses, overtime pay, tips, or commissions? [Show to respondents the five sectors they are the most likely to transition into based on their current sector of employment]

8. How certain are you of your answers to the previous question?

A.8 Past Coworkers at Current Job

1. In this section, we would like to ask you about your past coworkers at your current job, that is, coworkers who have worked with you at your current employer at some point in the past, but now work for another employer.

2. How many of your past coworkers who now work for another employer do you feel comfortable reaching out to for a career-related advice?

3. For how many of all your past coworkers at the current job do you know the industry where they are currently employed?

4. To the best of your knowledge, how many of your past coworkers who have changed job now work at employers in each one of these industries? [Show to respondents the five sectors they are the most likely to transition into based on their current sector of employment]

A.9 Past Coworkers at Past Job

1. How many people worked for your previous employer at your usual place of work, during your time of employment there?

2. Counting all locations where your previous employer operates, what is your best guess for the total number of persons who worked for your previous employer, during your time of employment there?

3. How many of your past coworkers at your previous job had also worked in your current industry of employment in the past, before working for your previous employer?

4. Did any of your past coworkers at your previous job mention to you any employer within the industry of your current employment?

5. How many of your past coworkers at your previous job have also worked at your current employer in the past?

6. Did any of your past coworkers at your previous job mention to you the company where you are currently employed?

7. Did any of them refer you to the position?

8. How many of your past coworkers at your previous job do you still maintain regular contact with? Please include those you feel comfortable reaching out to for networking and referrals.

A.10 Future Employment

1. Over the next year, what is the percent chance that you will voluntarily look for a new job at a different employer?

   Please move the slider to select a percentage from 0 to 100, where 0 means that there is no chance you will look for a new job and 100 means that it is absolutely certain that you will look for a new job. Please note that if you plan to switch position or branch within the same company, this does not count as looking for new jobs.

2. Would you consider switching to a job in another industry?

3. When considering which industry you'd like to switch into next, what are the factors that matter for your decisions? Please, select the three most important ones that influence your industry-switching decisions:

4. Would you consider looking for a job in each of the industries listed below? [Show to respondents the five sectors they are the most likely to transition into based on their current sector of employment]

5. What is the percent chance that you will receive a job offer if you apply to work for another employer within your current industry right now? Remember that a job offer is not necessarily a job you will accept.

6. In your opinion, what is the percent chance that you will receive a job offer if you apply to each one of the industries listed below? Remember that a job offer is not necessarily a job you will accept. Please move the sliders to select a percentage, where 0 on the far left means that it absolutely impossible that you will receive an offer and 100 on the far right means that you will receive an offer for sure. [Show to respondents the five sectors they are the most likely to transition into based on their current sector of employment excluding own sector]

7. What do you anticipate would be your annual starting salary if you were to switch to another employer within your current industry right now? Please include bonuses, overtime pay, tips, or commissions you expect.

8. For each of the listed industries, what do you anticipate would be your annual starting salary if you were to switch to that industry right now? Please include bonuses, overtime pay, tips, or commissions you expect. [Show to respondents the five sectors they are the most likely to transition into based on their current sector of employment]

9. How certain are you about your answers for the previous question?

A.11 Conjoint Experiment

Details are outlined in Section 4.

A.12 Additional Personal Characteristics

1. Are you currently married or living with a partner?

2. During the last year, how many months in total did you spend without a job?

A.13 Verification of Occupation and Industry

1. The next question addresses the following problem again. In surveys like this one, there are sometimes participants who don't read the questions carefully and just "click" through the questionnaire quickly. As a result, there are many random answers that falsify the results of the study. That's why we ask you to write your occupation at your main job below again, as you wrote it at the beginning of the survey.

2. Please select again the industry of your main employer from the list below, as you selected it at the beginning of the survey.

3. Please feel free to give us any feedback or impression regarding this survey.

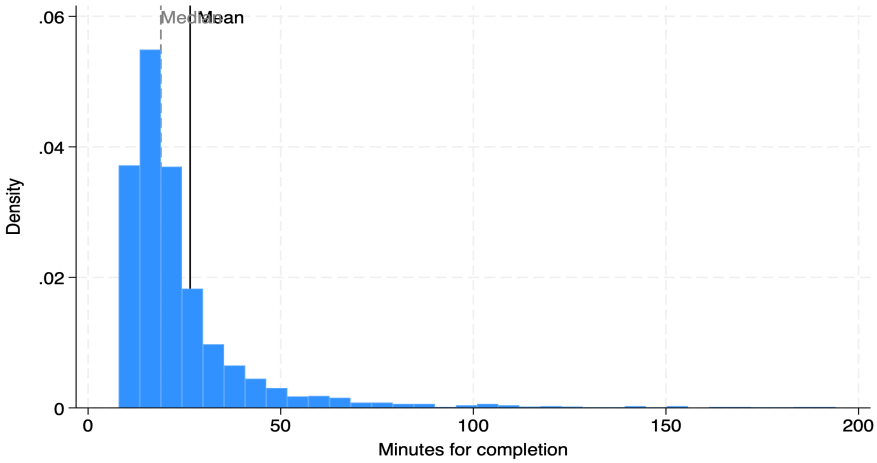
B Occupation Classifications

Although our paper focuses primarily on industry switching as opposed to occupation switching, we aim to collect accurate information on workers' occupations to best proxy for their skills and obtain a good sense of their outside options. Therefore, we adopt the occupational classification method outlined in Miano (2023), aiming to assign respondents to a 6-digit Standard Occupational Classification (SOC) group. Initially, respondents are shown a prompt, as seen in Figure A-4, asking them to describe their occupation in their own words using a text box. As they begin typing, a drop-down menu with suggested occupation titles appears, from which respondents are instructed to choose. If none of the options match their occupation, they are encouraged to modify their input.

To generate these suggestions, we begin with the "alternate occupation titles" compiled by ONET. These titles provide more accessible, everyday terminology for occupations compared to the official SOC titles. We perform a light cleanup of the original ONET list by removing overly broad titles (e.g., "Supervisor") and overly narrow ones (e.g., "Visiting Teacher"). We then use the O*NET mapping of alternate titles to 6-digit SOC codes to assign an occupation code based on the respondent's selection. Some alternate titles map directly to a single SOC code (e.g., "Accountant" corresponds to "Accountants and Auditors," SOC code 13-2011), while others correspond to multiple codes. For instance, "Secretary" could refer to "Legal Secretary," "Medical Secretary," or "Secretary, except medical and legal," which are distinct groups. When respondents select a title that corresponds to multiple SOC codes, they are prompted with a follow-up question.

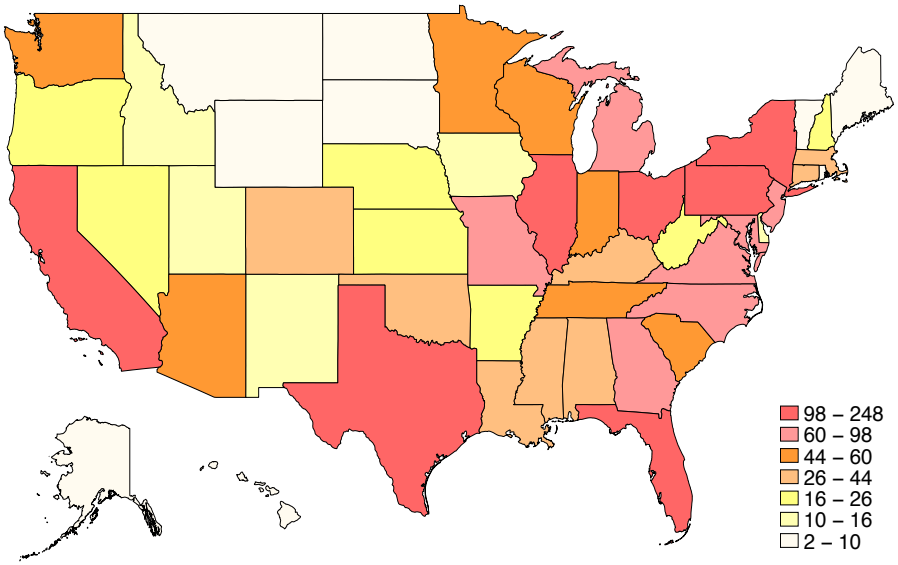
B.1 Sample Quality

Figure B1: Distribution of Time for Survey Completion



Note: Figure shows the distribution of the time respondents spent on the survey (truncated at 200 minutes). The mean duration is 26 minutes, the median 19, and the 25th and 75th percentiles are 14 and 27.

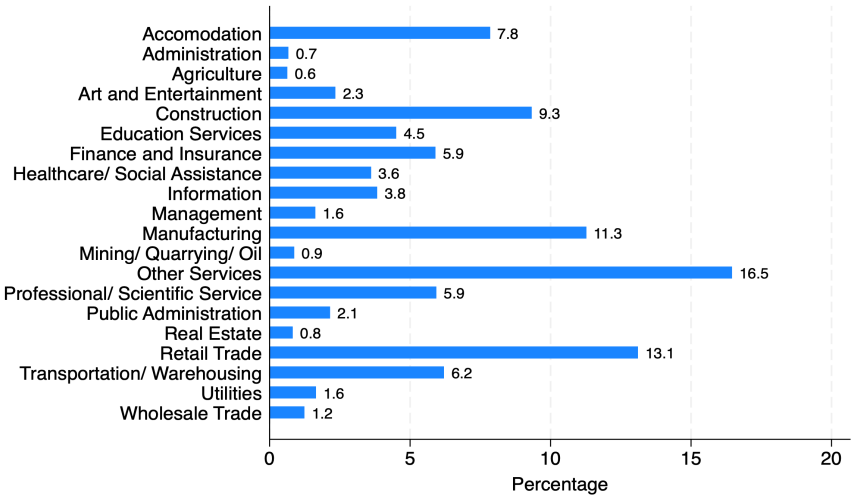
Figure B2: Geographic Distribution of Respondents in Sample



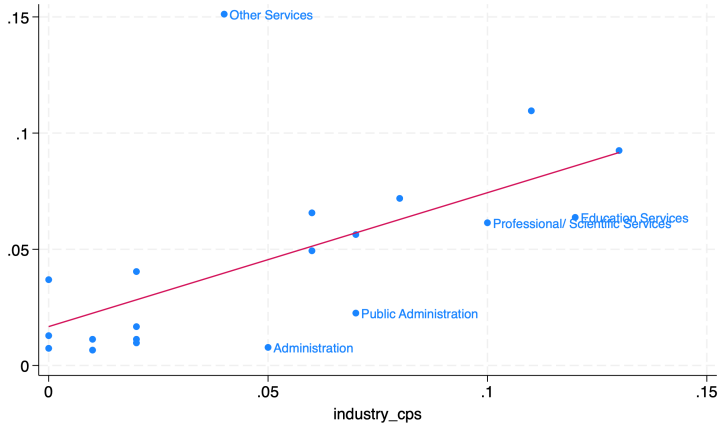
Note: Figure plots the number of respondents for the main survey sample by state.

Figure B3: Distribution of Sector

(A) Share of Respondents by Sector



(B) Share of Respondents by Sector in Survey vs. in CPS

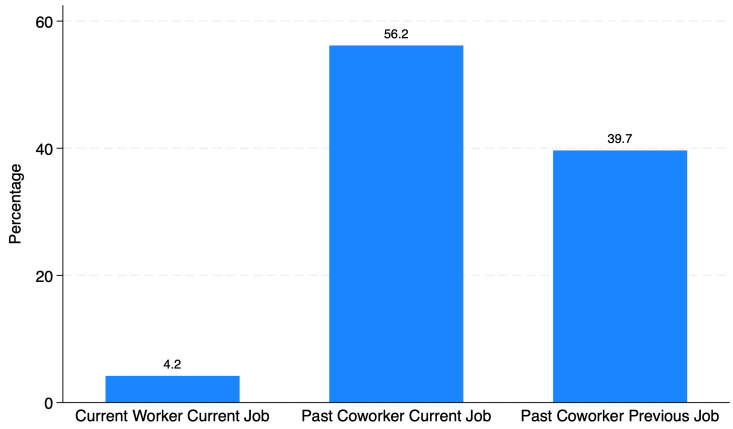


Note: Panel (A) plots the share of respondents in the main survey sample by 2-digit NAICS sector. Panel (B) plots the share of respondents by 2-digit NAICS sector in CPS data against the share of respondents by 2-digit NAICS sector in the main survey sample (on the y axis). The sectors that feature a higher difference in their shares between the survey and the CPS sample are labeled.

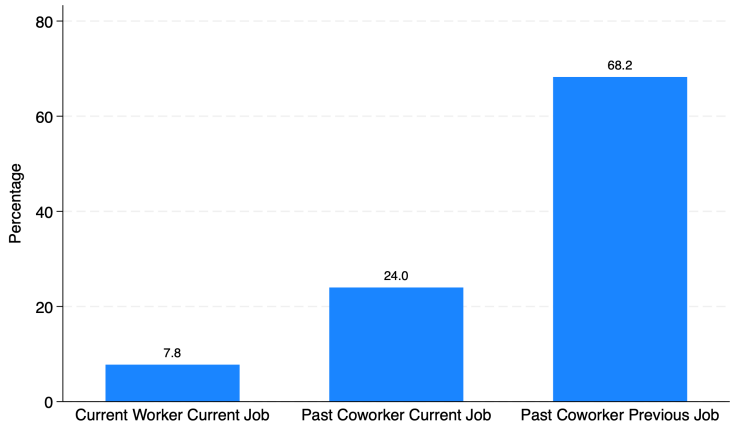
B.2 Survey Design

Figure B4: Distribution of Answers for the Coworker Type Question

(A) First Question



(B) Second Question

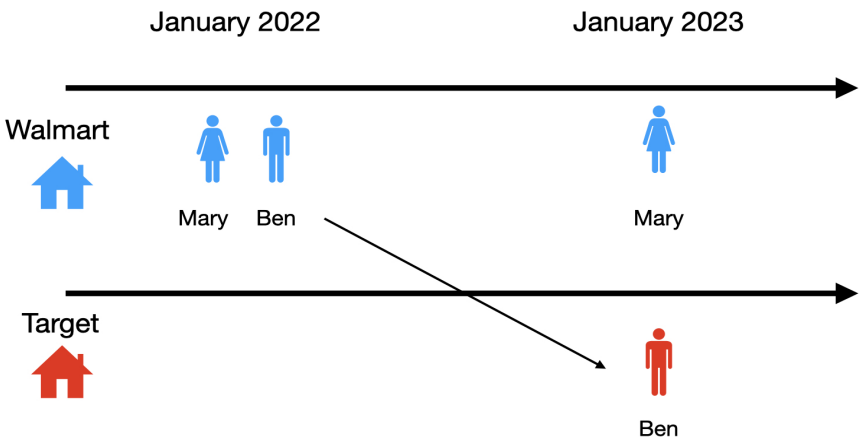


Note: Figure shows the percentage of respondents selecting into each category for the questions testing understanding of different types of coworkers. For those who answered the first question wrongly, a follow-up question is shown. The answer to the first question is "Past coworker at current job" and the answer to the second question is "past coworker at previous job." N = 2,721 for the question, and 1,177 for the second question.

C Additional Results

C.1 Additional Figures and Tables

Figure B5: Illustrative Graph for Coworker Types



Note: Figure show what the conjoint experiment on job-offer probabilities looks like for survey respondents. The features for Job 1 and 2 are randomly generated.

Figure B6: Conjoint Experiment

	Job 1	Job 2
Schedule Flexibility	Entirely flexible and able to work from home	No personal influence over work times and unable to work from home
Skill Level	5% lower than your current job	5% lower than your current job
Personal Connection	You know and have reached out to at least one employee at the job. He or she has good opinions of you but is incapable of referring you or helping your application in any concrete way	You know at least one employee at the job but have not reached out for any discussion of the job
Monthly Wage	10% more than your current wage	10% more than your current wage

0 10 20 30 40 50 60 70 80 90 100

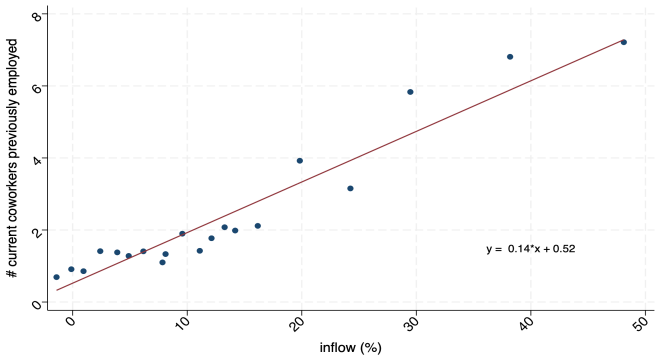
Job 1

Job 2

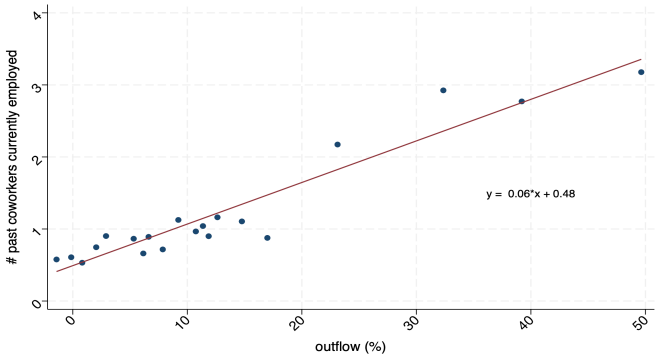
Note: Figure show what the conjoint experiment on job-offer probabilities looks like for survey respondents. The features for Job 1 and 2 are randomly generated.

Figure C1: Number of Coworkers

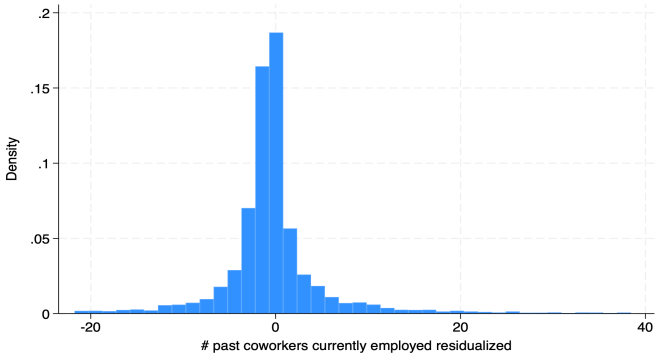
(A) Current Coworkers' Past Industries



(B) Past Coworkers' Current Industries



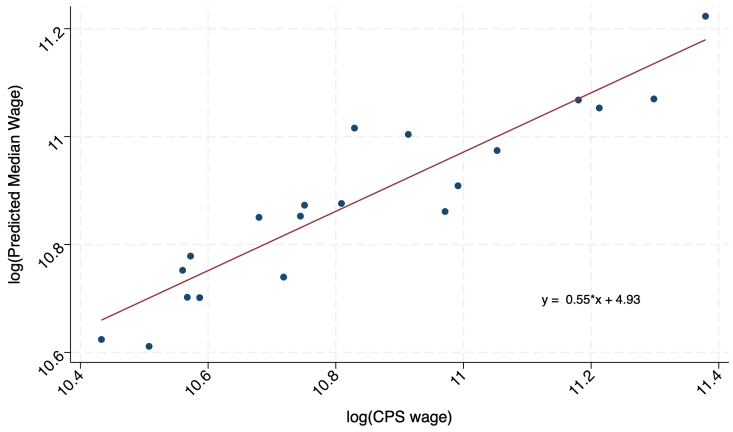
(C) Variation in Residualized Past Coworkers' current Industries



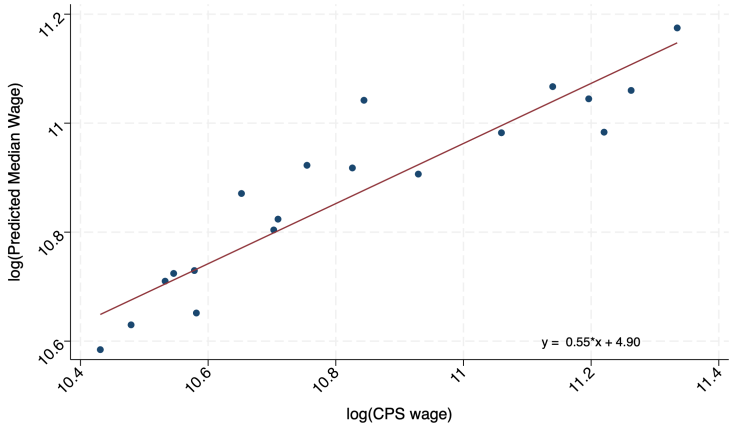
Note: Panel (A) is a binscatter plot of respondents' reported number of current coworkers last employed in an industry against the flow from that industry to her current industry of employment. Panel (B) is a binscatter plot of respondents' reported number of past coworkers currently employed in an industry against the outflow from her current sector of employment to that industry. Panel (C) is a histogram of the residualized number of past coworkers currently employed in each industry, obtained by regressing the number of past coworkers currently employed in an industry on the number of current coworkers previously employed in an industry and respondent fixed effect. Data is winsorized at the 99th percentile.

Figure C2: Predicted vs. Actual Median Wage (Own Sector vs. Other Sector)

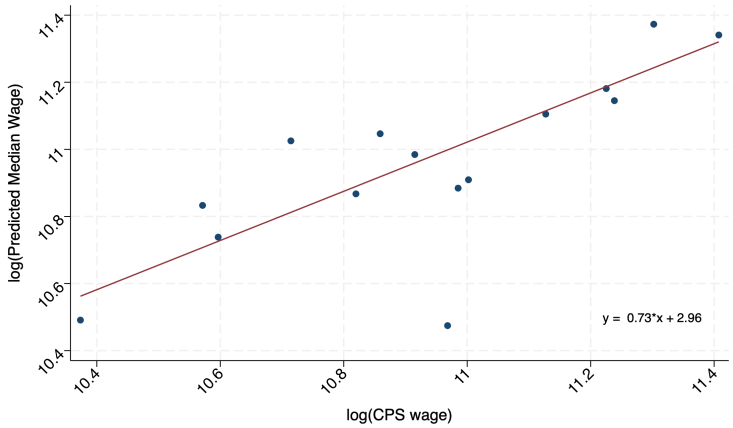
(A) All industries



(B) Other industries



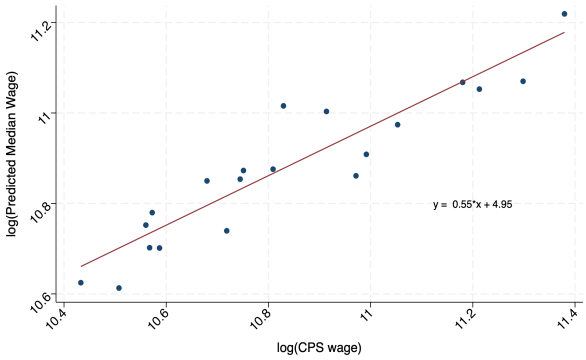
(C) Own industries



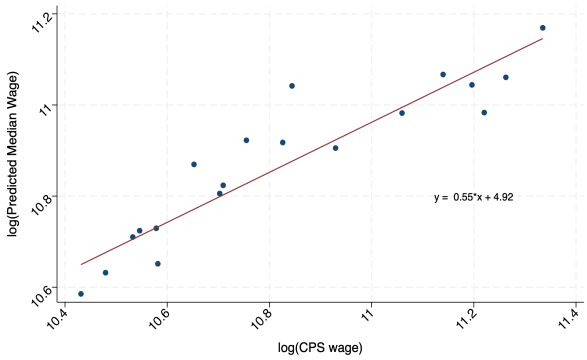
Note: Figure plots respondents' predicted median wage against the actual median wage by industry reported in CPS. Panel (A) pools across all industries. Panel (B) includes only estimates for industries other than their own. Panel (C) includes estimates for only their current industries. Sample consists of 2,721 individuals, each answering for the same list of five industries: the three with the highest out-transition flows from their current industry (which include their own industry) and the two with the lowest.

Figure C3: Predicted vs. Actual Median Wage (Controlling for Occupations)

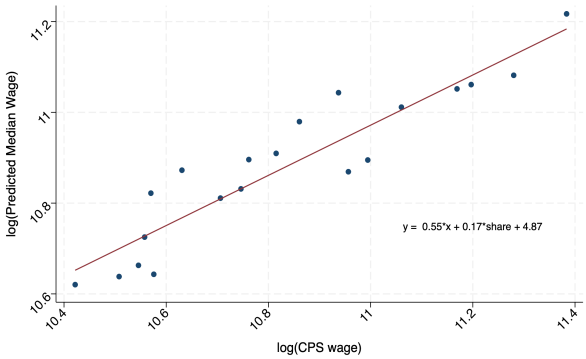
(A) All industries



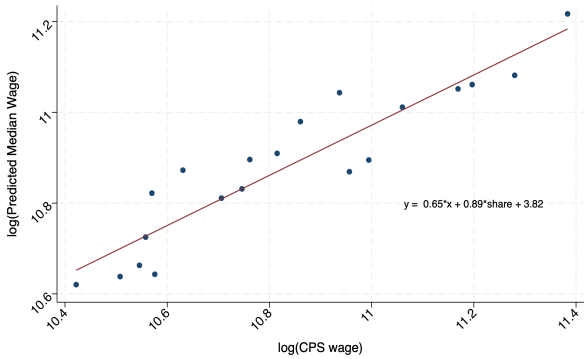
(B) Excluding own industries



(C) Controlling for transition share; all industries

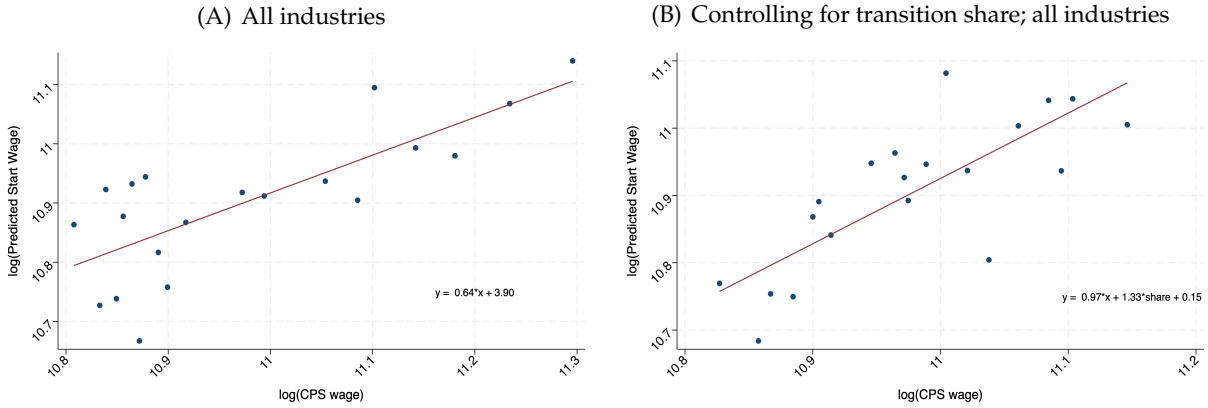


(D) Controlling for transition share; excluding own industries



Note: Figure plots respondents' predicted median wage against the actual median wage by industry reported in CPS, controlling for individuals' occupation at the 2-digit SOC level. Panel (A) and (B) include individual fixed effects. Panel (C) and (D) include individual fixed effects and control for the transition share from the listed sector to the participants' own sector. Sample consists of 2,721 individuals, each answering for the same list of five industries: the three with the highest out-transition flows from their current industry (which include their own industry) and the two with the lowest.

Figure C4: Predicted vs. Actual Personal Wage



Note: Figure plots respondents' predicted own wage against the wage predicted by personal characteristics using data from by industry reported in CPS. Panel (A) includes individual fixed effects and do not control for the transition shares between industry pairs. Panel (B) includes individual fixed effects and control for the transition share from the listed sector to the participants' own sector. Sample consists of 2,721 individuals, each answering for the same list of five industries: the three with the highest out-transition flows from their current industry (which include their own industry) and the two with the lowest.

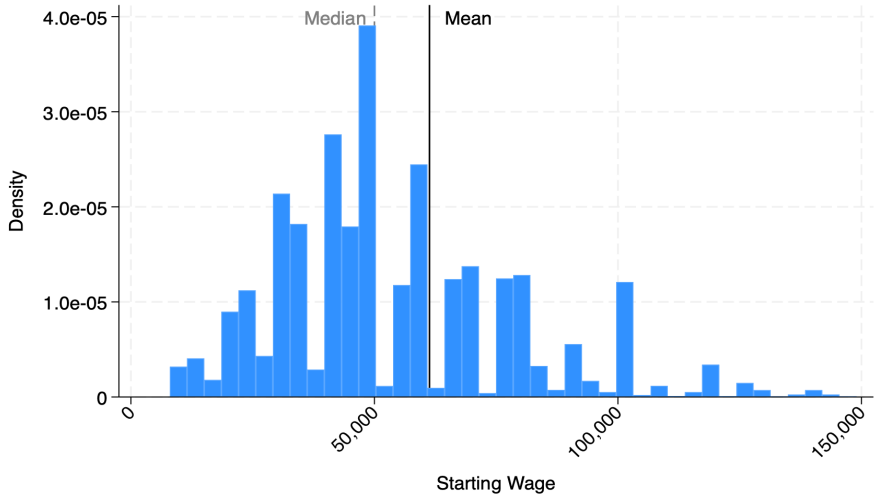
Table C1: Design of the Personal-Connection Attribute

Attribute shown	Level	Pr(shown)	Analysis category
Connection strength	No connection	0.22	Omitted (reference)
	Know an employee, no outreach	0.22	Separate dummy
	Reached out, no concrete help	0.22	Separate dummy
	Active help (referral or advocacy)	0.33	$Connection_{ijt}$
Active-help vignette (conditional on active help)	Referral agreement	1/3	Pooled into $Connection_{ijt}$
	Networking support	1/3	
	Informal mention	1/3	

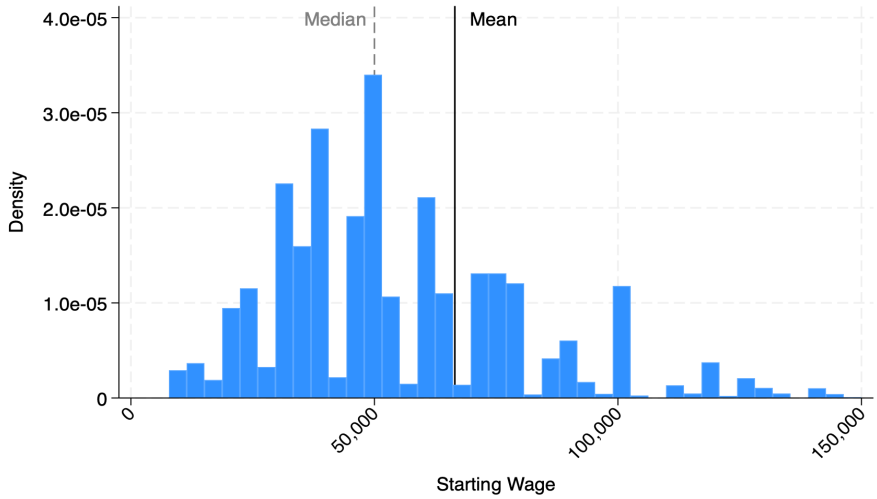
Note: The connection attribute has four ordered levels, shown with the probabilities in the third column. When the active-help level is drawn, one of three vignettes is shown with equal probability. In equation (7), $Connection_{ijt}$ equals one if any of the three active-help vignettes is shown; the two intermediate levels enter as separate dummies in $\boldsymbol{X}_{ijt}$, and "no connection" is the omitted category.

Figure C5: Distribution of Perceived Median and Starting Wages

(A) Perceived Median Wage

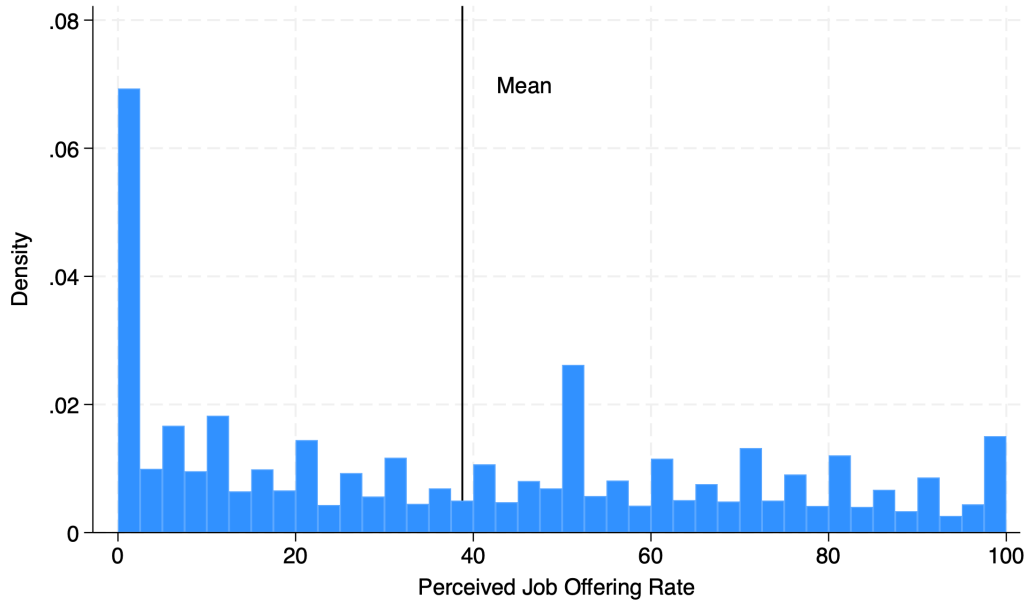


(B) Perceived Personal Starting Wage



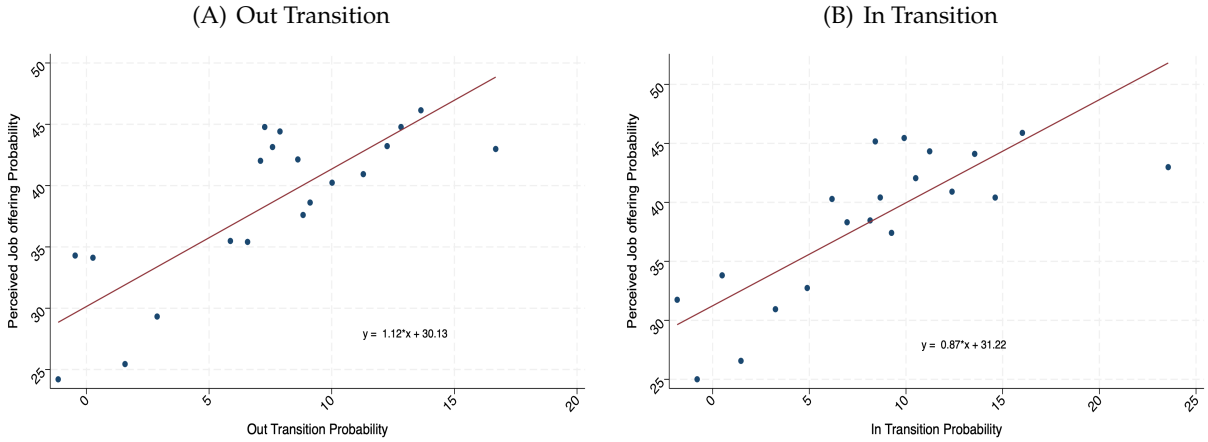
Note: Panel (A) plots the distribution of perceived median wages, pulled across all industries and respondents. Panel (B) plots the distribution of perceived personal starting wages.

Figure C6: Distribution of Perceived Job-Offer Probabilities



Note: Figure plots the distribution of respondents' perceived job-offer rate in other industries.

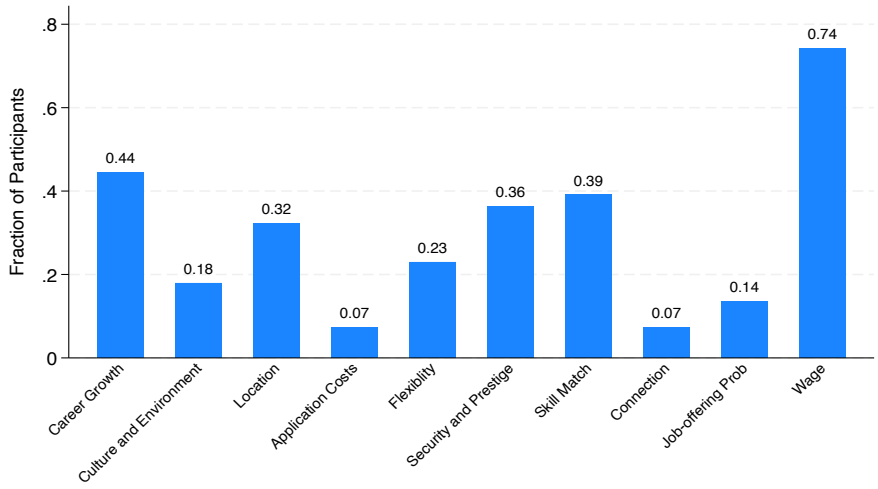
Figure C7: Perceived Job Offering Probability vs. Flow



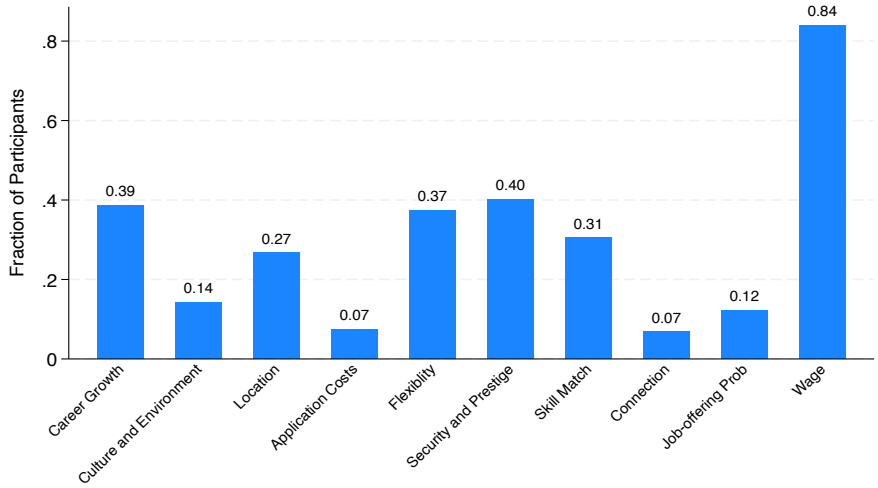
Note: Figure plots respondents' predicted job-offer rate in an industry against the actual transition probability. Panel (A) plots the perceived job-offer probability against the transition from their current industry to another industry. Panel (B) plots the perceived job-offer probability against the transition from another industry into their current industry.

Figure C8: Factors Influencing Job- and Sector Switching Decisions

(A) Reasons for Choosing the Current Job

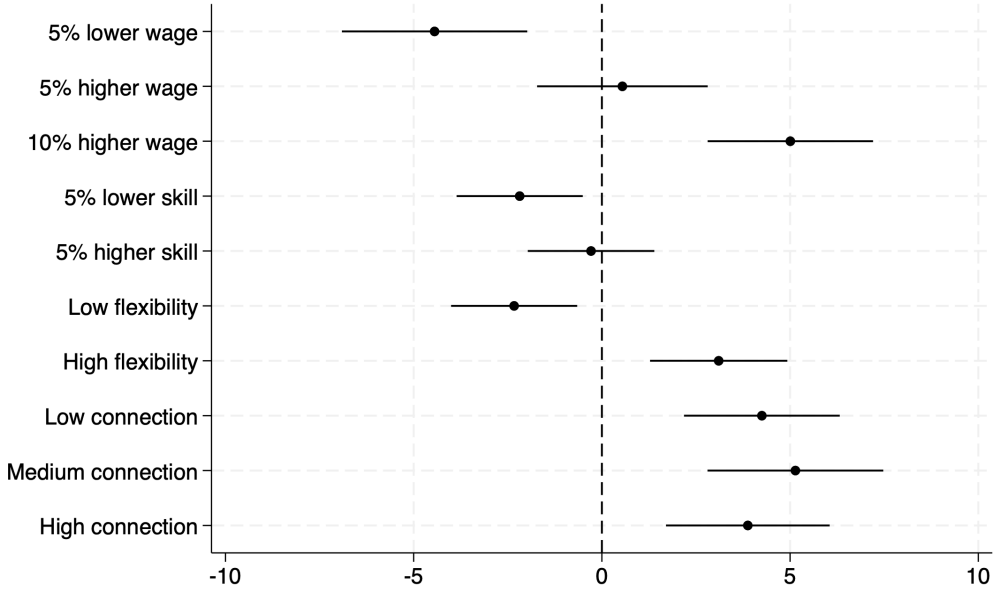


(B) Reasons for Choosing the Next Industry



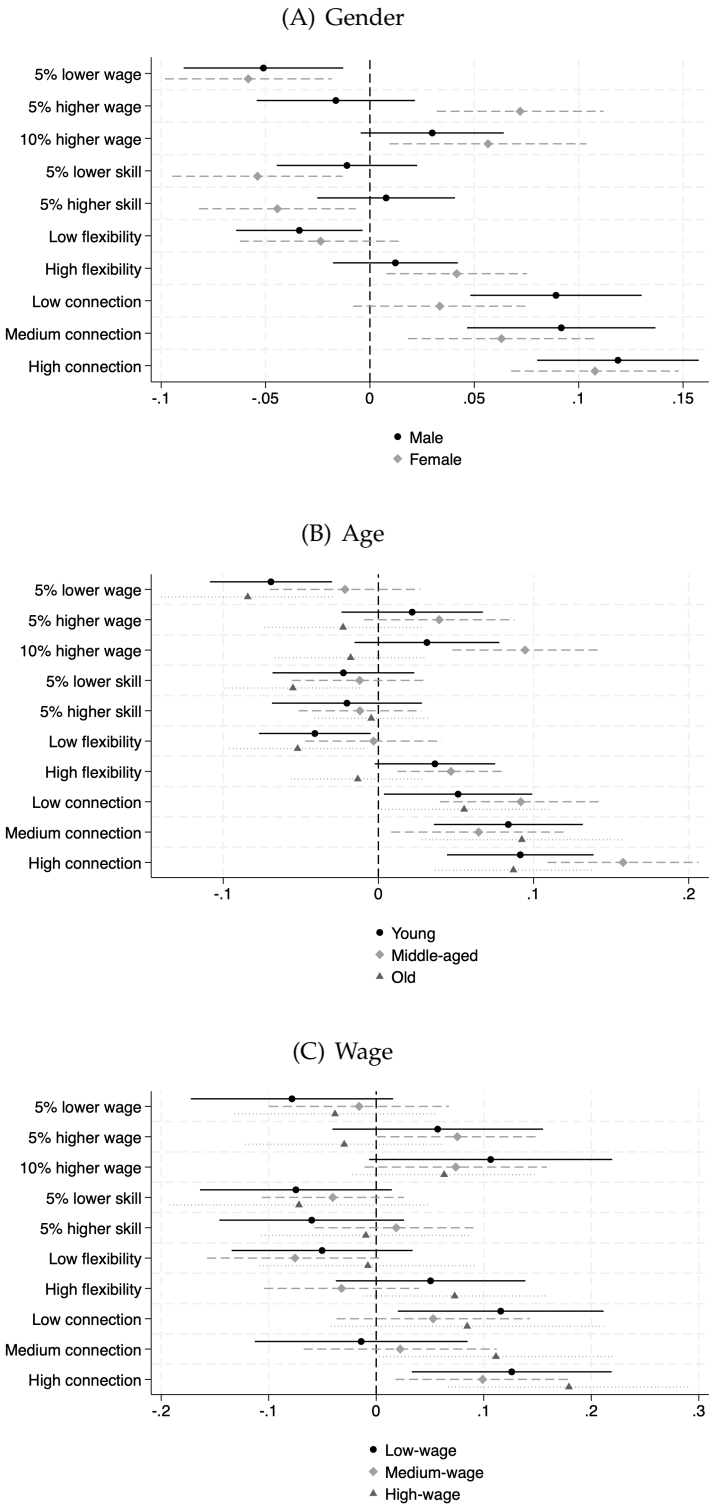
Note: Panel (A) plots the share of respondents choosing each reason for the question, "Why did you apply to your main job?" Panel (B) plots the share of respondents choosing each reason for the question, "When considering which industry you'd like to switch into next, what are the factors that matter for your decisions?" Each respondent selected three reasons, and some respondents chose the category "Other" which is excluded here.

Figure C9: Estimates of the Importance of Job Features in influencing Job-Offer Probability



Note: 95% confidence intervals are adjusted for clustering by respondent. Each respondent provided responses to four experiments on job-offer probabilities. The dependent variable is the respondent's perceived probability of receiving an offer. N = 2,721.

Figure C10: Estimates of the Importance of Job Features in influencing Job-Offer Probability



Note: 95% confidence intervals are adjusted for clustering by respondent. Each respondent provided responses to four experiments on job-offer probabilities. The dependent variable is the respondent's perceived probability of receiving an offer. Panel (A) plots separately for male vs. female. Panel (B) plots separately for young, middle-aged, and old workers. Panel (C) plots separately for low-wage, medium-wage, and high-wage workers. N = 2,721.

Table C2: Summary Statistics of Current Coworkers and Past Coworkers at Current Jobs

	Mean	Median	P25	P75	Obs.
Current coworkers with interaction	19.76	12.00	5.00	25.00	2,721
Current coworkers mention wages	4.11	1.00	0.00	4.00	2,721
Current coworkers mention amenities	3.78	0.00	0.00	3.00	2,720
Current coworkers mention satisfaction	4.61	1.00	0.00	5.00	2,721
Current coworkers know past industry	8.07	4.00	1.00	10.00	2,721
Past coworkers able to reach out to	3.43	1.00	0.00	4.00	2,721
Past coworkers know current industry	4.23	2.00	0.00	4.00	2,721

Note: Table reports the summary statistics regarding current and past coworkers at the current jobs. The first row indicates the number of current coworkers the respondents are interacting with every month. The second to fourth rows indicate the number of current coworkers that mention the wages, amenities, and their levels of satisfaction with their previous jobs. The fifth row indicates the number of current coworkers for whom the respondents are aware of their previous industry of employment. The sixth row indicates the number of past coworkers that left the respondents' current employers and to whom the respondents feel comfortable enough to reach out. The last row indicates the number of past coworkers that left the respondents' current employers and for whom the respondents know their current industry of employment.

Table C3: Perceived and Actual Wages by Worker Characteristics

	(1)	(2)	(3)	(4)
	Panel (A): Gender and Search Status			
	Male	Female	Job Seekers	Job Stayers
log(Predicted Median Wage)	0.50***	0.62***	0.49***	0.60***
	(0.04)	(0.05)	(0.05)	(0.04)
Constant	5.45***	4.16***	5.58***	4.43***
	(0.40)	(0.52)	(0.50)	(0.41)
Mean of Dep. Var	10.91	10.82	10.87	10.88
R^2	0.57	0.50	0.56	0.53
	Panel (B): Income Groups			
log(Predicted Median Wage)	0.50***	0.54***	0.57***	
	(0.05)	(0.04)	(0.05)	
Constant	5.05***	5.01***	4.85***	
	(0.49)	(0.41)	(0.55)	
Mean of Dep. Var	10.50	10.83	11.02	
R^2	0.53	0.42	0.54	

Notes: Table displays the coefficient estimates from regressing logged predicted median wage on logged actual median wage in the CPS data, while controlling for respondent fixed effects. The dependent variable in all regressions is the respondent's logged predicted median wage in the listed industry. Standard errors are clustered by the industry elicited. In Panel (A), first and second columns use the samples of males and females only. Third and Fourth columns use the samples of job seekers and non-seekers only. Job seekers are defined as individual who think of themselves as having higher than 40 percent of chance of looking for a job in a company different from the ones that they are currently employed at within the next year. Panel (B) displays the estimates obtained separately for different income groups, with columns (1) - (3) being low income, middle income, and high income, respectively.

C.2 Reliability of the Estimates from our Survey

Our conjoint analysis results are consistent across workers with varying personal and employment characteristics. However, the measures derived from the conjoint experiment may be subject to hypothetical bias. Survey experiments can yield less reliable data when respondents face choices that are unrealistic or unfamiliar to them (Mas and Pallais (2020)). We addressed this by using plain language in the vignettes to present concrete scenarios where coworker connections could influence job-offer rates, which should help respondents better understand and relate to the situations, thus mitigating this concern. To further address this form of bias, we conducted some more direct tests by splitting the sample into groups based on their expected familiarity with coworker relationships. We anticipate higher familiarity with referral mechanisms or other implicit connection-induced job-offer

Table C4: Current Coworkers' Prior Industries and Prediction Errors in Logged Median Wages

	Wage Error	Wage Error	Wage Error	Wage Error	Wage Error
CoworkerShare	-0.05	-0.06*	-0.06*	-0.06*	-0.05
	(0.03)	(0.03)	(0.03)	(0.03)	(0.04)
OutTransition	0.00	0.00	0.00	0.00	0.00
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
PerceivedProb		0.00	0.00	0.00	0.00
		(0.00)	(0.00)	(0.00)	(0.00)
PastCoworkerShare				0.02	
				(0.04)	
PositiveError					0.00
					(0.02)
PositiveError × *CoworkerShare*					-0.01
					(0.04)
Occupation FE	No	No	Yes	Yes	Yes
Mean of Dep. Var	0.40	0.40	0.40	0.40	0.40
Adjusted R^2	0.35	0.35	0.35	0.35	0.35

Note: Table displays the coefficient estimates for equation (2). The outcome variable for all columns is the absolute value of the logged difference between respondents' predicted median wage in another industry and the actual median wage observed in CPS. The second to column adds the control for the respondent's perceived probability of receiving an offer in that industry. The third additionally controls for occupational group fixed effects. The fourth column controls for the share of coworkers at the last job that are currently employed at the reference industry. The last column studies the asymmetric effect of current coworkers on errors in wage prediction, separately for respondents who overestimate and underestimate wages, by including an indicator variable *PositiveError* for overestimation, and including its interaction with the current coworker shares.

advantages among respondents who are more in touch with their past coworkers, as well as those who are actively seeking new job opportunities or who are working in person.

A few survey questions enable us to categorize respondents based on their interactions with past coworkers and their personal experiences. Specifically, we divide workers into binary groups using four different criteria. First, we classify workers as either "in touch with coworkers" or "not in touch with coworkers" based on whether they report a number higher than 1 for the question: "How many of your past coworkers who now work for another employer do you feel comfortable reaching out to for career-related advice?" To enhance robustness, we repeat the classification using responses to a similar question. The results, shown in Figure C11, reveal that the importance respondents place on connections is consistent across these subsamples, contradicting the notion that hypothetical bias drives our findings. The estimates for low, medium, and high connections are comparable between workers who are in contact or comfortable reaching out to their previous coworkers and those who are not. Moreover, respondents that work in person place a lower importance on the "high flexibility

Table C5: Current Coworkers' Prior Industries and Prediction Errors in Starting Wages

	Wage Error	Wage Error	Wage Error	Wage Error	Wage Error
CoworkerShare	-3,203.75	-3,724.97	-3,557.09	-2,883.35	-8,157.85*
	(3,370.58)	(3,441.64)	(3,475.12)	(3,572.57)	(4,672.58)
OutTransition	-546.58**	-604.54**	-599.55**	-593.69**	-559.93**
	(244.74)	(255.78)	(257.38)	(256.46)	(265.07)
PerceivedProb		43.33	43.42	45.23	40.66
		(41.01)	(41.42)	(41.85)	(41.73)
PastCoworkerShare				-3,564.25	
				(4,105.47)	
PositiveError					-4,570.62*
					(2,484.47)
PositiveError × *CoworkerShare*					7,667.84
					(5,541.70)
Occupation FE	No	No	Yes	Yes	Yes
Mean of Dep. Var	33,499.49	33,508.45	33,364.08	33,364.08	33,364.08
R^2	0.43	0.43	0.43	0.43	0.43

Note: Table displays the coefficient estimates for equation (2). The outcome variable for all columns is the absolute value of the difference between the respondents' predicted starting wage for themselves in another industry and the starting wage for them predicted by CPS data, obtained using a Mincer-style regression with individual characteristics. The second to column adds the control for the respondent's perceived probability of receiving an offer in that industry. The third additionally controls for occupational group fixed effects. The fourth column controls for the share of coworkers at the last job that are currently employed at the reference industry. The last column studies the asymmetric effect of current coworkers on errors in wage prediction, separately for respondents who overestimate and underestimate wages, by including an indicator variable *PositiveError* for overestimation, and including its interaction with the current coworker shares.

in work schedule" category, providing evidence for that workers who tend to associate jobs that are similar to their current ones higher probabilities of receiving an offer.

In addition, we classify workers into groups based on their labor market experiences and situations. Workers who interact more with their coworkers at their current workplace may have a more accurate perception of the benefits personal connections can provide. In addition, remote workers may be more reliant on their connections in searching for new jobs because of the lack of opportunity to engage with other individuals in other workplace-related platforms. Thus, we categorize workers by whether their main job is currently remote/hybrid or in-person. Moreover, job seekers may provide more accurate estimates, given their direct involvement in the job search process. We therefore classify workers into those who are not searching for a job and those actively looking for a new job—whether to replace their primary or secondary jobs or to prepare for potential layoffs. Figure C12 shows that the 95% confidence intervals for the connection categories overlap across subsamples of in-person vs. remote/hybrid workers and job seekers vs. non-job seekers, indicating consistency in the perceived importance of connections across these groups. Moreover, the importance workers

Table C6: Past and Current Coworker Counts and the Intention to Switch Jobs

	Switch Job	Switch Job	Switch Job	Switch Job
PastCoworkers	0.50***	0.49***	0.51***	0.50***
	(0.08)	(0.08)	(0.08)	(0.08)
CurrentCoworkers	0.08	0.07	0.10*	0.10*
	(0.06)	(0.05)	(0.05)	(0.05)
OutTransition		-0.14	-0.13	-0.15*
		(0.10)	(0.09)	(0.08)
log(*CurrentWage*)			-4.13*	-4.67**
			(2.05)	(1.76)
Occupation FE	No	No	No	Yes
Mean of Dep. Var	38.41	38.41	38.41	38.43
Adjusted R^2	0.02	0.03	0.03	0.06

Note: Table displays the coefficient estimates for equation (4), estimated at the respondent level. The outcome variable across all columns is the respondent's stated percent chance of voluntarily looking for a new job at a different employer within the next year, ranging from 0 to 100. Because this outcome is not industry-specific, the coworker regressors are counts rather than industry shares: *PastCoworkers* is the number of past coworkers at the respondent's current job whose current industry of employment the respondent knows, and *CurrentCoworkers* is the number of current coworkers whose previous industry of employment the respondent knows. *OutTransition* is the average CPS out-transition rate from the respondent's current industry across the five listed industries. Column (2) adds *OutTransition*. Column (3) controls for the respondent's logged current wage. Column (4) additionally controls for the occupational group fixed effects.

with flexible work arrangements place on the "high-flexibility" category is high, while the importance workers with in-person jobs place on the "high-flexibility" category is low, suggesting that workers are likely to associate higher job-offer rates with job features similar to their own.

Table C7: Current Coworkers' Prior Industries and Wage Beliefs

Specification	Weight on the true destination wage $\hat{\phi}_0$ (baseline)	$\hat{\phi}_{cw}$ (per unit coworker share)
Individual fixed effects	0.5148	0.1288
	(0.0169)	(0.0597)
	[0.482, 0.547]	[0.016, 0.239]
Pooled	0.6623	0.2307
Observations	6,031	
Respondents	1,580	

Note: Table reports the belief-weight parameters of equation (9), estimated from equation (15). The dependent variable is the log difference between the respondent's predicted median wage in a listed industry and the actual CPS median in her own industry; the regressors are the log wage differential and its interaction with the share of current coworkers previously employed in that industry. The belief weight is $\omega = \phi_0 + \phi_{cw}s$. The fixed-effects specification (bold) identifies both parameters from within-respondent variation across the industries each respondent was shown and is the one used in the calibration; the pooled specification additionally uses between-respondent variation, which is larger but confounded with respondent-level differences in connectedness and attentiveness.

Table C8: Current Coworkers' Prior Industries and Wage Beliefs

	(1)	(2)	(3)
$CoworkerShare \times \log(DestWage)$	0.48***	0.56***	0.77***
	(0.04)	(0.03)	(0.14)
$CoworkerShare \times \log(OrigWage)$	-0.49***	-0.56***	-0.78***
	(0.04)	(0.03)	(0.14)
Mean of Dep. Var	-0.05	-0.05	-0.06
R^2	0.53	0.57	0.54

Note: Table displays the coefficient estimates, ϕ_{cw}, for equation (16). Column (1) defines coworker share as the number of current coworkers last employed in an industry over the total number of current coworkers of which the respondents know their last industry of employment. Column (2) defines coworker share as the number of current coworkers last employed in an industry over the total number of current coworkers who mentioned their wage associated with their last job during their interactions with the reference individual. Column (3) defines coworker share as the number of current coworkers last employed in an industry over the sum of all current coworkers last employed in one of the three closest or two farthest industries (the industries we asked about in the survey).

Table C9: Summary Statistics of Past Coworkers at Past Jobs

	Mean	Median	P25	P75	Obs.
Past coworkers in contact	2.60	1.00	0.00	3.00	2,308
Past coworkers employed in current industry	8.67	2.00	0.00	8.00	2,300
Past coworkers employed in current firm	1.63	0.00	0.00	1.00	2,300
Past coworkers mentioned current industry	0.27	0.00	0.00	1.00	2,329
Past coworkers mentioned current firm	0.28	0.00	0.00	1.00	2,329
Past coworkers referral	0.18	0.00	0.00	0.00	2,329

Note: Table reports some summary statistics related to the past coworkers at the previous job. The variables, past coworkers mentioned current industry, past coworkers mentioned current firm, past coworkers referral are dummies equal to 1 if the respondent has at least one past coworker at their previous job that, respectively, mentioned to them their current industry of employment, mentioned to them their current firm of industry, and provided them with a referral for the current position.

D Additional Survey Evidence

This subsection collects descriptive material on the accuracy of wage beliefs, which does not feed the model in Section 5.

E Model Appendix

E.1 Model Derivation

Recall that the Bellman equation is:

$$V_t^n = \ln w_t^n + \max_k \left\{ \mathbb{E}_t^n \left[\Gamma^{n,k} \beta V_{t+1}^k + (1 - \Gamma^{n,k}) \beta V_{t+1}^n - \kappa^{n,k} \right] + \nu \epsilon_t^k \right\}$$

We can write the sector-choice probabilities from origin sector k to destination sector m from the perspective of workers in sector k as:

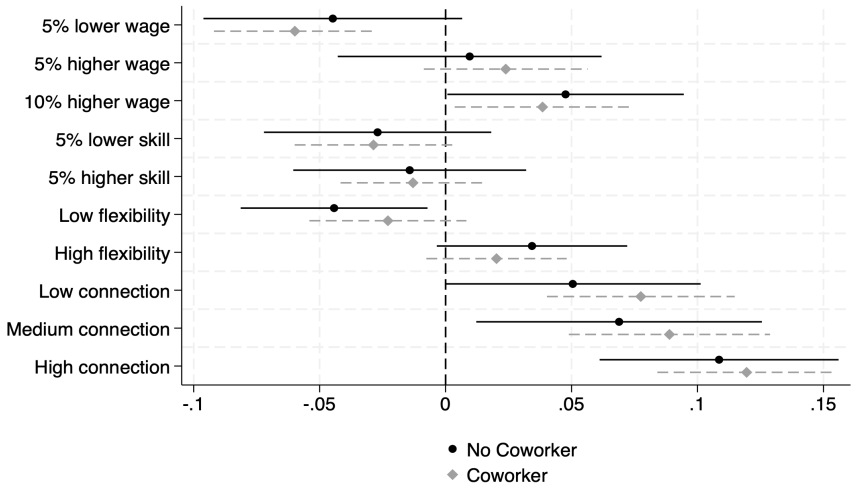
$$\begin{aligned}
&\mu_t^n(k,m) \\
=& \Pr\left(\mathbb{E}_t^n \left[\Gamma^{k,m} \beta V_{t+1}^m + \left(1 - \Gamma^{k,m}\right) \beta V_{t+1}^k - \kappa^{k,m} \right] + \nu \epsilon_t^m \right. \\
\geq& \mathbb{E}_t^n \left[\Gamma^{k,j} \beta V_{t+1}^j + \left(1 - \Gamma^{k,j}\right) \beta V_{t+1}^k - \kappa^{k,j} \right] + \nu \epsilon_t^j, \quad \forall j \neq m \Big) \\
=& \Pr\left(\nu \left(\epsilon_t^m - \epsilon_t^j \right) \geq \mathbb{E}_t^n \left[\Gamma^{k,j} \beta V_{t+1}^j - \Gamma^{k,m} \beta V_{t+1}^m + \left(\Gamma^{k,m} - \Gamma^{k,j} \right) \beta V_{t+1}^k - \left(\kappa^{k,j} - \kappa^{k,m} \right) \right], \quad \forall j \neq m \right) \\
=& \Pr\left(\epsilon_t^j \leq \epsilon_t^m - \frac{1}{\nu} \mathbb{E}_t^n \left[\Gamma^{k,j} \beta V_{t+1}^j - \Gamma^{k,m} \beta V_{t+1}^m + \left(\Gamma^{k,m} - \Gamma^{k,j} \right) \beta V_{t+1}^k - \left(\kappa^{k,j} - \kappa^{k,m} \right) \right], \quad \forall j \neq m \right) \\
=& \frac{\exp\left(\mathbb{E}_t^n [\Gamma^{k,m} \beta V_{t+1}^m + (1 - \Gamma^{k,m}) \beta V_{t+1}^k - \kappa^{k,m}] \right)^{\frac{1}{\nu}}}{\sum_j \exp\left(\mathbb{E}_t^n [\Gamma^{k,j} \beta V_{t+1}^j + (1 - \Gamma^{k,j}) \beta V_{t+1}^k - \kappa^{k,j}] \right)^{\frac{1}{\nu}}}
\end{aligned}$$

where the last line uses the property of the Gumbel distribution.

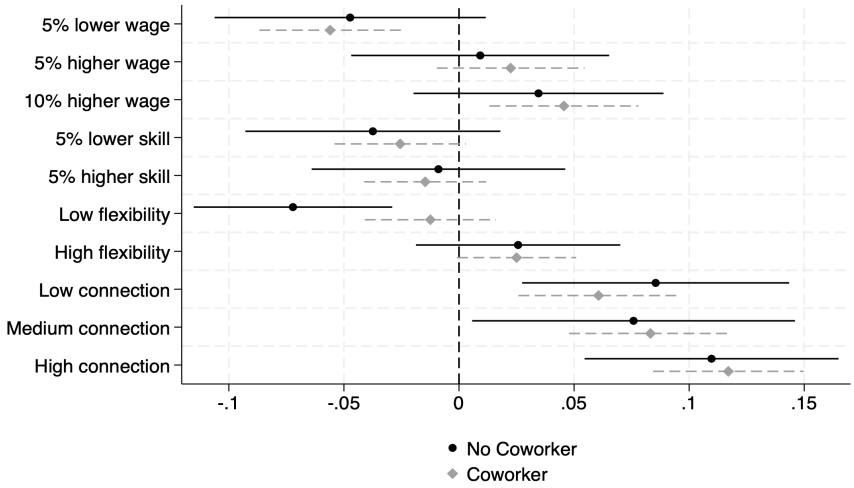
In addition, using the property of the Gumbel distribution, we can also write the expected utility

Figure C11: Estimates of the Importance of Job Features in influencing Job-Offer Probability

(A) Coworkers for Reaching Out



(B) Coworkers in Contact



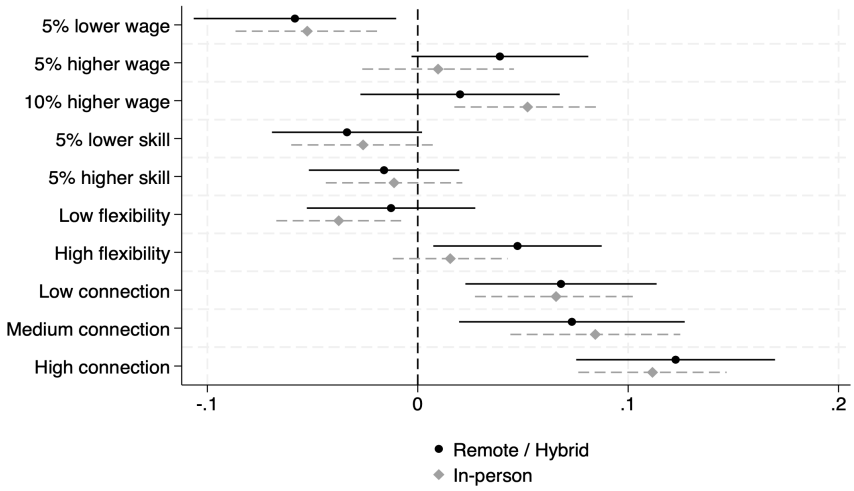
Note: 95% confidence intervals are adjusted for clustering by respondent. Each respondent provided responses to four experiments on job-offer probabilities. The dependent variable is the respondent's perceived probability of receiving an offer. Panel (A) plots separately for those who have at least one previous coworker with whom they are comfortable reaching out vs. those who do not. Panel (B) plots separately for those who have at least one previous coworker with whom they are in regular contact vs. those who do not. N = 2,721.

of being employed in sector n:

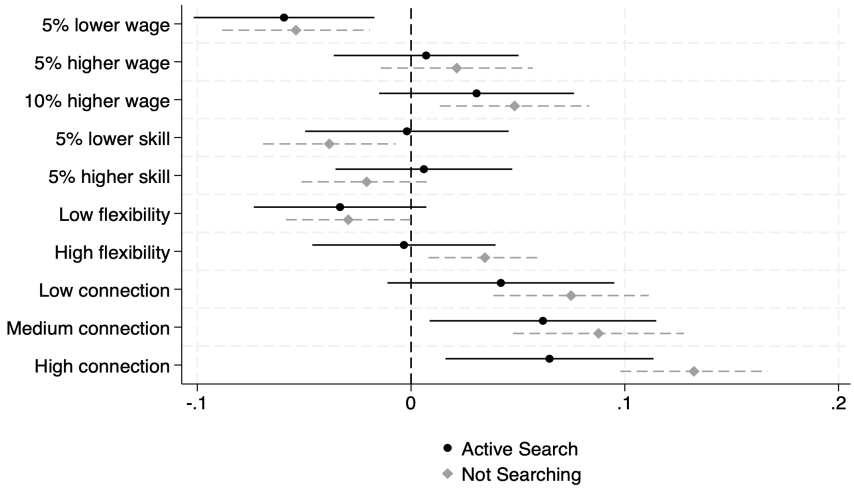
$$\mathbb{E}_t^n[V_t^k] = \mathbb{E}_t^n[\ln w_t^k] + \nu \ln\left(\sum_j \exp\left(\frac{1}{\nu}\mathbb{E}_t^n\left[\Gamma^{k,j}\beta V_{t+1}^j + \left(1-\Gamma^{k,j}\right)\beta V_{t+1}^k - \kappa^{k,j}\right]\right)\right)$$

Figure C12: Estimates of the Importance of Job Features in influencing Job-Offer Probability

(A) Remote Work



(B) Searching for Job



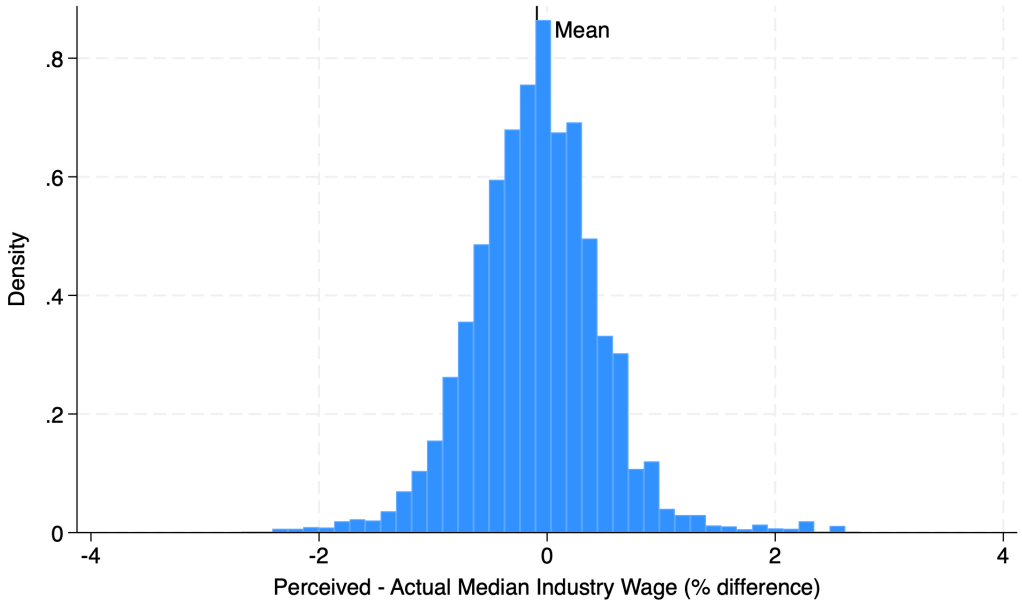
Note: 95% confidence intervals are adjusted for clustering by respondent. Each respondent provided responses to four experiments on job-offer probabilities. The dependent variable is the respondent's perceived probability of receiving an offer. Panel (A) plots separately for those working a remote or a hybrid job vs. those who work in person. Panel (B) plots separately for workers who are searching for a job vs. those who are not. N = 2,721.

E.2 Estimation Equation

The belief on flows coincides with the actual flow, $\mu_t(n,m)$, when $n = k$, i.e., $\mu_t(n,m) = \hat{\mu}_t^n(n,m)$. Therefore,

$$\mu_t(n,m) = \hat{\mu}_t^n(n,m) = \frac{\exp\left(\mathbb{E}_t^n[\Gamma^{n,m}\beta V_{t+1}^m + (1-\Gamma^{n,m})\beta V_{t+1}^n - \kappa^{n,m}]\right)^{\frac{1}{\nu}}}{\sum_j \exp\left(\mathbb{E}_t^n[\Gamma^{n,j}\beta V_{t+1}^j + (1-\Gamma^{n,j})\beta V_{t+1}^n - \kappa^{n,j}]\right)^{\frac{1}{\nu}}} \tag{D.2}$$

Figure D1: Misperception: Median Industry Wage



Note: Figure plots the distribution of respondents' misperception of the median wage industries other than their own. Misperception is defined as the difference between the answers respondents' put down for the median wage in the industries minus the median industry wage as reported in CPS.

Based on equation (D.2), we can obtain the following equation:

$$\ln \frac{\mu_t(n,m)}{\mu_t(n,n)} = \frac{1}{\nu}\mathbb{E}_t^n[\Gamma^{n,m}\beta(V_{t+1}^m - V_{t+1}^n) - \kappa^{n,m}] \tag{D.3}$$

Moreover, the extreme value algebra also yields the following equation:

$$\mathbb{E}_t^n[V_t^n] = \ln w_t^n + \beta \mathbb{E}_t^n[V_{t+1}^n] - \nu \ln \mu_t^{n,n} \tag{D.4}$$

And more generally we have

$$\mathbb{E}_t^n[V_{t+1}^m] = \mathbb{E}_t^n \mathbb{E}_{t+1}^n[V_{t+1}^m] = \mathbb{E}_t^n \ln w_{t+1}^m + \beta \mathbb{E}_t^n[V_{t+2}^m] - \nu \mathbb{E}_t^n \ln \mu_{t+1}^{m,m} \tag{D.5}$$

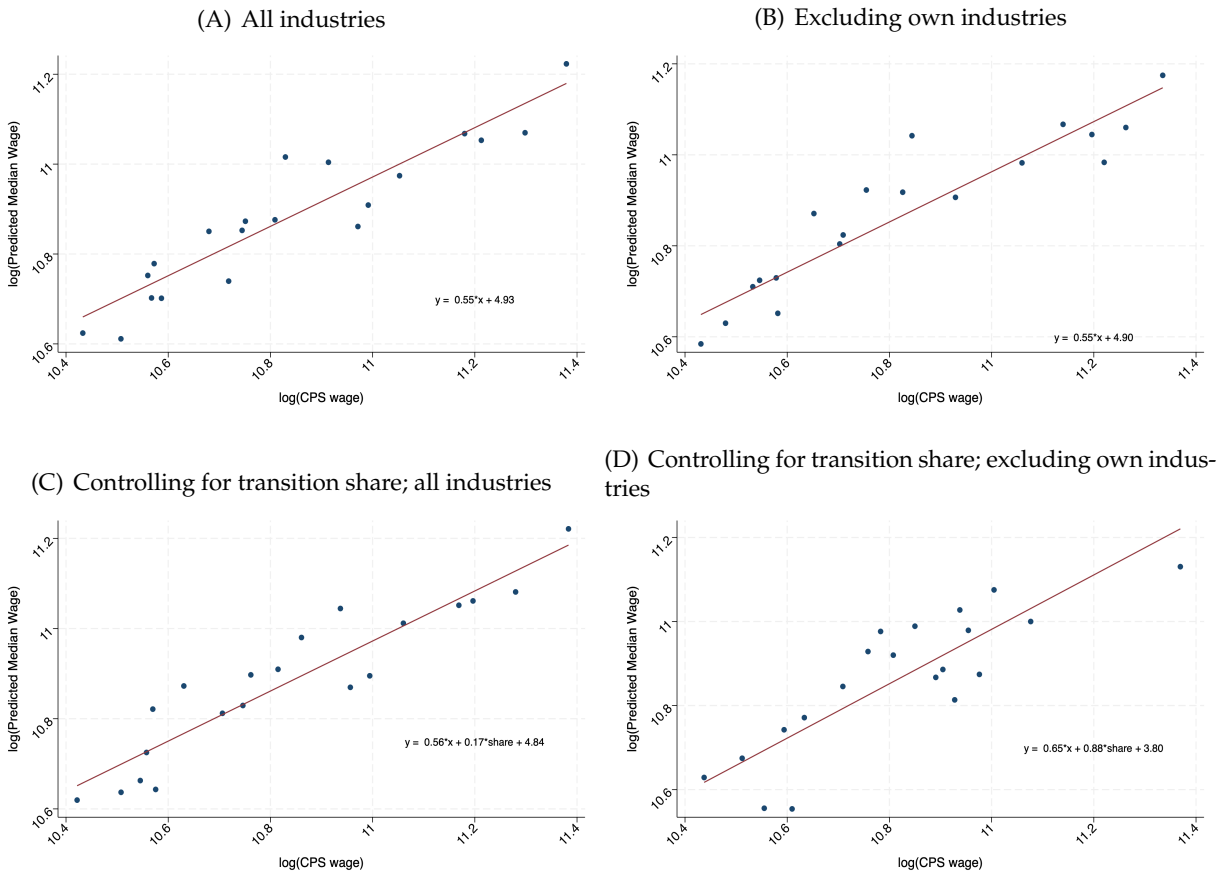
Combining equations (D.3), (D.4), and (D.5), we arrive at the estimation equation (12).

E.3 Solving the model at steady state

We solve for the steady-state distribution of labor, flows, and welfare in the following steps.

- **Step 1 (survey moments).** Estimate ϕ_0 and ϕ_{cw} from the wage-belief regression (15) and ψ_{past}, ψ_{cw}, ψ_{wage} from the offer-probability regression (17). Set $\phi_{past} = 0$ and normalize $\psi_0 = 0$.
- **Step 2 (offer probabilities).** Construct the 19×19 matrix $\mathbf{\Gamma}$ from equation (10), with $\Gamma^{nn} = 1$.

Figure D2: Predicted vs. Actual Median Wage



Note: Figure plots respondents' predicted median wage against the actual median wage by industry reported in CPS. Panel (A) and (B) include individual fixed effects. Panel (C) and (D) include individual fixed effects and control for the transition share from the listed sector to the participants' own sector. Sample consists of 2,721 individuals, each answering for the same list of five industries: the three with the highest out-transition flows from their current industry (which include their own industry) and the two with the lowest.

The sector-level network objects are taken from the CPS job-to-job flow matrices: the inflow shares proxy s^{nk}_{cw}, the share of a sector's current workers who arrived from k, and the outflow shares proxy s^{nk}_{past}, the share of a sector's former workers now employed in k.

- **Step 3 (application probabilities).** For a candidate cost vector, solve the fixed point in equation (12) for the application probabilities $\boldsymbol{\mu}$, and form realized transitions $\boldsymbol{P} = \boldsymbol{\mu} \odot \boldsymbol{\Gamma}$ as in (11).

- **Step 4 (calibrate the cost profile).** Choose $\{\tilde{\kappa}_k\}$ with $\sum_k \tilde{\kappa}_k = 0$ to minimize the employment-weighted squared distance between the model-implied and LEHD conditional destination distributions across the 342 off-diagonal pairs. Estimation uses L-BFGS-B with finite-difference gradients.

- **Step 5 (calibrate the cost level).** At the optimal profile, solve for the scalar κ_0 that reproduces the aggregate cross-sector exit rate of 0.10, using Brent's method on the employment-weighted

exit rate. Steps 4 and 5 are iterated to convergence. Set $\kappa^k = \kappa_0 + \tilde{\kappa}_k$.

- **Step 6 (welfare).** Compute $\vec{W}$ from equation (13) and the stationary distribution $\vec{\pi}$ as the unit eigenvector of $\boldsymbol{P}^\top$.

- **Step 7 (counterfactuals).** Recompute Steps 2, 3, and 6 under each counterfactual $\boldsymbol{\Gamma}$ or belief-weight matrix, **holding $\{\kappa^k\}$ at the values calibrated in Steps 4–5**. Report consumption-equivalent variation from equation (14), evaluated at the baseline weights $\vec{\pi}$.

E.4 Identification

It is worth being explicit about which data identify which parameters, since the model draws on two independent sources.

The five behavioral parameters are identified entirely by the survey. ϕ_0 and ϕ_{cw} come from the within-person relationship between wage prediction errors and coworker shares; ψ_{past}, ψ_{cw}, and ψ_{wage} from the cross-sector variation in perceived offer probabilities within respondent. No transition data enter their estimation.

The 19 adjustment costs are identified by the CPS. The profile $\{\tilde{\kappa}_k\}$ is over-identified: 18 free parameters against 342 conditional destination shares, leaving 324 overidentifying restrictions once the level and the exit-rate moment are counted. Figure E1 plots the model-implied conditional shares against their LEHD counterparts. The fit is good but not mechanical, which is what over-identification should deliver: a correlation of 0.84 and a root mean squared error of 0.029, with the model somewhat compressing the largest destination shares.

One caveat deserves statement. Because the model is specified at the sector level, we proxy the network objects s^{nk}_{cw} and s^{nk}_{past} with LEHD job-to-job flow shares rather than with the survey's own coworker shares, which are measured at the respondent level and for a five-industry choice set rather than all 19. The outflow matrix therefore appears both inside $\boldsymbol{\Gamma}$, as the proxy for where a sector's past coworkers went, and as the source of the calibration target in Step 4. These are different functions of the same matrix—a network intensity in the first case, a conditional destination distribution the model must reproduce through the full $\boldsymbol{\Gamma}$-to-$\boldsymbol{\mu}$ mapping in the second—and the ψ parameters that govern the first are estimated from survey data alone. But the arrangement means the transition fit in Figure E1 should be read as a consistency check rather than as an out-of-sample test.

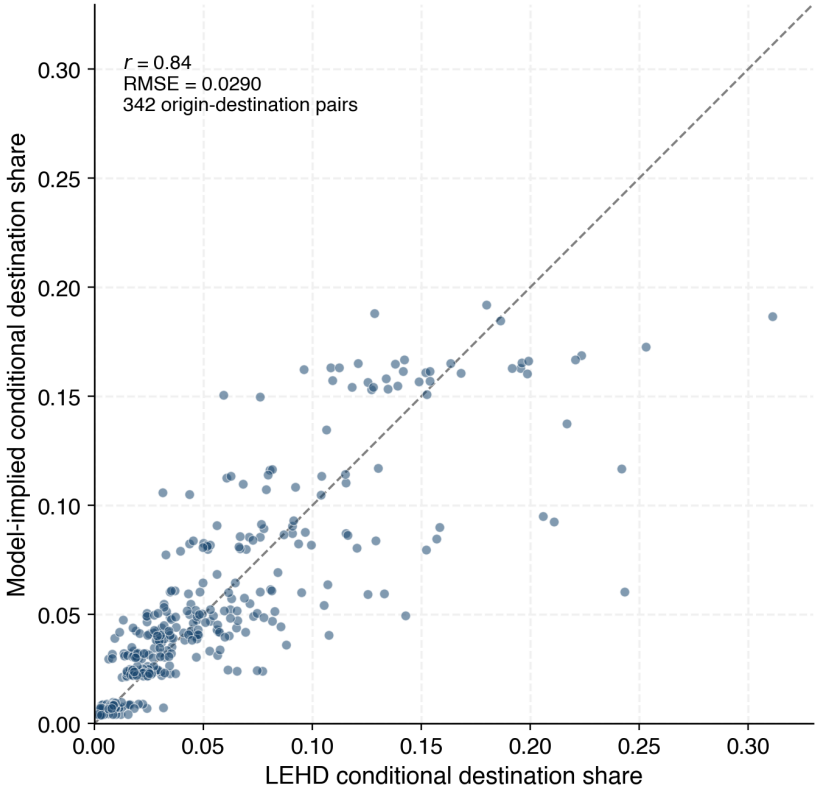
The transition target warrants a full description, since it is the one input not drawn from the survey or the CPS. It is constructed from the LEHD Job-to-Job Flows Origin-Destination (J2JOD) public-use files, state-level, unadjusted, tabulated by origin sector and destination sector. These files are published quarterly. We aggregate the 51 state files to the national level, average quarters within each year, average across years 2000–2019, and take the flow measure to be employer-to-employer transitions plus hires from persistent nonemployment ($EE + AQHire$). Each origin row is then normalised to sum to one. Two features follow. First, sector-by-sector cells are published only in the state files, not

the national ones, which is why the aggregation runs through the states. Second, the resulting object is a destination composition conditional on a move, with an average of 33.7% of moves remaining within the origin sector; it therefore carries no frequency of its own and is close to invariant to the counting interval. The model period is set instead by the discount factor $\beta = 0.95$ and the aggregate annual cross-sector exit rate used to calibrate κ_0, both annual, consistent with the CPS reference period from which sector wages are constructed.

E.5 Bootstrap Procedure

All confidence intervals in Section 5 are percentile bootstrap intervals from 200 resamples. Each replicate draws respondents with replacement from the survey, re-estimates the belief regression and the fractional logit, rebuilds $\mathbf{\Gamma}$, re-calibrates the full cost vector $\{\kappa^k\}$ against the CPS targets, and recomputes every counterfactual. The bootstrap therefore propagates survey sampling uncertainty through the entire calibration chain rather than treating the calibrated parameters as known. It does not incorporate sampling uncertainty in the LEHD transition matrix or in CPS sector wages, so the reported intervals should be read as understating total uncertainty by an unknown but probably modest amount.

Figure E1: Model Fit: Model-Implied vs. LEHD Conditional Destination Shares



Note: Each point is one of the 342 off-diagonal origin-destination pairs. The horizontal axis is the LEHD share of movers from an origin who go to a given destination; the vertical axis is the model-implied share. The dashed line is the 45-degree line. The cost profile $\{\tilde{\kappa}_k\}$ is calibrated to minimize the employment-weighted distance between the two, with the level κ_0 set separately by the aggregate exit rate.

E.6 Sensitivity to the Exit-Rate Target

The transition data identify the composition of cross-sector moves but not their rate, so the calibration requires one moment stated in levels to pin down the cost level κ_0. Table E1 varies that moment. The cost profile $\{\tilde{\kappa}_k\}$ is held at its baseline calibration and κ_0 is re-solved at each target, which isolates the channel through which the assumption operates.

Each individual channel keeps its sign across the range. Correcting wage beliefs is worth between 0.70 and 1.28 percent, removing coworker wage information costs between 0.06 and 0.11 percent, removing coworker-linked access gains between 0.10 and 0.26 percent, and de-segregating the network gains between 0.16 and 0.22 percent. The net effect of removing all coworker channels, by contrast, moves from -0.005 to $+0.203$ and is not robustly signed. This is the basis for the statement in Section 5.4 that we report the magnitude of the net effect without interpreting its sign. Of the quantities we report, the belief benchmark and the access channel are the most sensitive to this moment, each varying by roughly a factor of two and a half across the range; they move in opposite directions, since a higher exit rate implies lower adjustment costs and therefore both cheaper reallocation and a smaller gain from correcting beliefs.

Table E1: Sensitivity to the Aggregate Exit-Rate Target

Exit rate target	κ_0	Perfect beliefs	Remove wage information	Remove access	Remove all channels	De-segregate
0.08	9.93	+2.757	-0.085	+0.103	+0.021	+0.052
0.10	9.38	+2.585	-0.078	+0.141	+0.066	+0.056
0.13	**8.71**	**+2.229**	**-0.066**	**+0.184**	**+0.121**	**+0.061**
0.16	8.14	+1.887	-0.056	+0.218	+0.164	+0.064
0.20	7.47	+1.513	-0.045	+0.252	+0.208	+0.068
Range		[+1.51, +2.76]	[-0.085, -0.045]	[+0.103, +0.252]	[+0.021, +0.208]	[+0.052, +0.068]

Note: Each row re-solves the cost level κ_0 to match the stated aggregate annual cross-sector exit rate, holding the cost profile $\{\tilde{\kappa}_k\}$ at its baseline calibration and all survey parameters fixed. Entries are consumption-equivalent variation relative to the coworker baseline at that exit rate, in percent. The baseline specification (0.13, in bold) is the one reported in Section 5.

E.7 Model Exhibits

This subsection collects the calibration inputs and the tabular counterparts of the figures reported in Section 5.

Table E2: Calibrated Parameter Values

Parameter	Description	Estimate	Std. Error	95% CI
Wage beliefs, equation (9)				
$\hat{\phi}_0$	Baseline weight on true destination wage	0.515	0.017	[0.482, 0.547]
$\hat{\phi}_{cw}$	Added weight per unit coworker share	0.129	0.060	[0.016, 0.239]
Perceived offer probability, equation (10)				
$\hat{\psi}_{past}$	Past-coworker share	0.316	0.230	[0.017, 0.884]
$\hat{\psi}_{cw}$	Current-coworker share	0.526	0.272	[0.196, 1.187]
$\hat{\psi}_{wage}$	Destination−origin log wage gap	-0.267	0.182	[-0.795, -0.105]
Adjustment costs, calibrated to CPS transitions				
$\bar{\kappa}$	Mean adjustment cost	8.710	0.056	[8.562, 8.778]
κ^k range	Across the 19 destination sectors	[5.20, 13.45]		
Fixed parameters				
β	Discount factor	0.95		
ν	Scale of the taste shock	2		

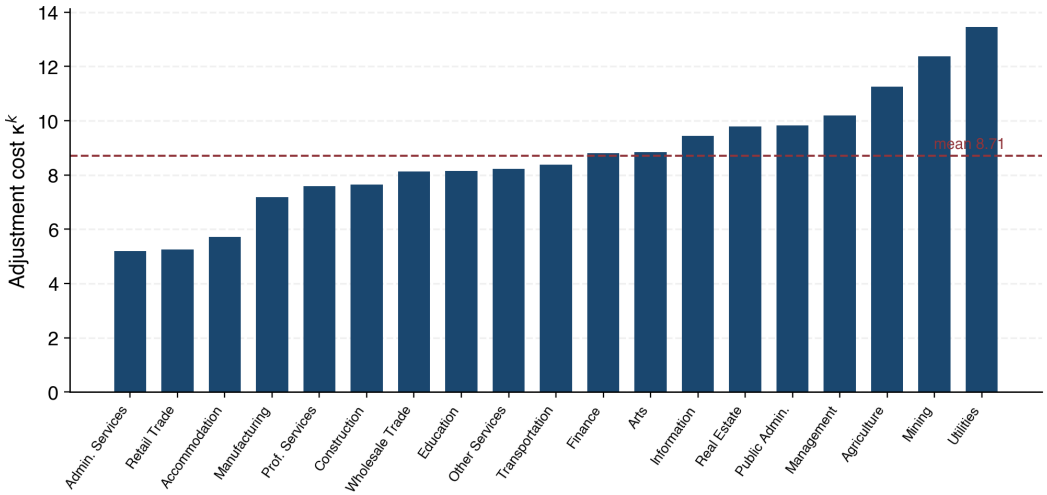
Note: Estimates are full-sample point estimates. Standard errors and 95% percentile confidence intervals come from 200 bootstrap resamples over survey respondents; each replicate re-estimates the survey moments, the $\mathbf{\Gamma}$ matrix, the cost vector $\{\kappa^k\}$, and welfare. Because the offer-probability equation is a fractional logit, the bootstrap distribution of the ψ parameters is right-skewed and the intervals are correspondingly asymmetric around the point estimates.

Table E3: Calibration Targets and Model Fit

Moment	Target	Model	Source
Conditional destination shares (342 off-diagonal pairs)			
RMSE		0.0290	LEHD J2J
Correlation (r)		0.84	LEHD J2J
Aggregate cross-sector exit rate			
Annual exit rate	0.100	0.100	Calibration target
Moments	342 conditional shares + 1 exit rate		
Parameters	19 destination costs $\{\kappa^k\}$ ($= \kappa_0 + \tilde{\kappa}_k$, $\sum_k \tilde{\kappa}_k = 0$)		
Overidentifying restrictions	324		

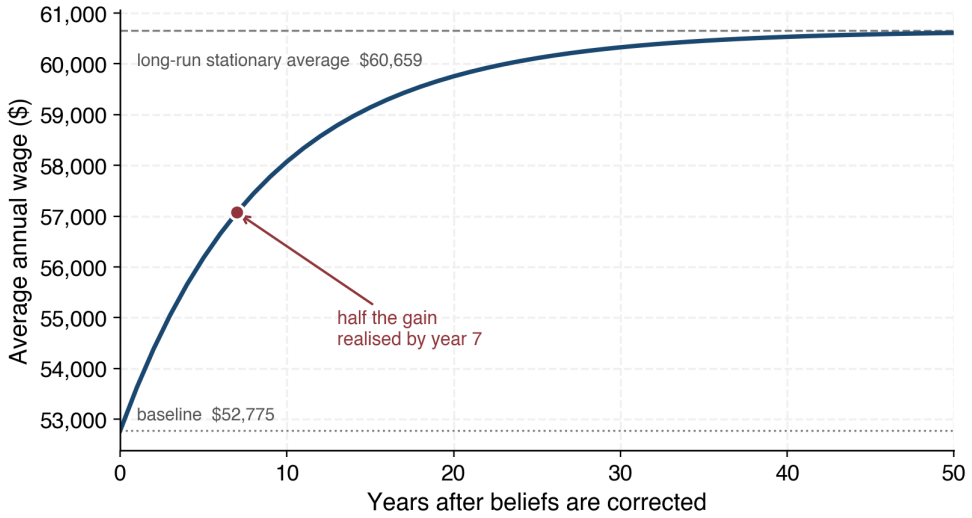
Note: The cost profile $\{\tilde{\kappa}_k\}$ is calibrated to the LEHD Job-to-Job Flows conditional destination distribution—the share of movers from each origin going to each destination—and the level κ_0 to the aggregate annual cross-sector exit rate, which is imposed as a calibration target rather than taken from the transition file, since a conditional distribution does not identify the rate of moving. RMSE and correlation are computed across the 342 off-diagonal origin-destination pairs. The exit rate is matched exactly by construction of κ_0.

Figure E2: Estimated Destination-Specific Adjustment Costs (κ^k)



Note: Estimated destination-specific adjustment cost κ^k for each of the 19 destination sectors, ordered from lowest to highest. The profile is calibrated to the LEHD conditional destination distribution and the level to the aggregate cross-sector exit rate. The dashed line marks the mean.

Figure E3: Transition Path of Average Wages After Beliefs Are Corrected



Note: Employment-weighted geometric mean sector wage, $\exp(\vec{\pi}_t^{\top} \ln \vec{w})$, in each year after wage beliefs are corrected at date zero. The economy starts from the baseline stationary employment distribution and evolves under the counterfactual transition matrix. The dotted line marks the baseline average and the dashed line the new long-run stationary average. Half the eventual gain is realized after roughly a decade.

Table E4: Channel Decomposition: Welfare Effects of Coworker Networks

Counterfactual	Total CEV (%)	Wage component	Search costs	Option value	95% CI on total
Remove coworker wage information ($\phi_{cw} = 0$)	-0.066	-0.138	-0.287	+0.360	[-0.130, -0.009]
Remove coworker-linked access ($\psi_{past} = \psi_{cw} = 0$)	+0.184	+0.102	-0.181	+0.264	[+0.079, +0.351]
Perfect wage beliefs (frontier)	+2.229				[+1.967, +2.547]

Note: Entries are consumption-equivalent variation relative to the coworker baseline, in percent, with $\{\kappa^k\}$ held at the baseline calibration. The total is decomposed into the three additive components of the welfare recursion: the wage flow, search costs paid on applications, and the logit option value $\nu H(\boldsymbol{\mu})$. Only the wage component reflects reallocation; each counterfactual flattens a matrix and therefore raises the option value mechanically, by a similar amount in both cases. Confidence intervals are percentile bootstrap intervals on the total from 400 resamples over survey respondents.

Table E5: Directing versus Broadening Coworker Networks

Design	Reach	CEV (%)	Wage component	Option value	Gain to low-wage 6	Low-to-high transition (%)	CEV per pt. treated
Baseline						2.948	
Directed	10%	+0.007	+0.019	-0.057	+0.018	2.958	0.0016
	25%	+0.031	+0.052	-0.090	+0.054	2.974	0.0028
	50%	+0.088	+0.114	-0.081	+0.131	3.004	0.0040
	100%	+0.252	+0.262	+0.081	+0.331	3.068	0.0057
Broad	10%	-0.000	+0.003	-0.016	-0.000	2.949	-0.0001
	25%	+0.002	+0.008	-0.033	+0.002	2.952	0.0001
	50%	+0.010	+0.019	-0.044	+0.011	2.956	0.0004
	100%	+0.042	+0.050	-0.020	+0.043	2.967	0.0010

Note: The six lowest-wage sectors, employing 44% of workers, are treated. Each intervention holds the treated sectors' total tie intensity fixed and changes only the destination mix; reach is the share of that mix redirected. "Directed" points ties toward the six highest-paying sectors; "broad" spreads them in proportion to destination employment. Entries are consumption-equivalent variation relative to the coworker baseline, in percent, with $\{\kappa^k\}$ held at the baseline calibration. Wage component and option value are two of the three additive terms of the welfare recursion; search costs make up the remainder. The final column divides the total by the share of employment actually treated.

Table E6: Sensitivity to the Perceived-Access Assumption

ρ	α	Mean q^{nk}	Remove access	De-segregate	Perfect beliefs
0.50	+0.0223	0.509	+0.086	+0.033	+2.200
0.75	+0.0111	0.509	+0.138	+0.047	+2.219
1.00	**-0.0000**	**0.508**	**+0.184**	**+0.061**	**+2.229**
1.25	-0.0111	0.507	+0.225	+0.074	+2.229
1.50	-0.0222	0.506	+0.260	+0.087	+2.219
Range			[+0.086, +0.260]	[+0.033, +0.087]	[+2.200, +2.229]

Note: Applications are directed by perceived offer probabilities $\tilde{\Gamma}^{nk}$; realized transitions use q^{nk} with $\text{logit}\, q^{nk} = \alpha + \rho\, \text{logit}\, \tilde{\Gamma}^{nk}$, where α is solved at each ρ to preserve the aggregate cross-sector exit rate. $\rho = 1$ (bold) reproduces the calibrated-beliefs baseline. Entries are consumption-equivalent variation relative to the coworker baseline at that ρ, in percent. $\{\kappa^k\}$ is held at the baseline calibration throughout.